



CAE REPORT

深度探索如何應用衛星資料輔助午後雷陣雨短延時預測

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BACKGROUND

午後雷陣雨（thunderstorm）是臺灣夏天常見的降雨型態之一，由於其發生區域小、事件短、強度高，預測延時及可以應變之時間非常短，容易造成都市淹水。此外，隨著氣候變遷，午後雷陣雨發生之頻率、強度都有提高之趨勢，因此如何提升午後雷陣雨之預測延時及準確度，對於臺灣治災防洪上極具重要意義。因此如何提升午後雷陣雨之預測延時及準確度，對於臺灣治災防洪上極具重要意義。

INTRODUCTION

- 午後雷陣雨為臺灣常見氣候
- 事件短、強度高且易造成淹水
- 研究透過雷達資料萃取出雨胞提供雷陣雨預測



PROCESS



STEP 1

取得台北地區近
幾年午後雷陣雨
事件

STEP 2

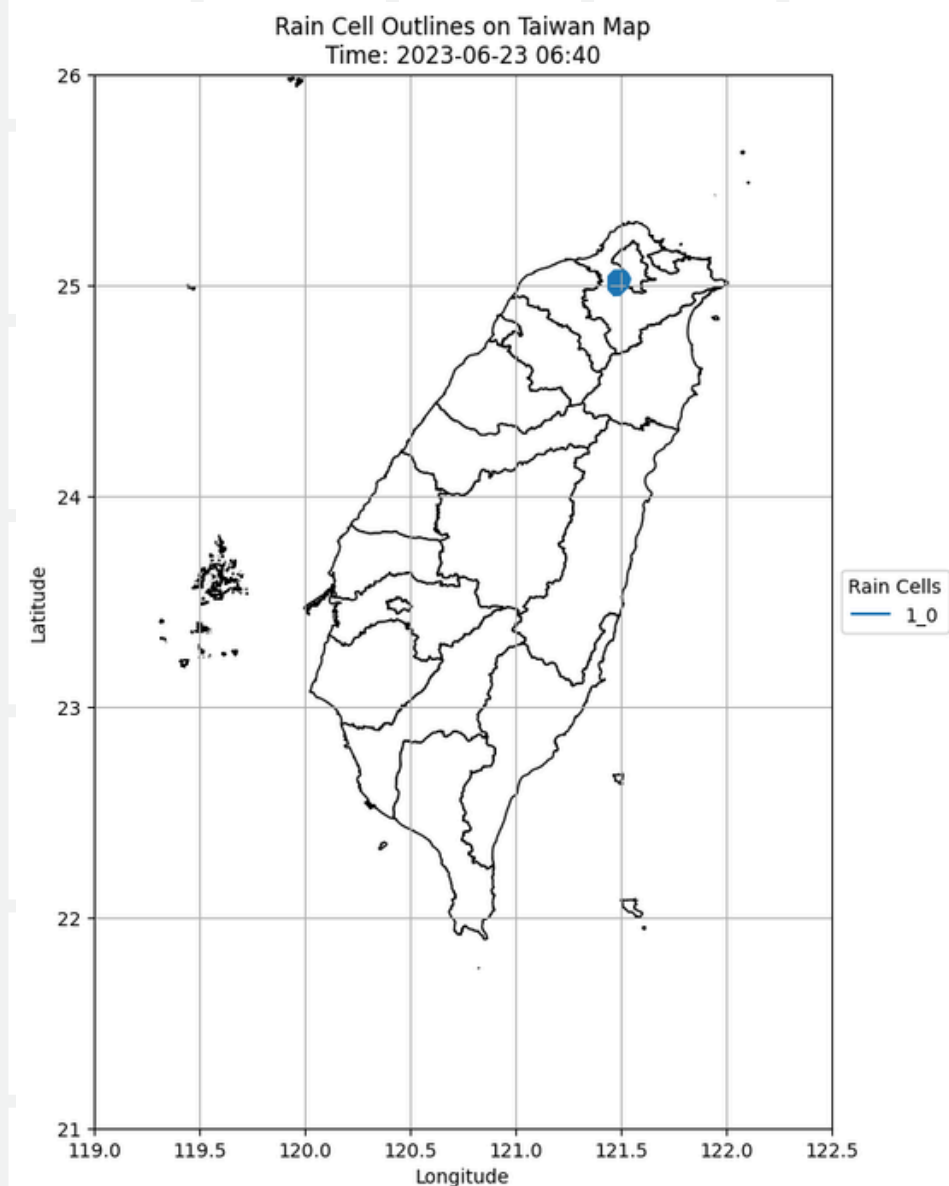
獲取各事件雷達
回波資料(中央
氣象署)並進行
雨胞辨識

STEP 3

將偵測到的雨胞
投影在臺灣地圖
上並追蹤

RADAR

- 從中央氣象署取的雷達資料(每10分鐘一筆)
- 設定事件時間(2023 6月~9月雷陣雨事件)
- 萃取雨胞資訊
- 根據雨胞資訊（時間、位置），將雨胞投影在臺灣地圖



RAIN CELL TRACKING

	英國	臺灣
最小降雨面積	3 pixel	1 pixel
VIL閾值	5 kgm ⁽⁻²⁾	3 kgm ⁽⁻²⁾
平滑寬度	1 pixel	3 pixel
資料時間	UTC+0	UTC+8

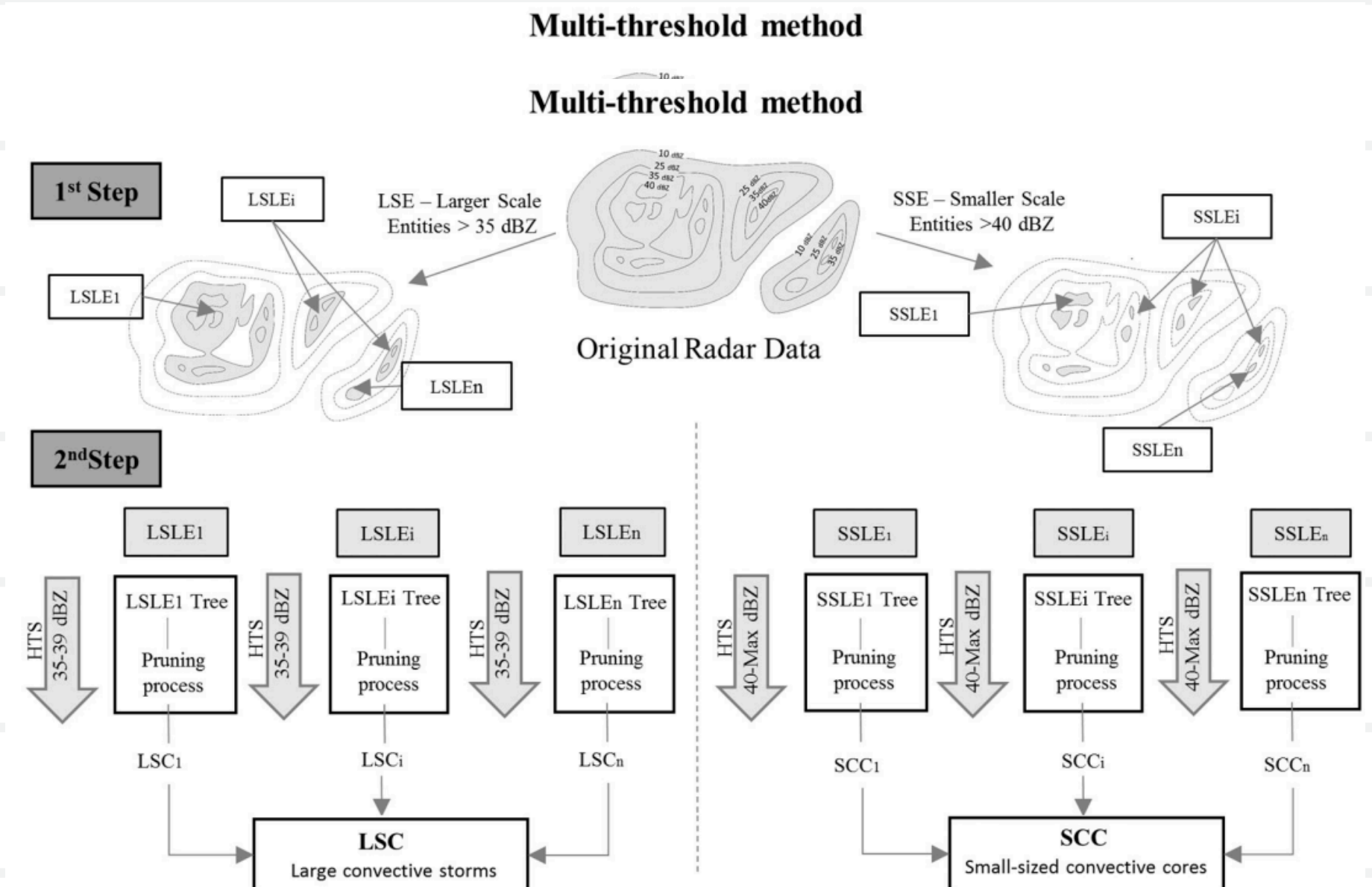
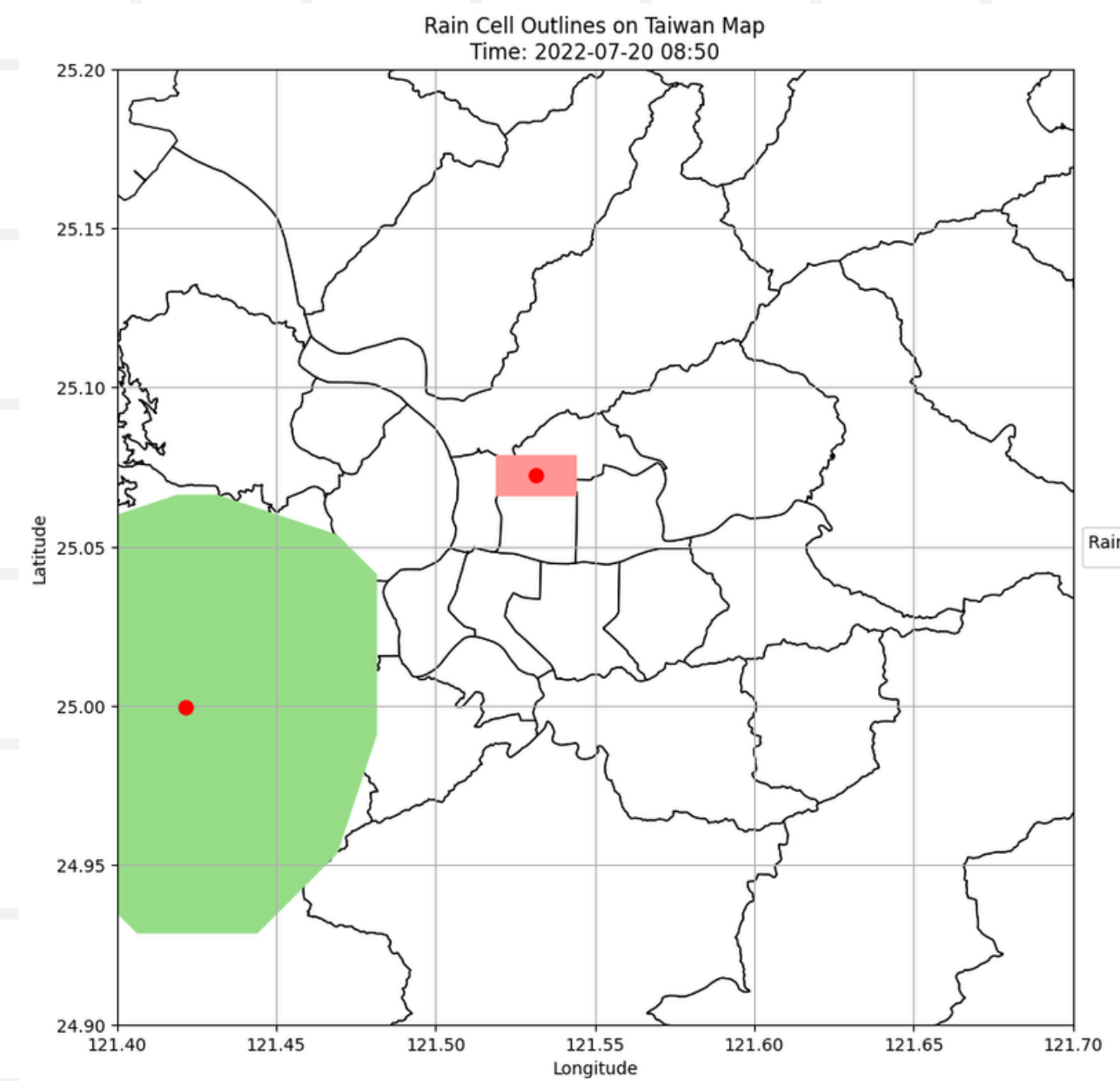
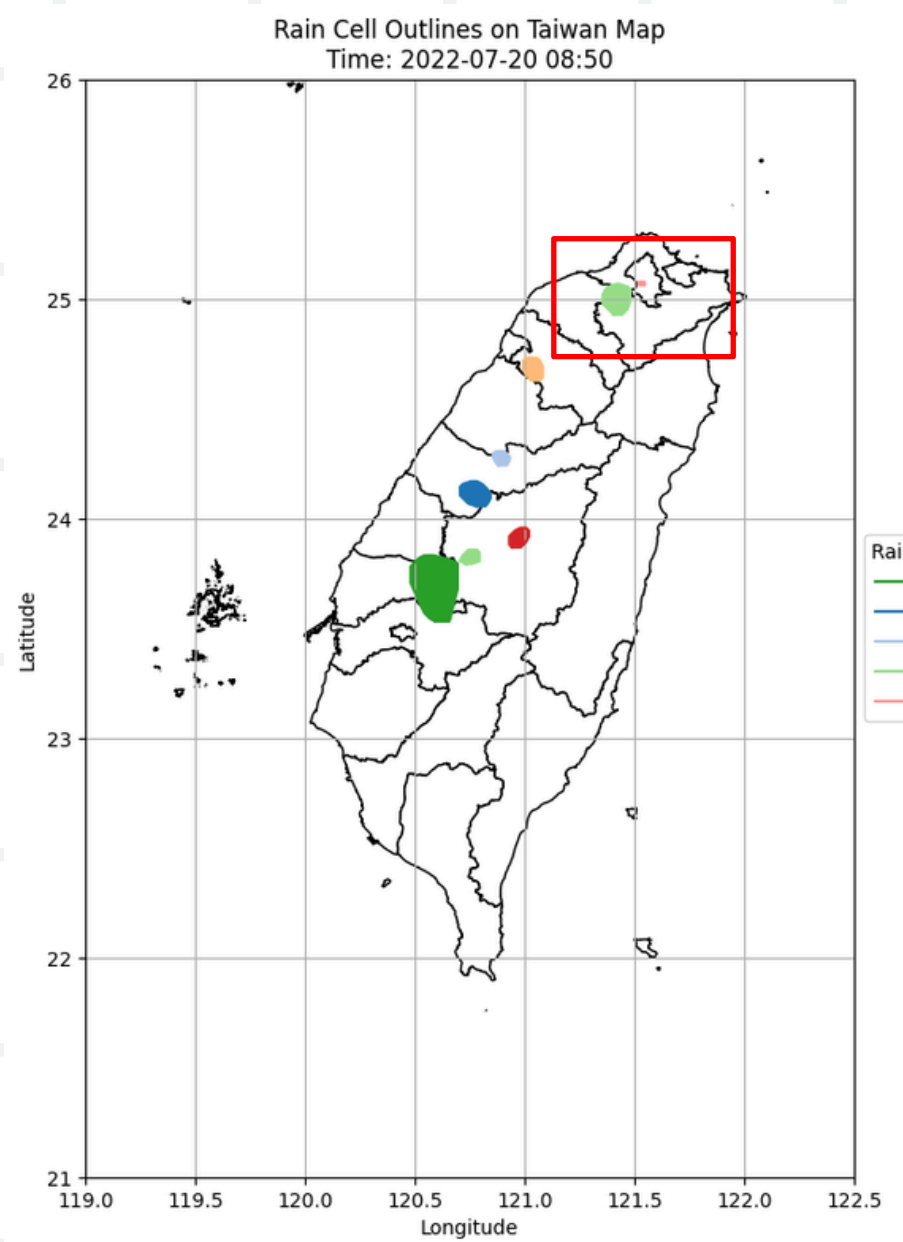


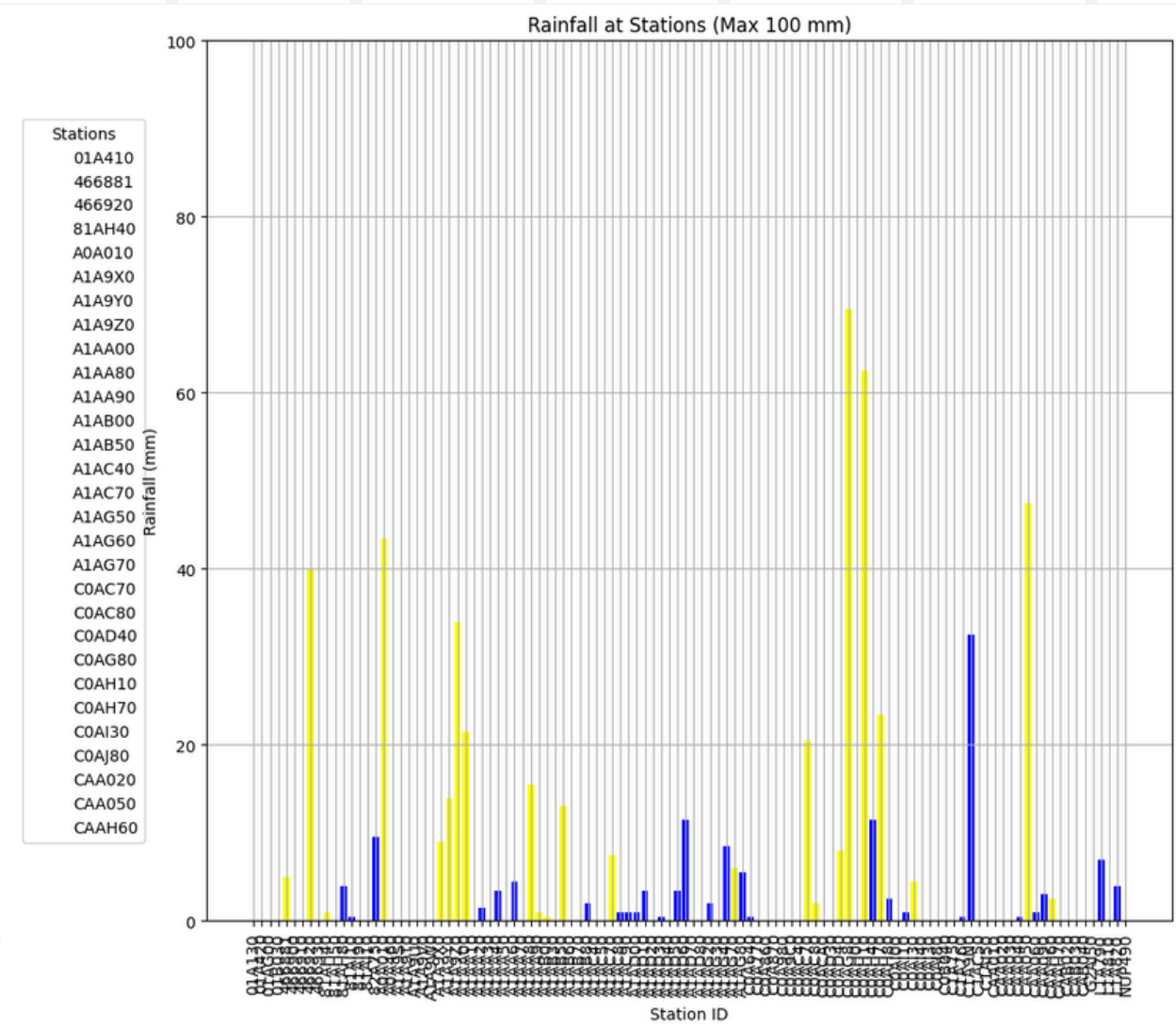
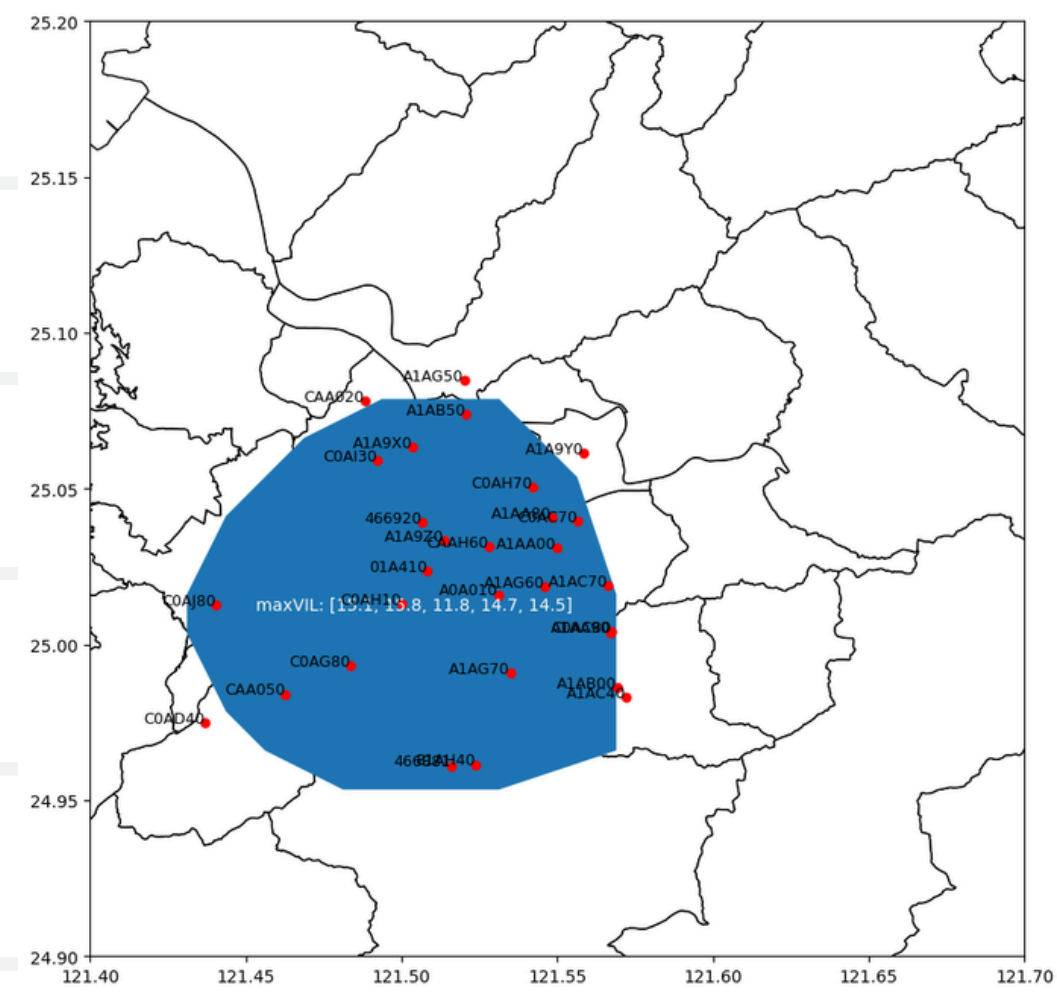
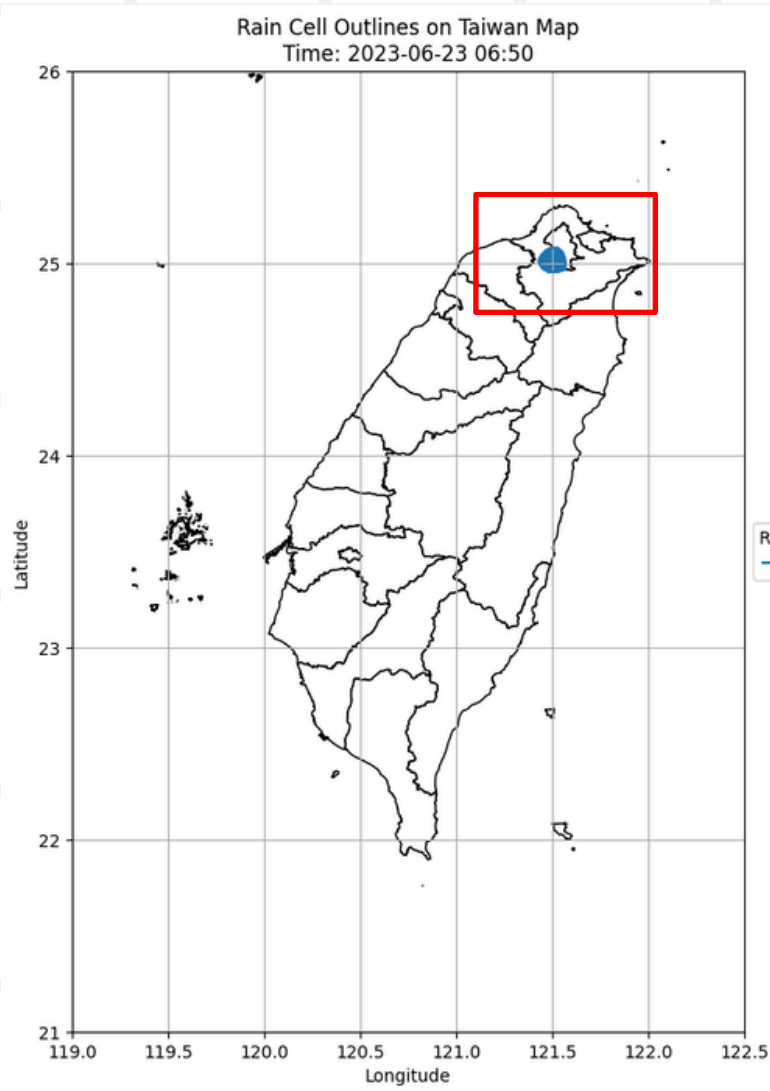
Fig1. Rain Cell Tracking Process

<https://www.sciencedirect.com/journal/atmospheric-research>

RESULT



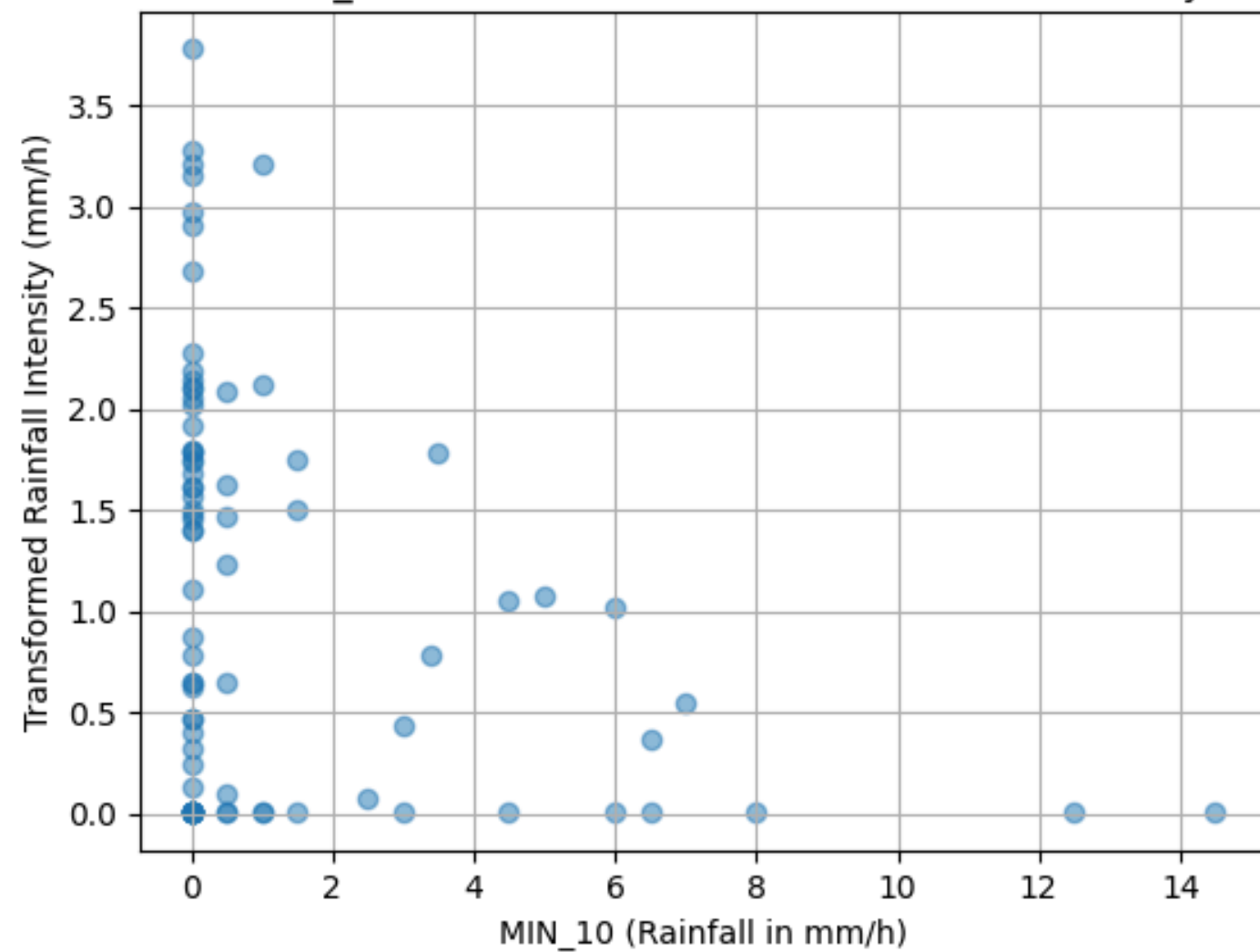
RESULT



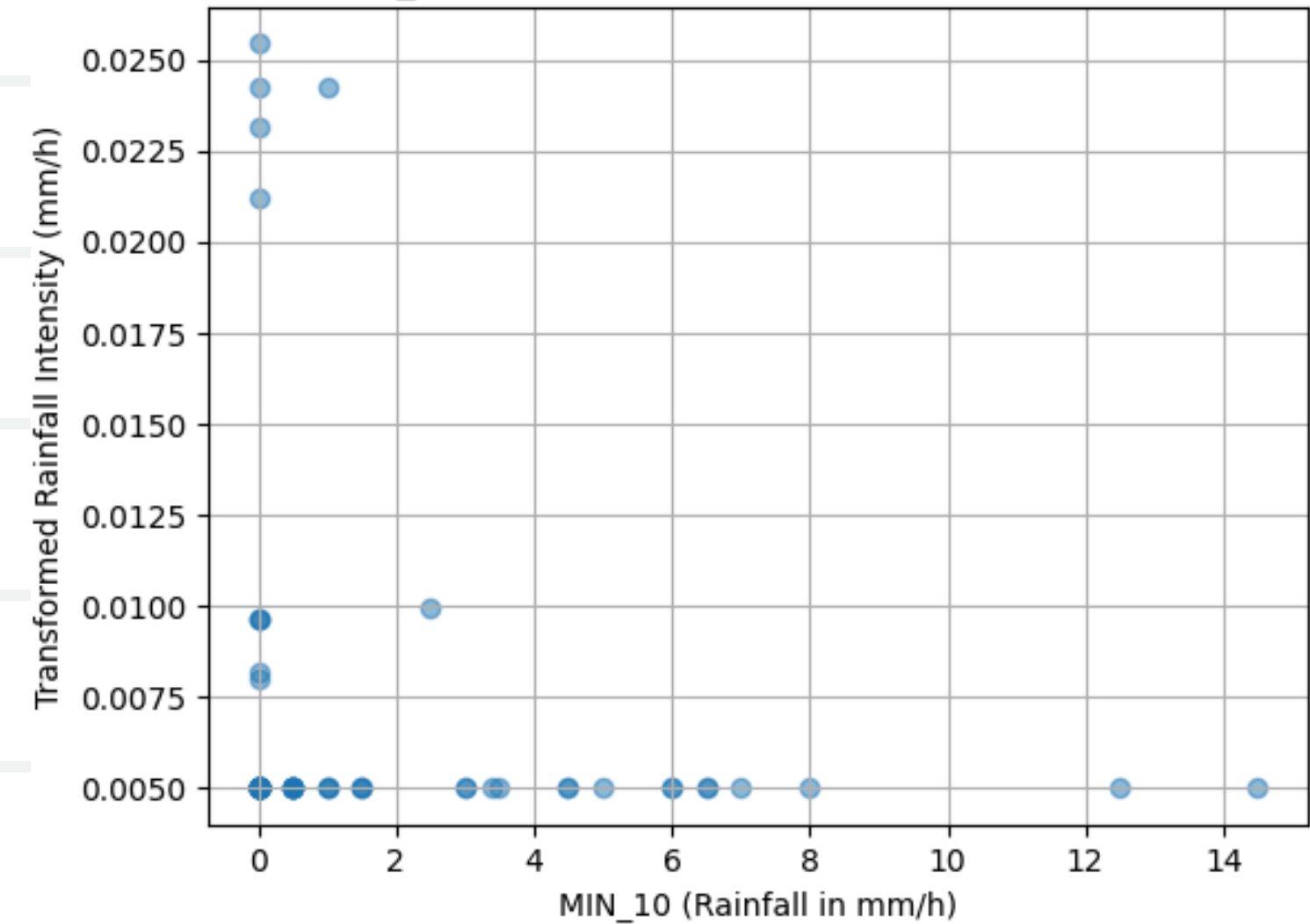
RADAR VS RAIN STATION



Station Rainfall (MIN_10) vs Nearest Transformed Rainfall Intensity (within 5km)



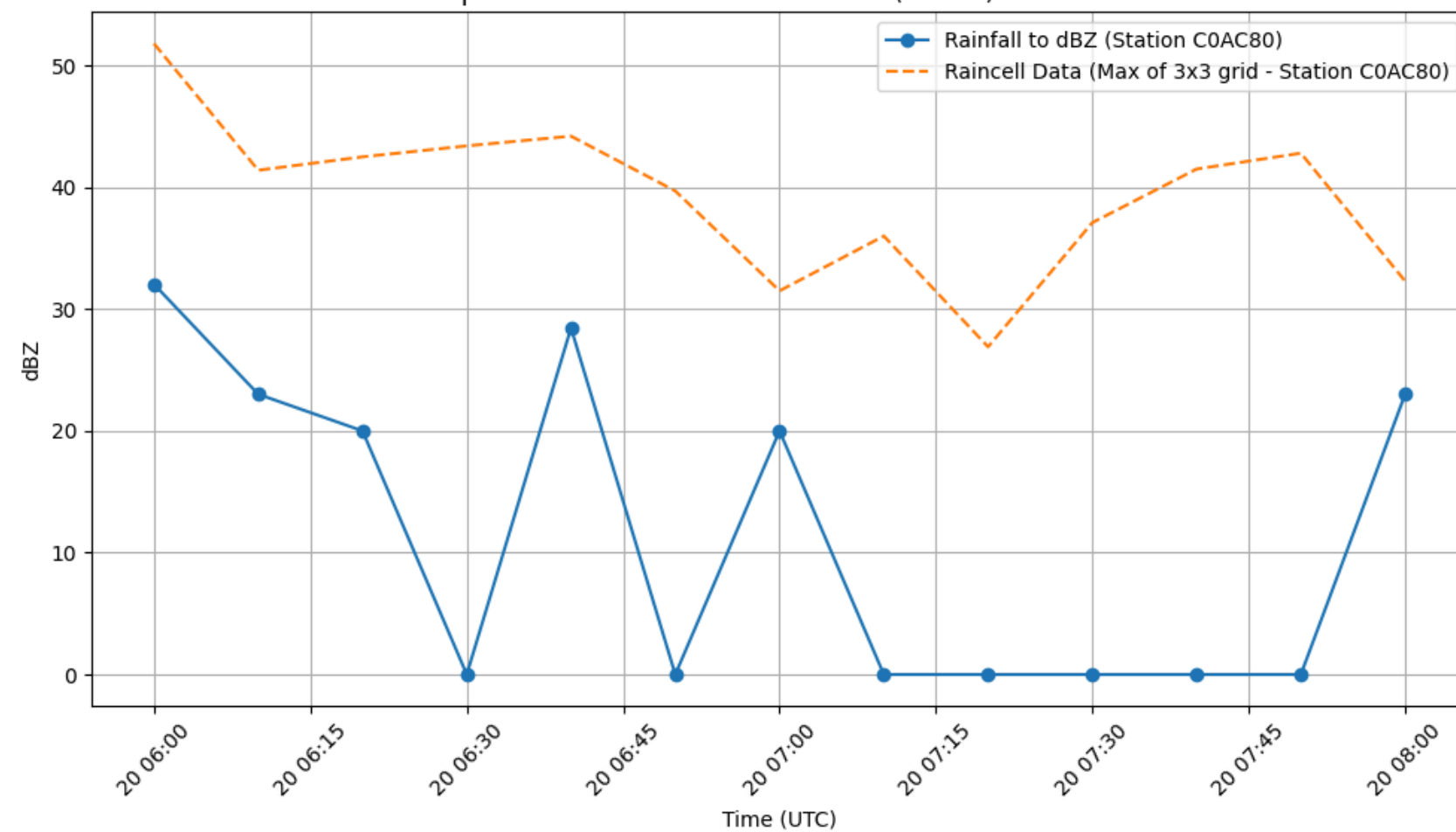
Station Rainfall (MIN_10) vs Nearest Transformed Rainfall Intensity (within 5km)



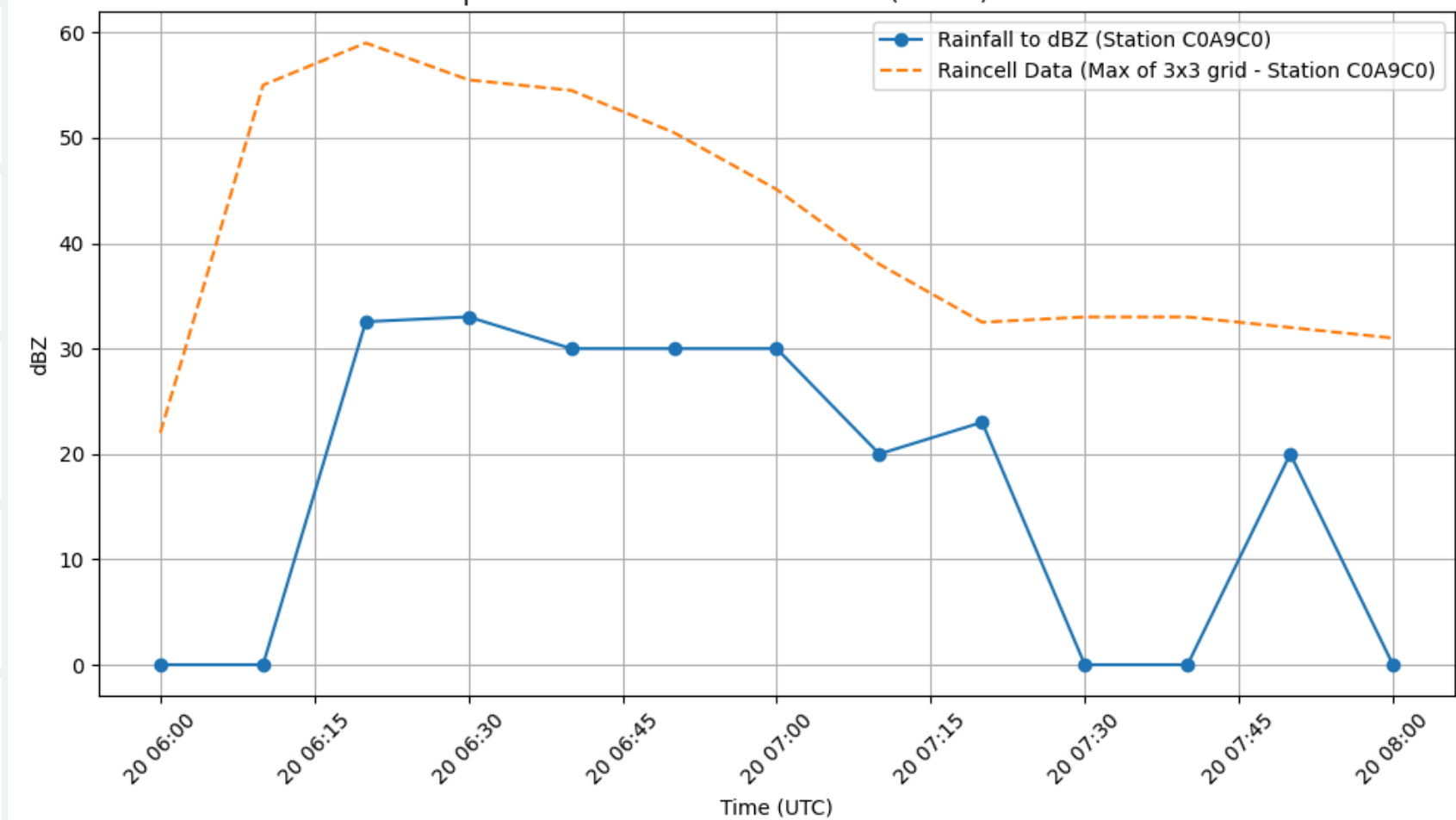
RADAR VS RAIN STATION



Comparison of Station C0AC80 Rainfall (to dBZ) and Raincell



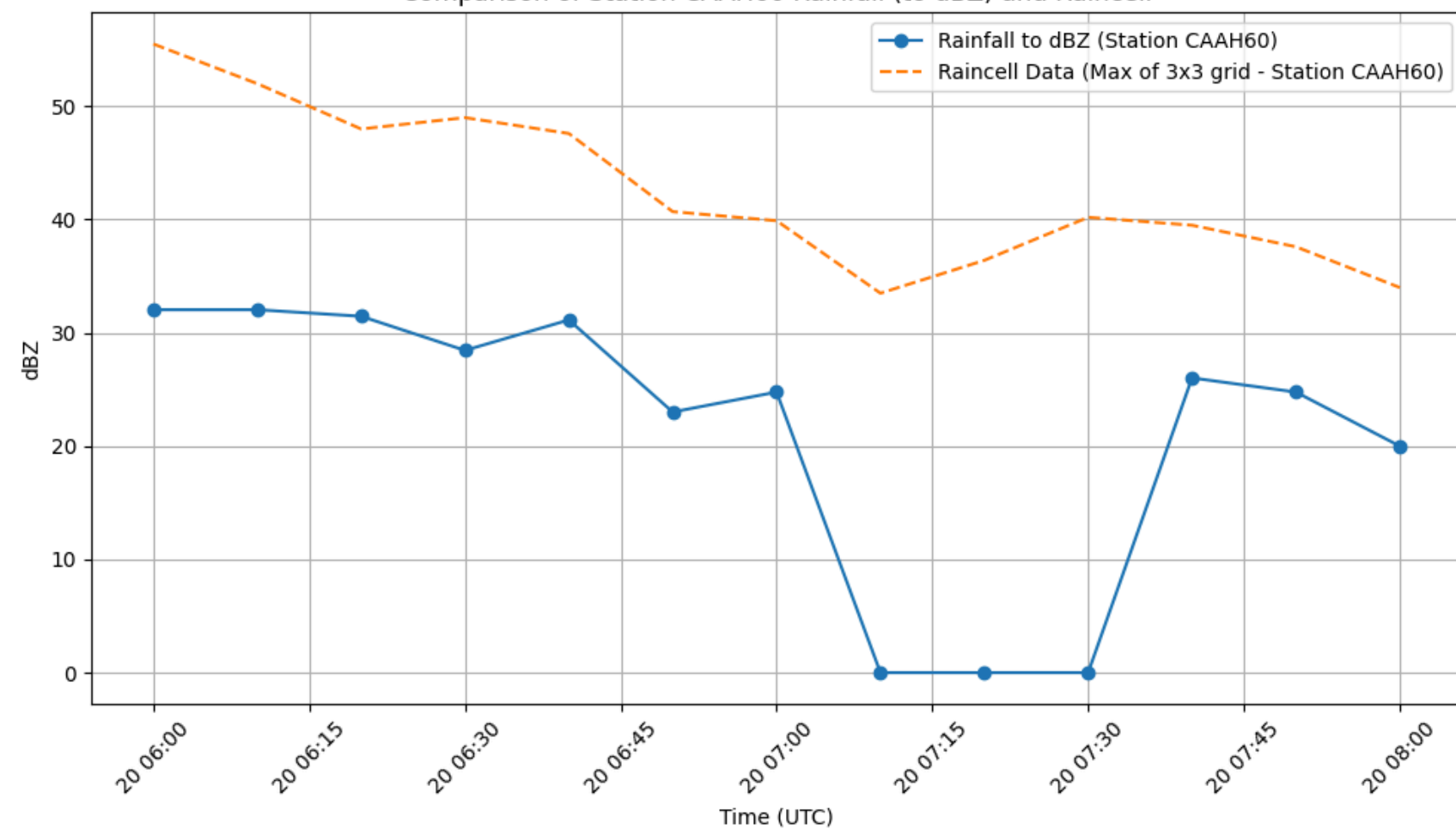
Comparison of Station C0A9C0 Rainfall (to dBZ) and Raincell



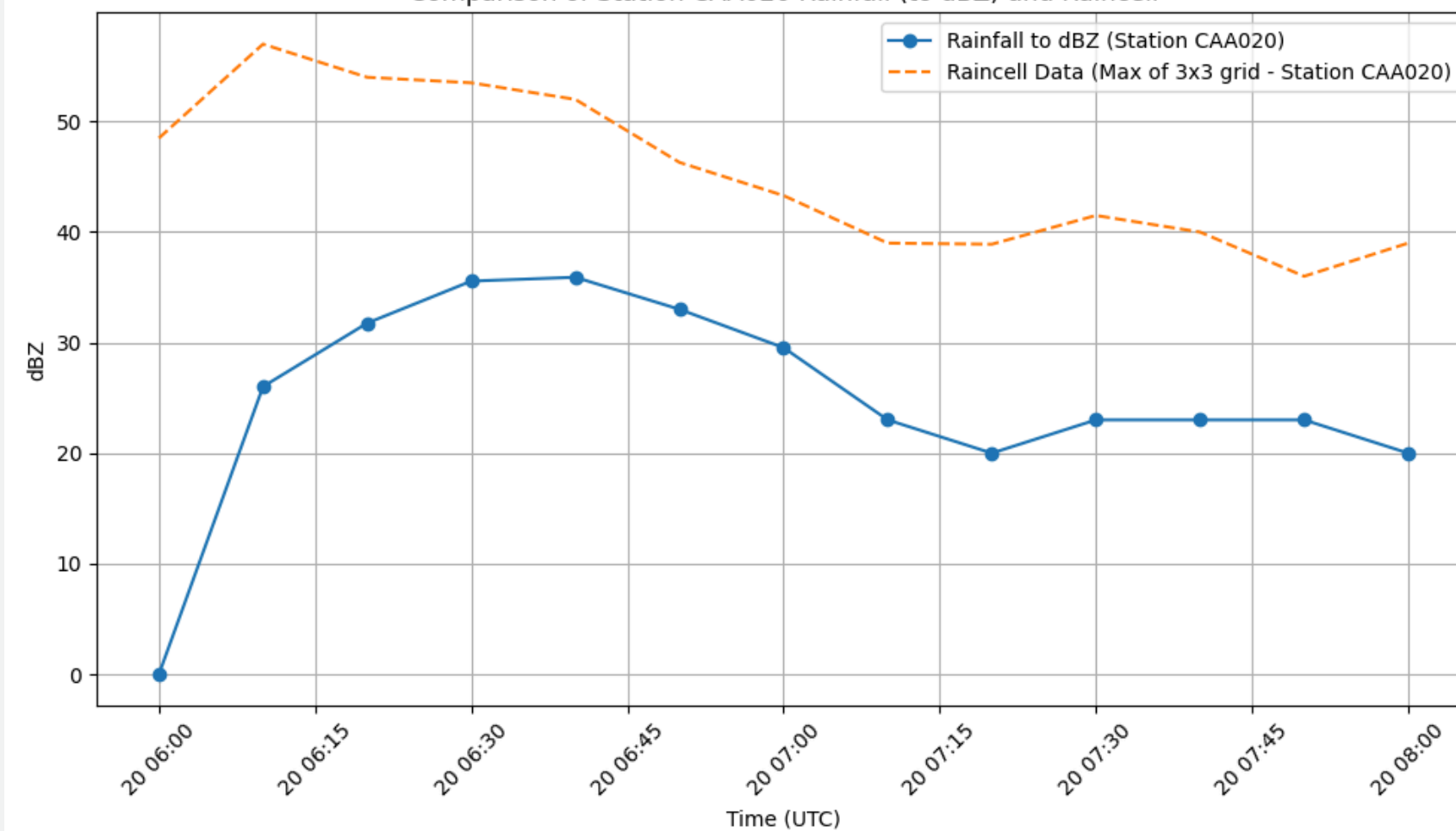
RADAR VS RAIN STATION



Comparison of Station CAAH60 Rainfall (to dBZ) and Raincell



Comparison of Station CAA020 Rainfall (to dBZ) and Raincell



RADAR VS RAIN STATION



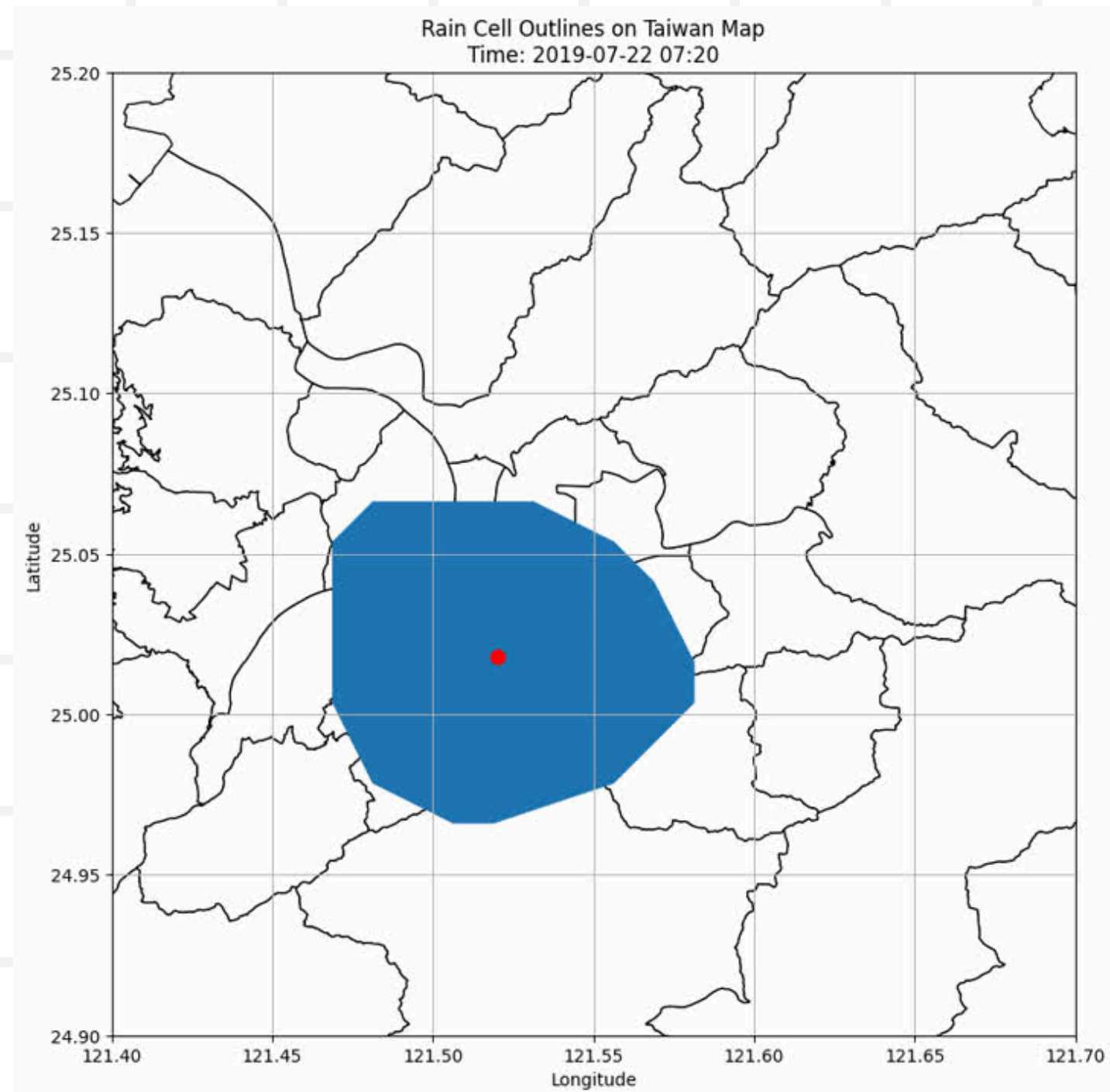
- 將雨量站座標轉移到矩陣
- 找出雨胞經過的雨量站
C0A9C0、C0AC80、CAA020、CAAH60
- 比較雨量站和雨胞DBZ
- 雨量站有大量缺值: 圖形不合
- 圖形趨勢相同，但有時間誤差
- 雨胞估計dbz>實際雨量站測值

RAIN CELL PARAMETER

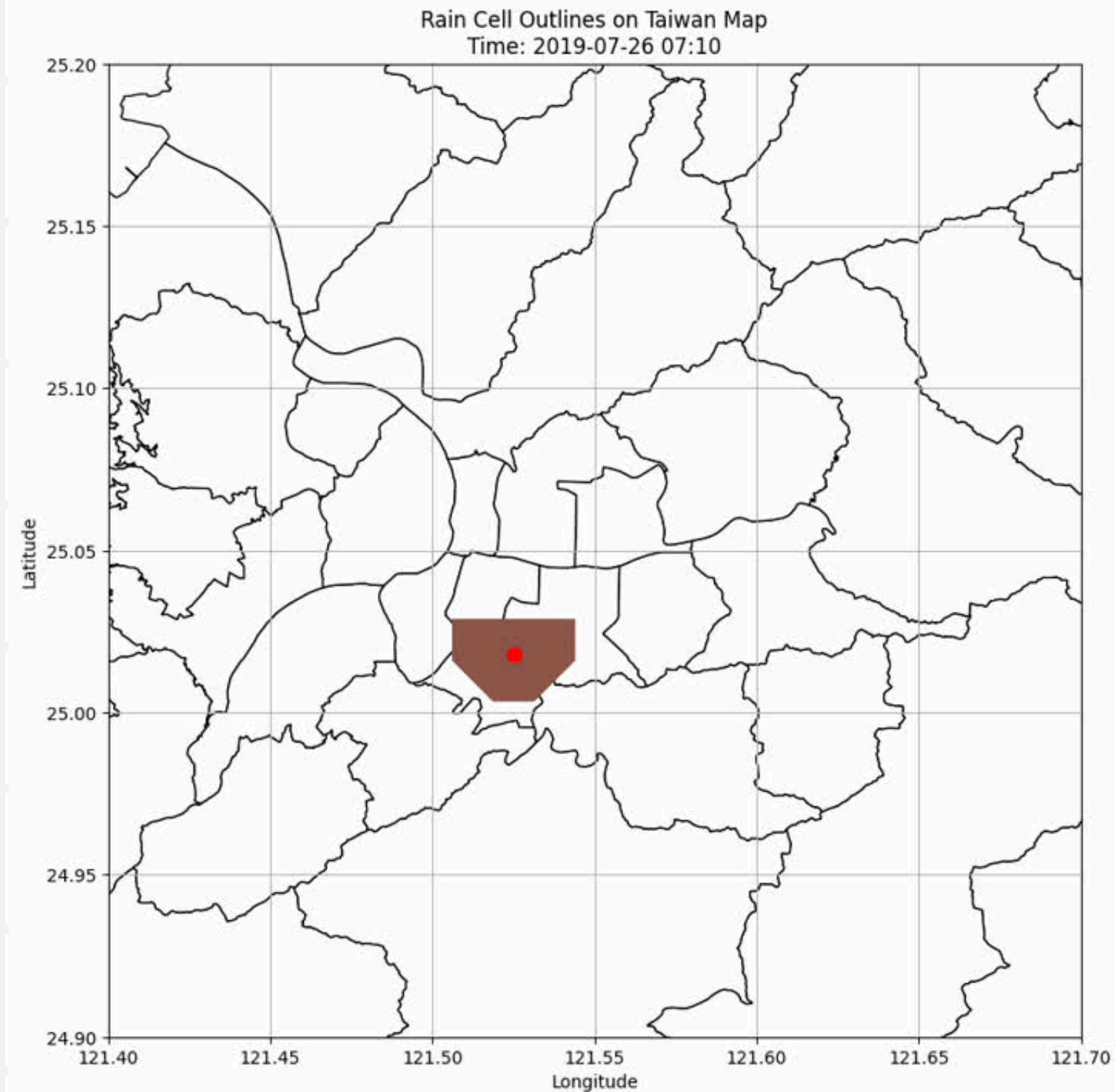


```
saliency = 1
print("Min area:", saliency)
sm_width_pixels = 3.0
print("Smooth width:", sm_width_pixels)
q_bin_min_dbz = 3
print("Min thresh:", q_bin_min_dbz)
q_bin_max_dbz = 20.0
print("Max thresh:", q_bin_max_dbz)
q_bin_width_dbz = 1.0
print("Bin width:", q_bin_width_dbz)
```

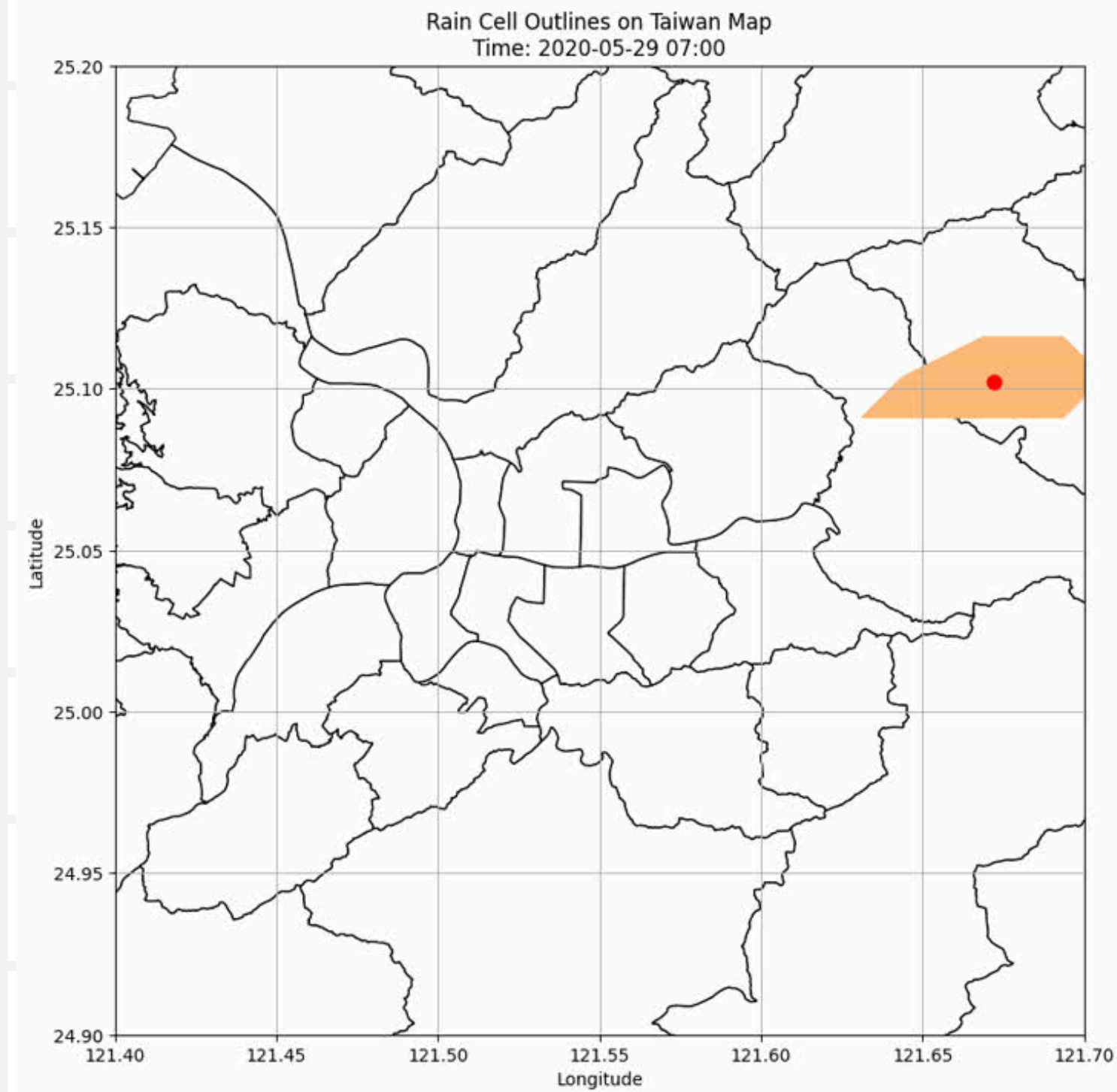
RAIN CELL GIF



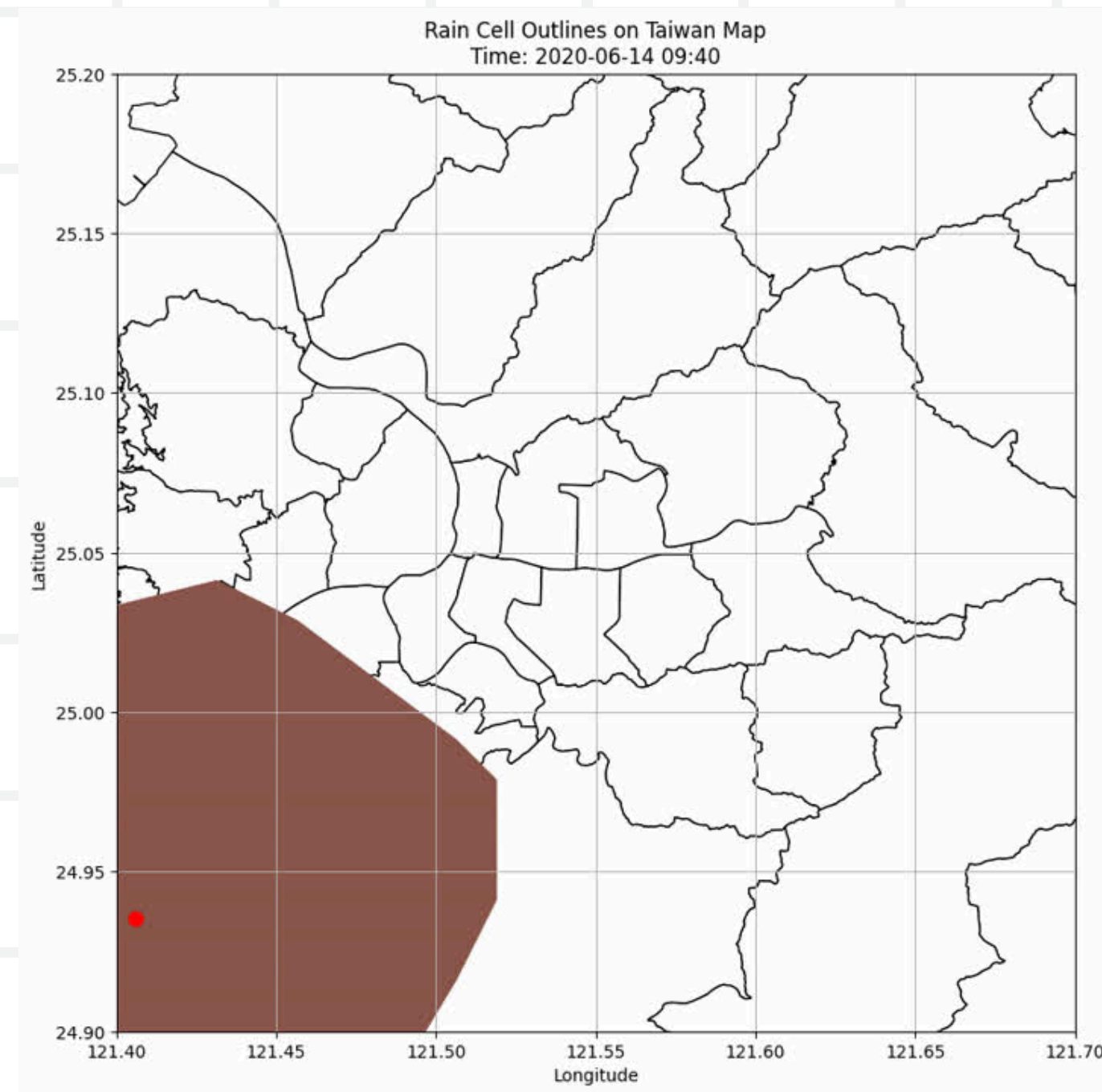
RAIN CELL GIF



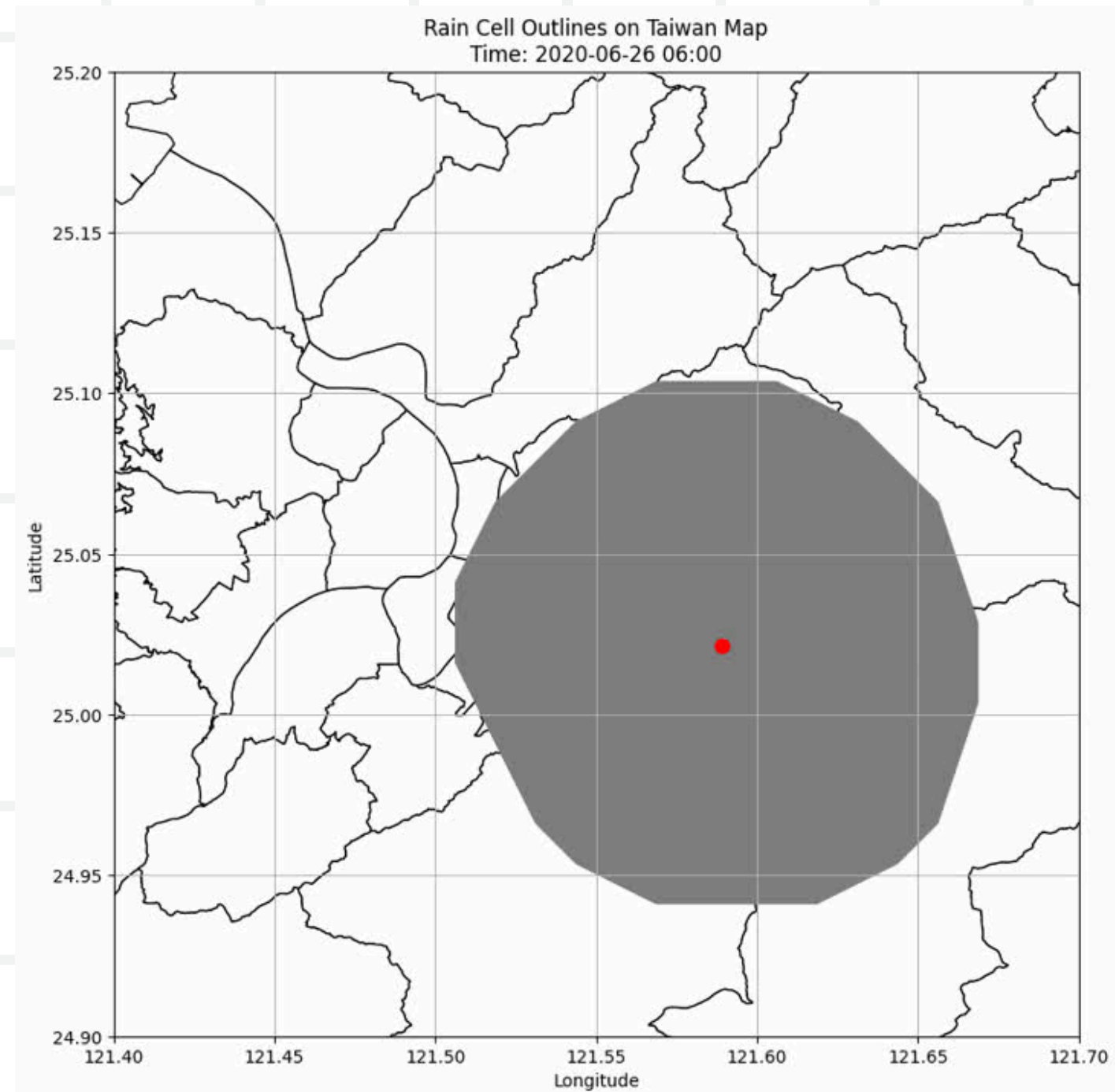
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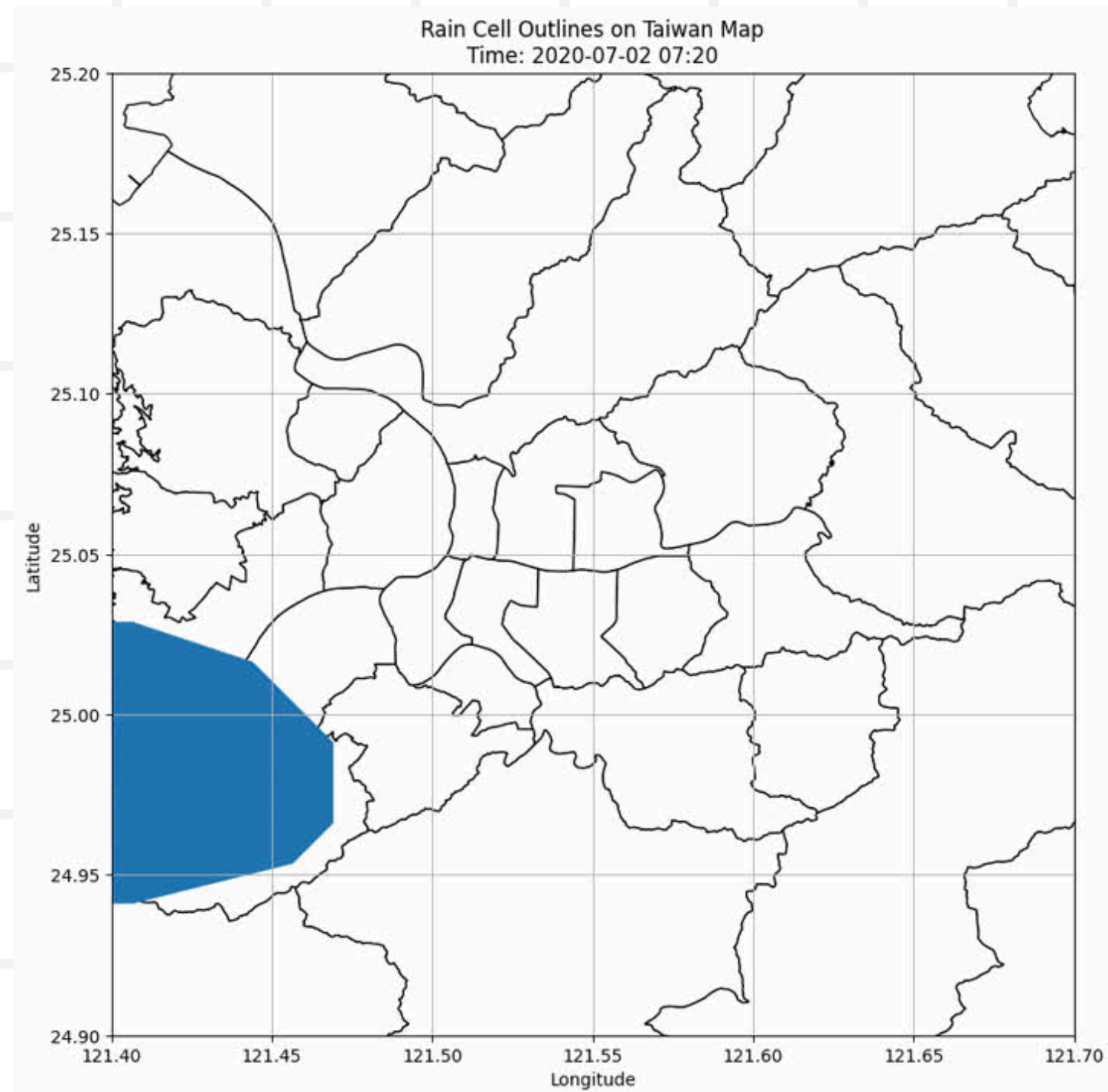
RAIN CELL GIF



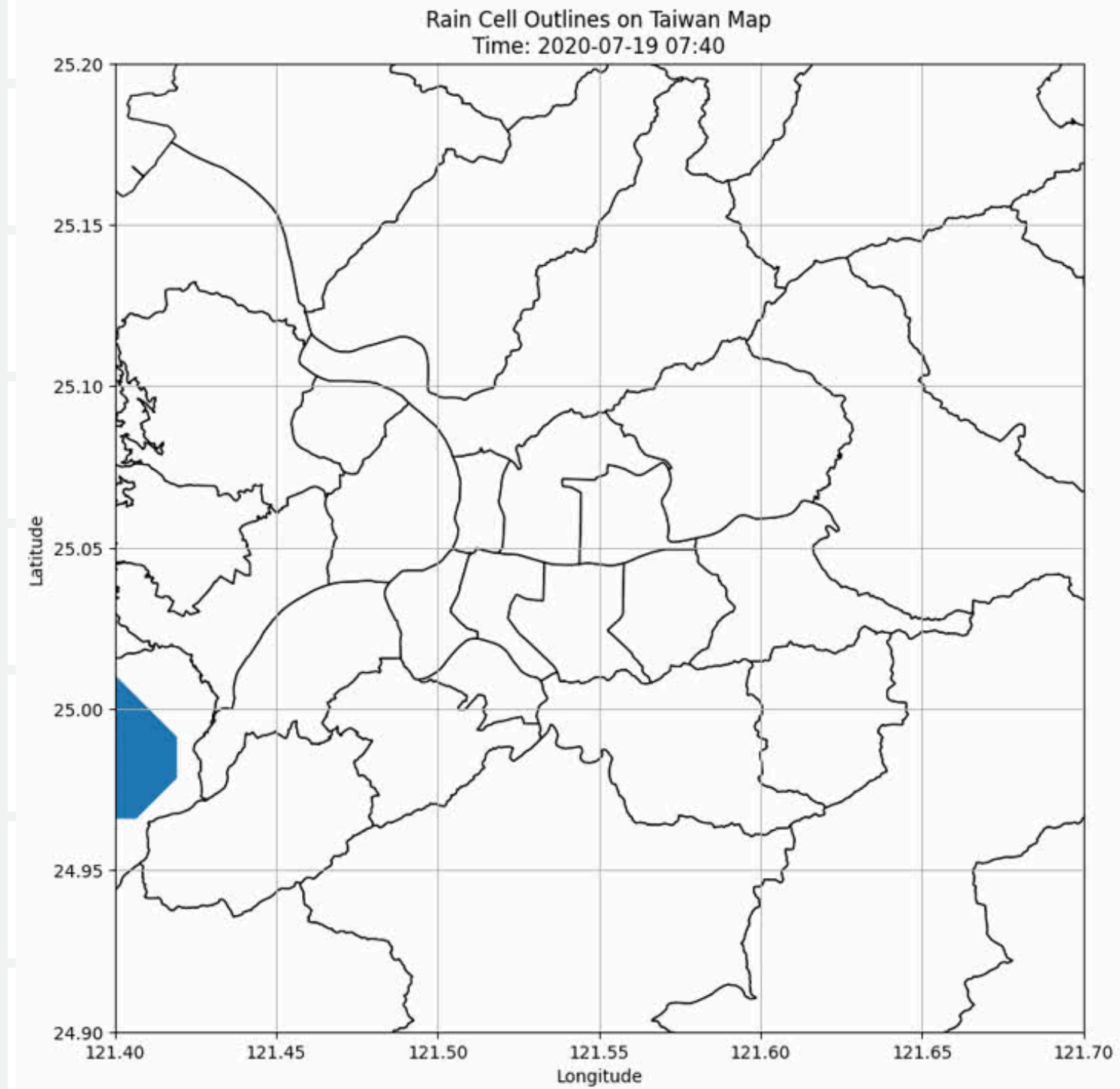
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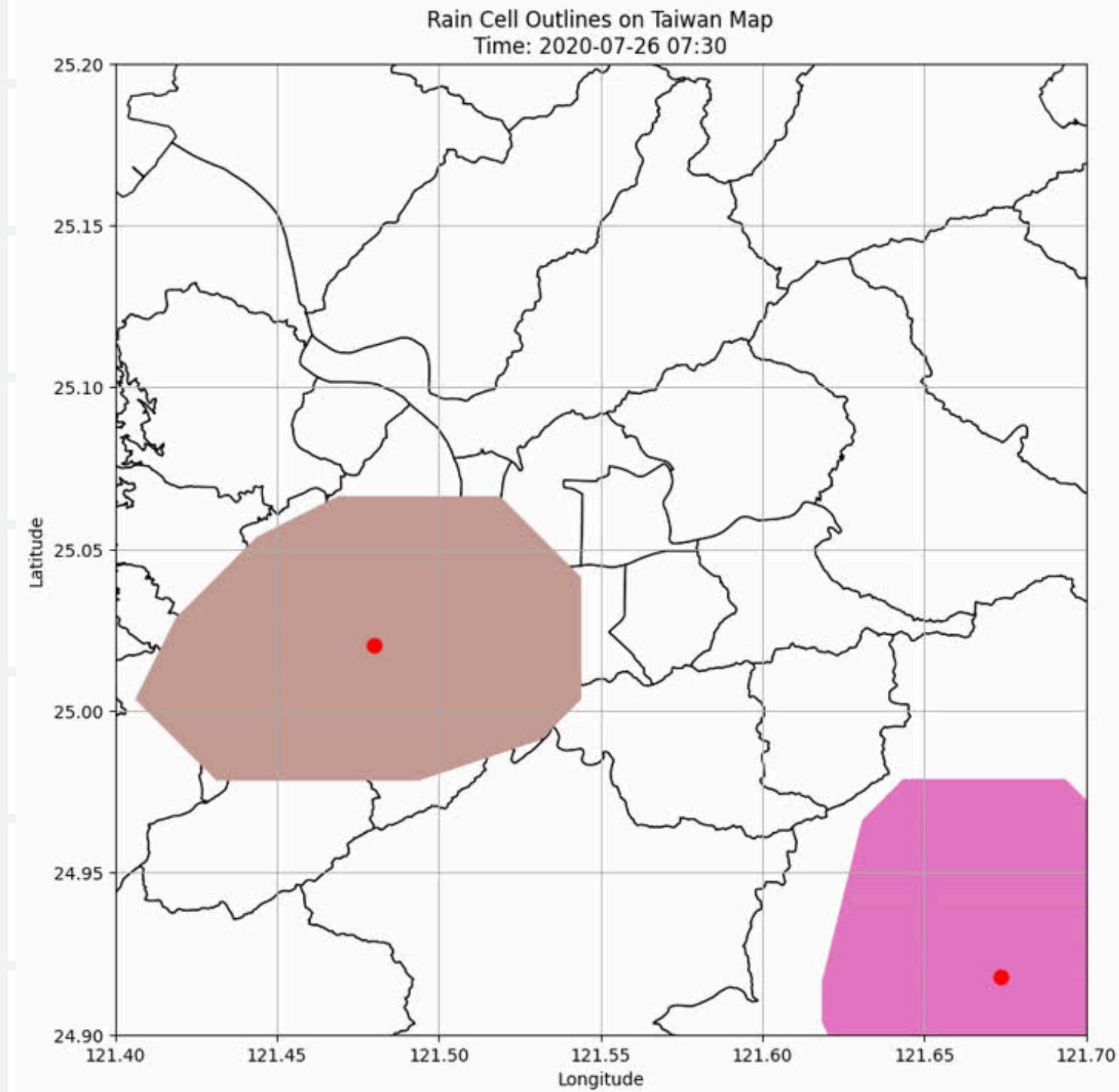
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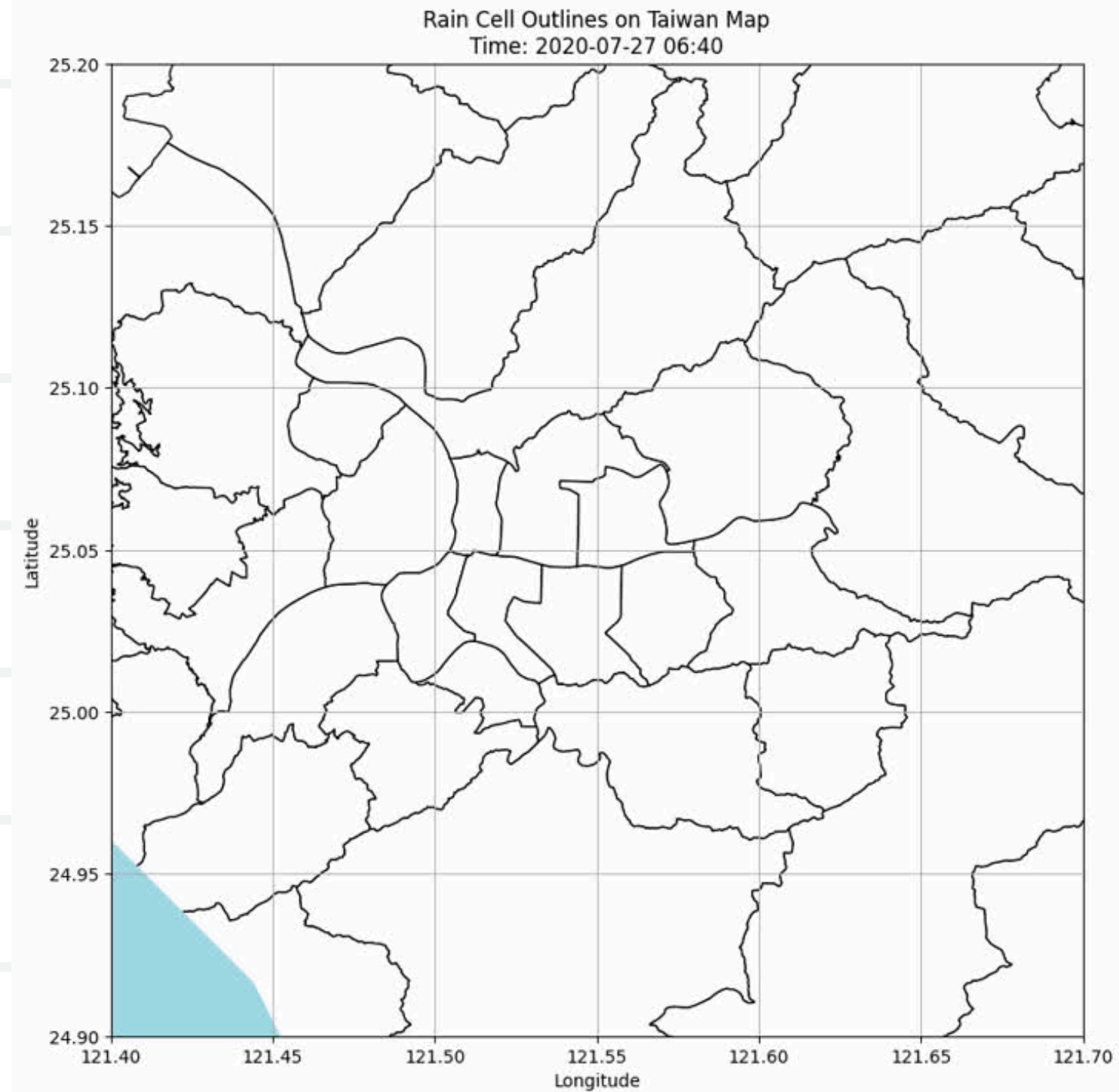
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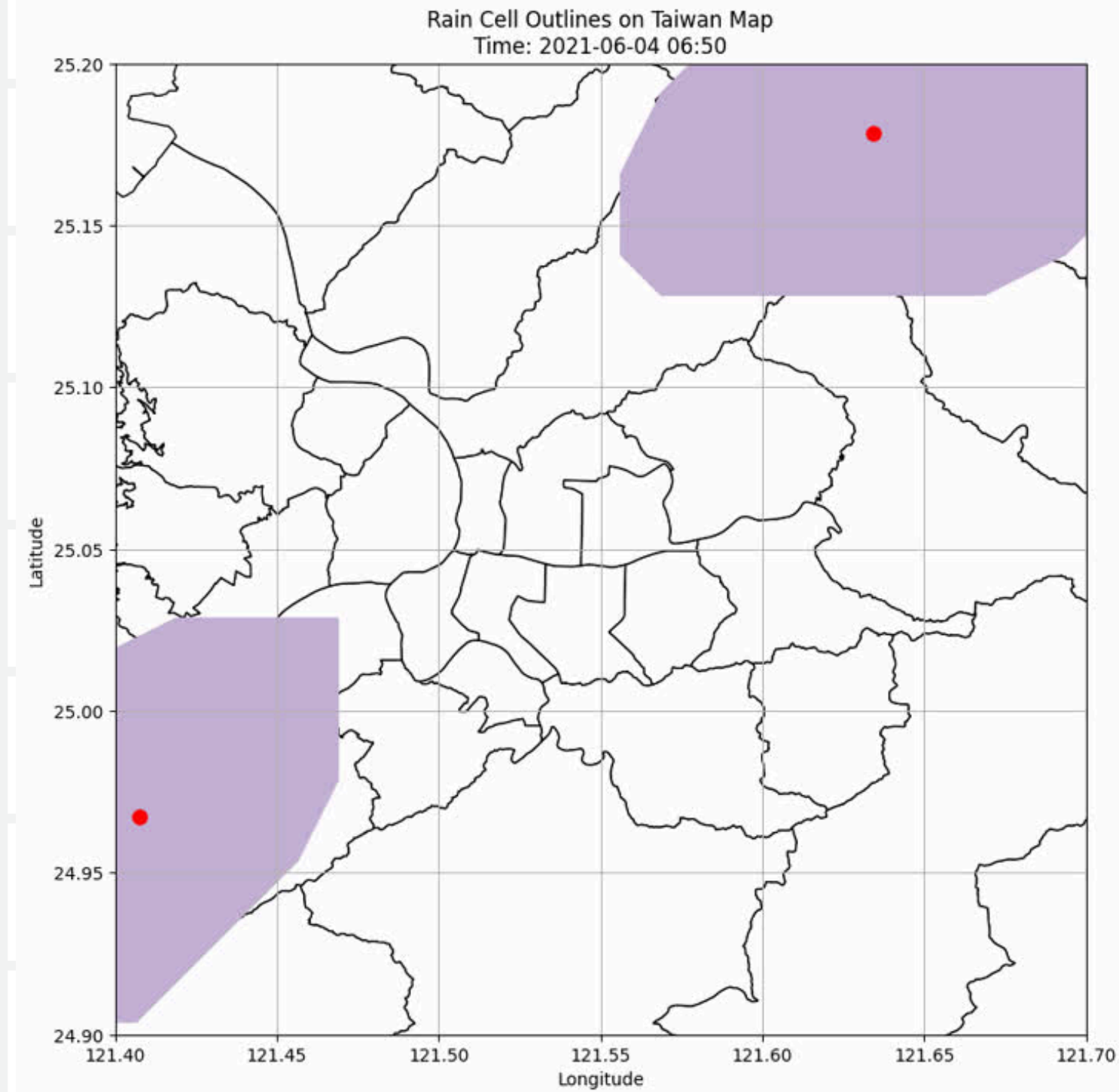
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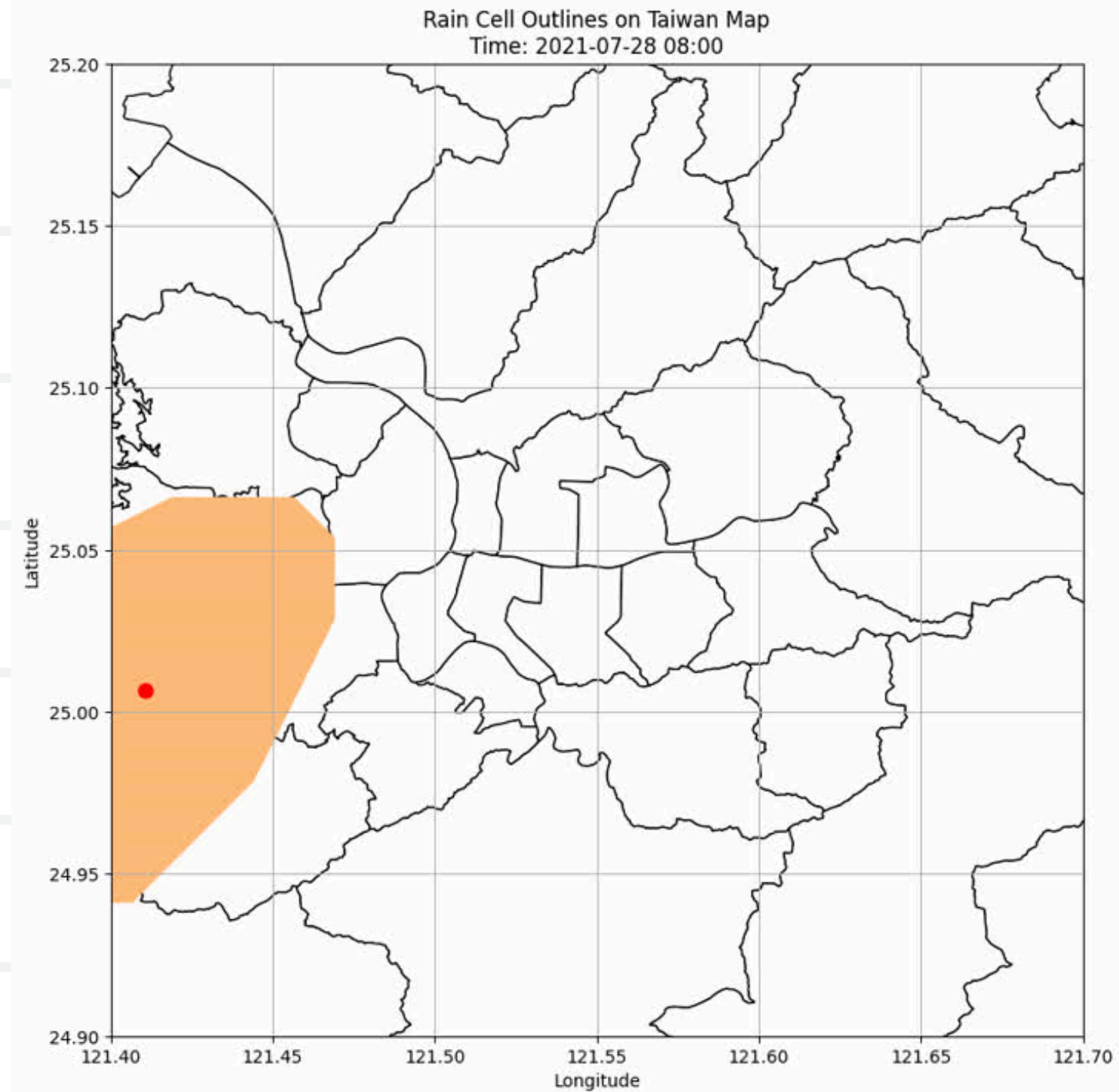
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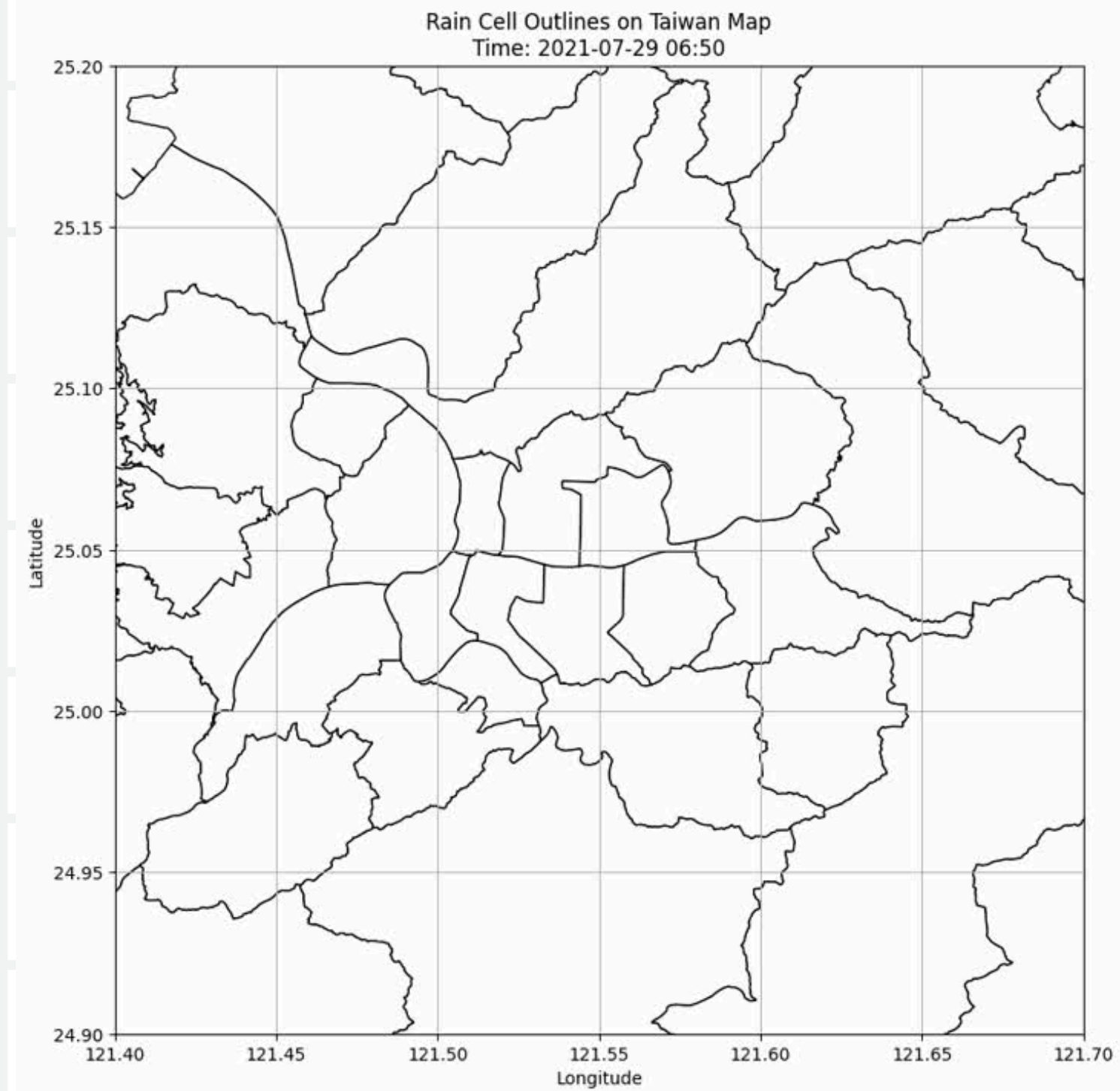
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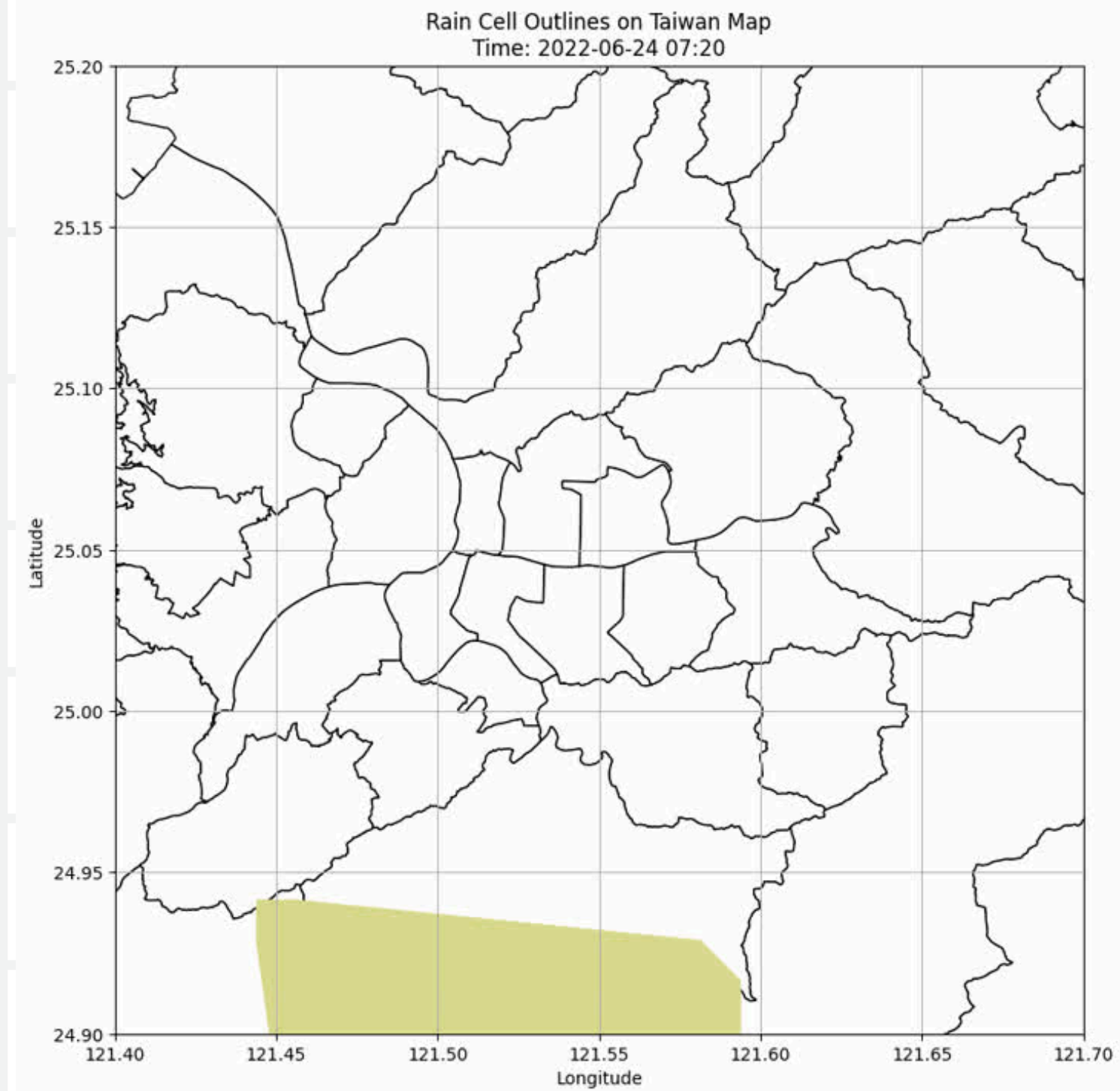
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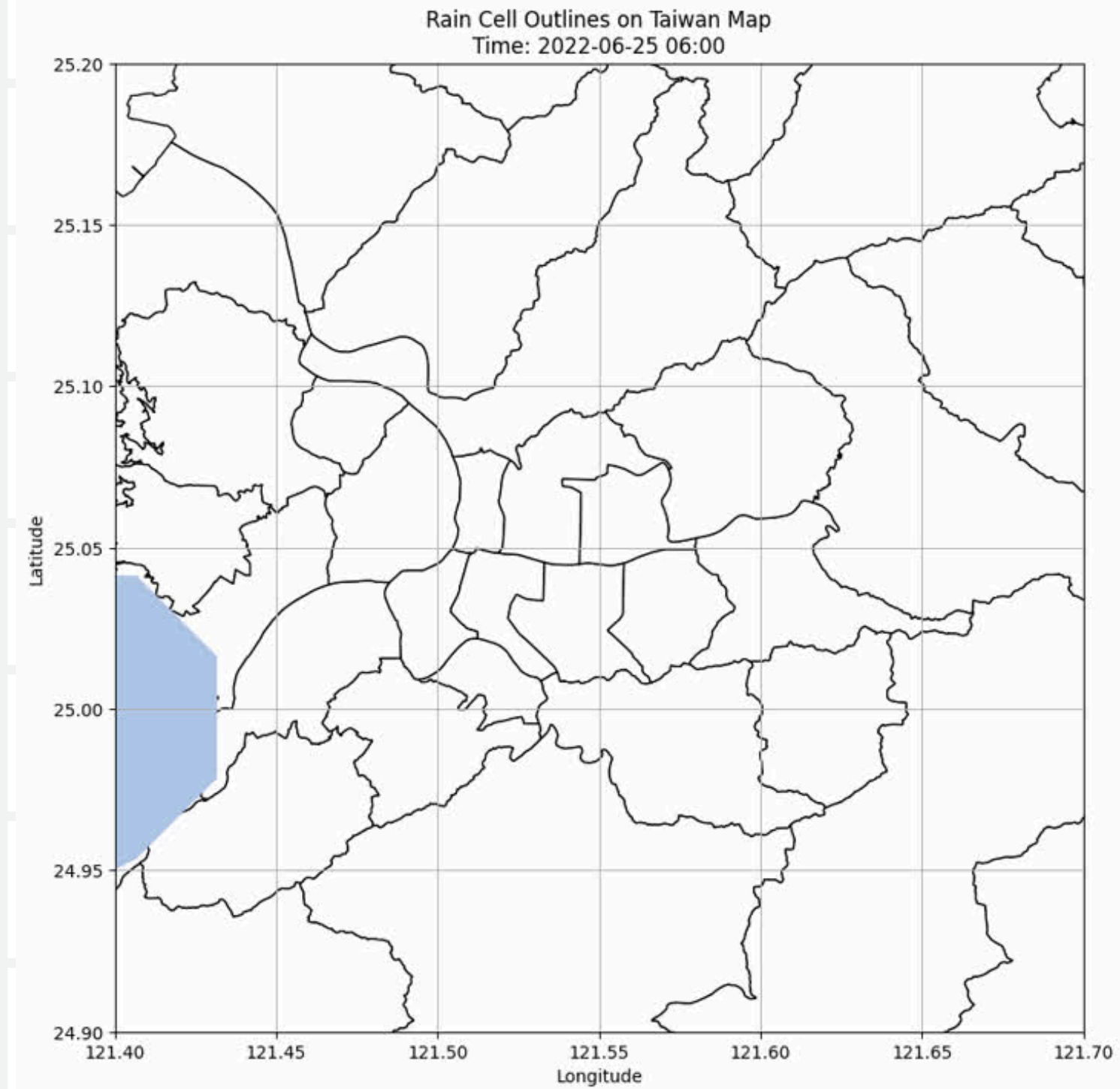
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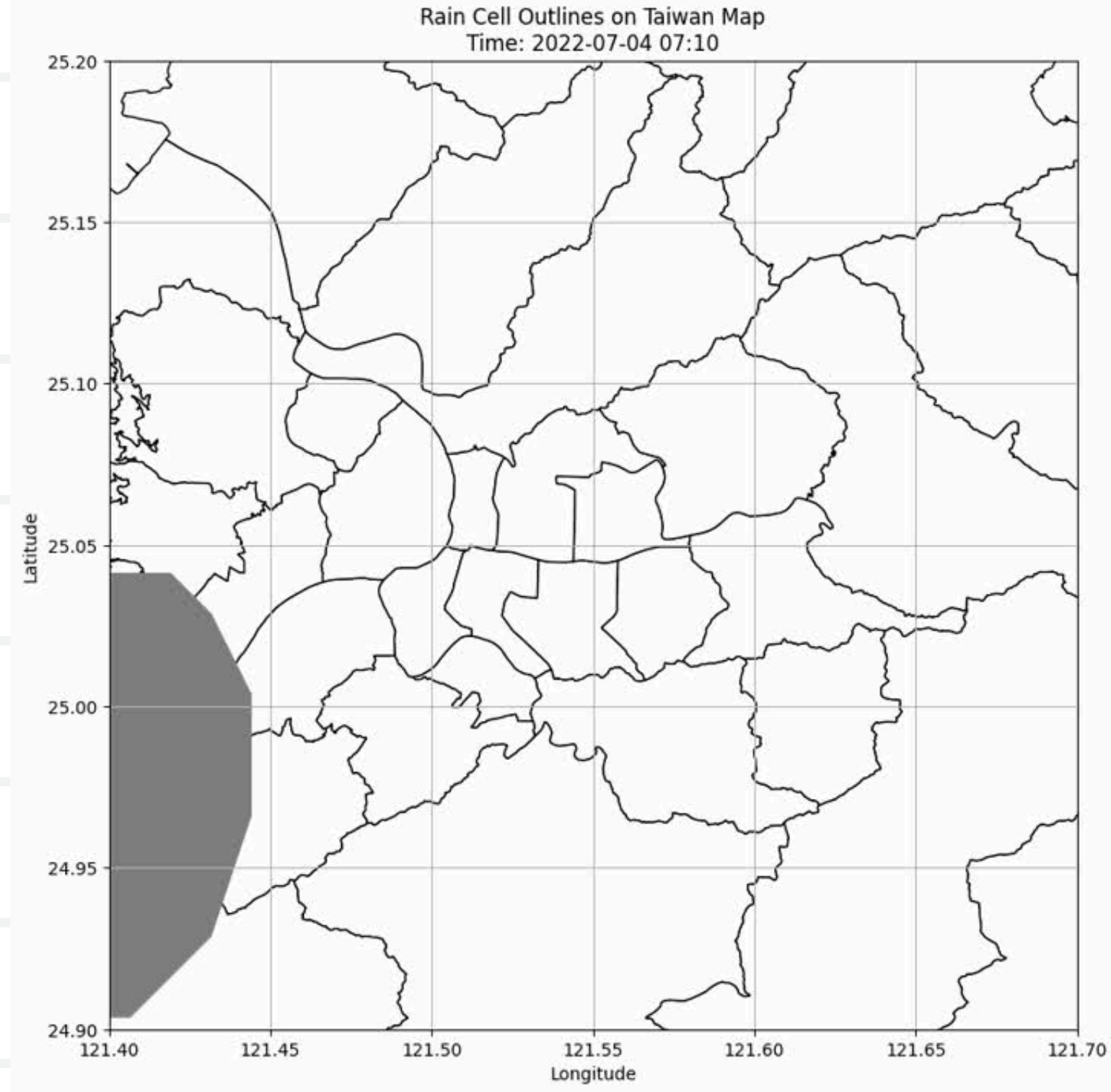
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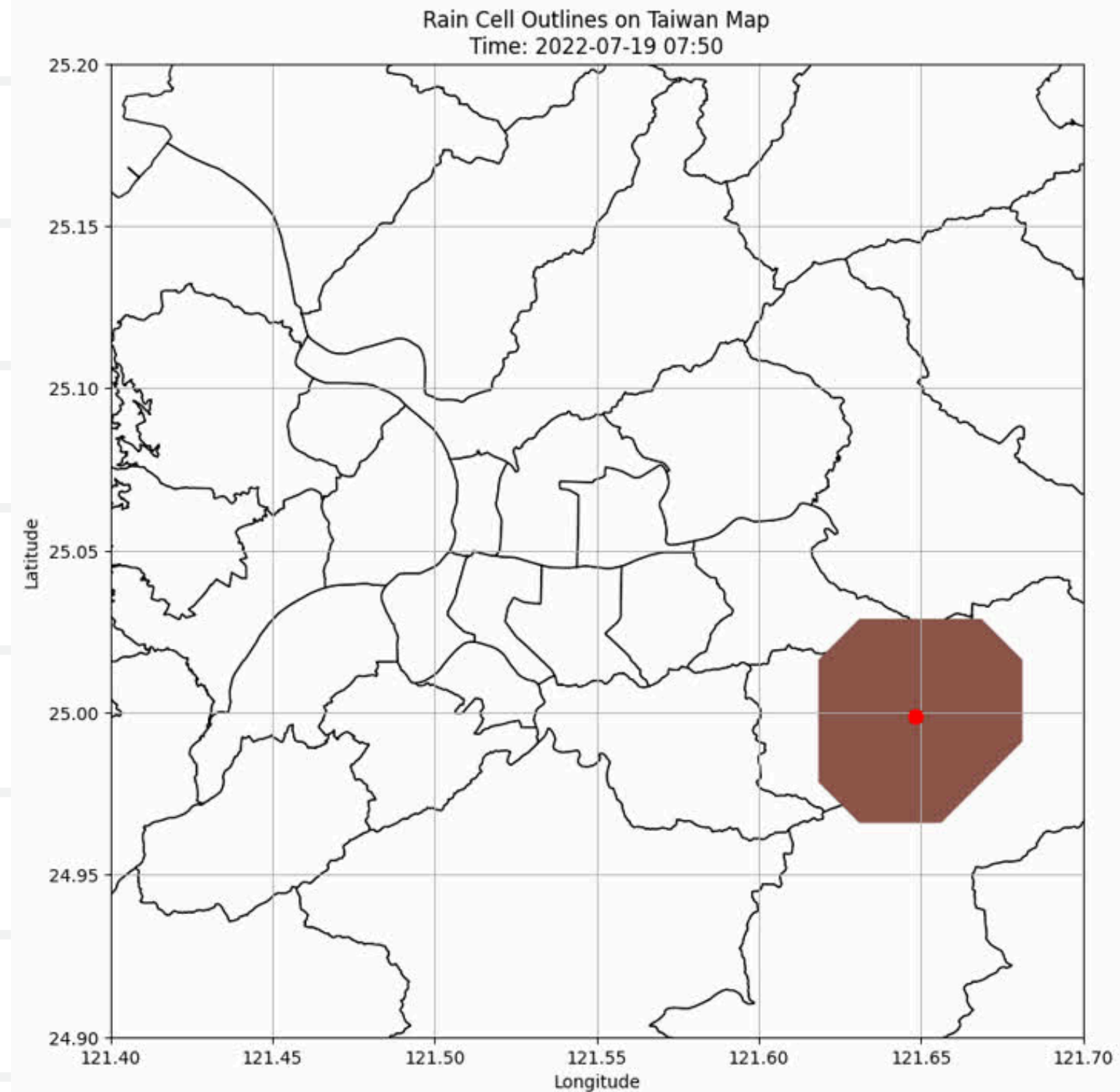
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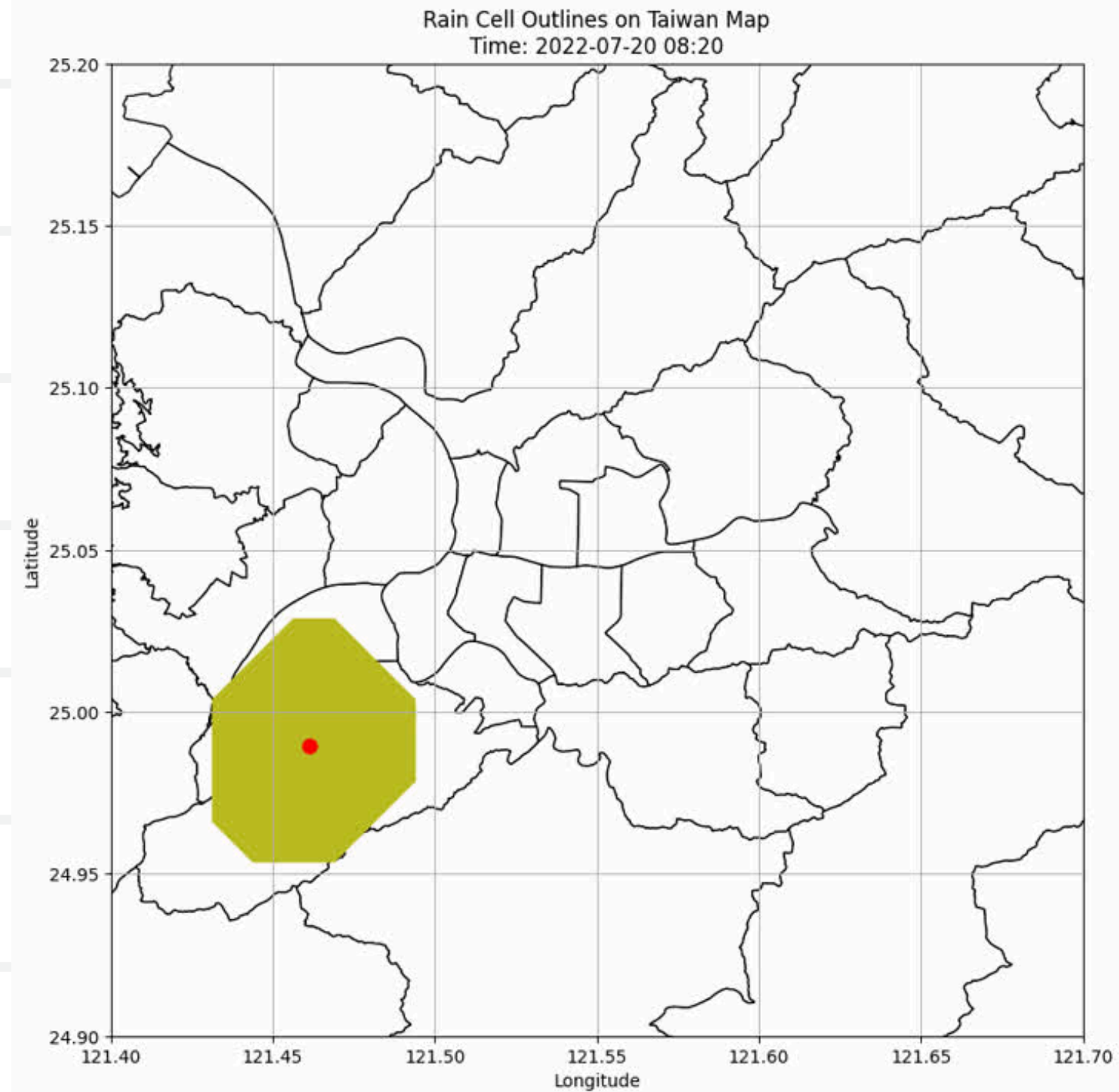
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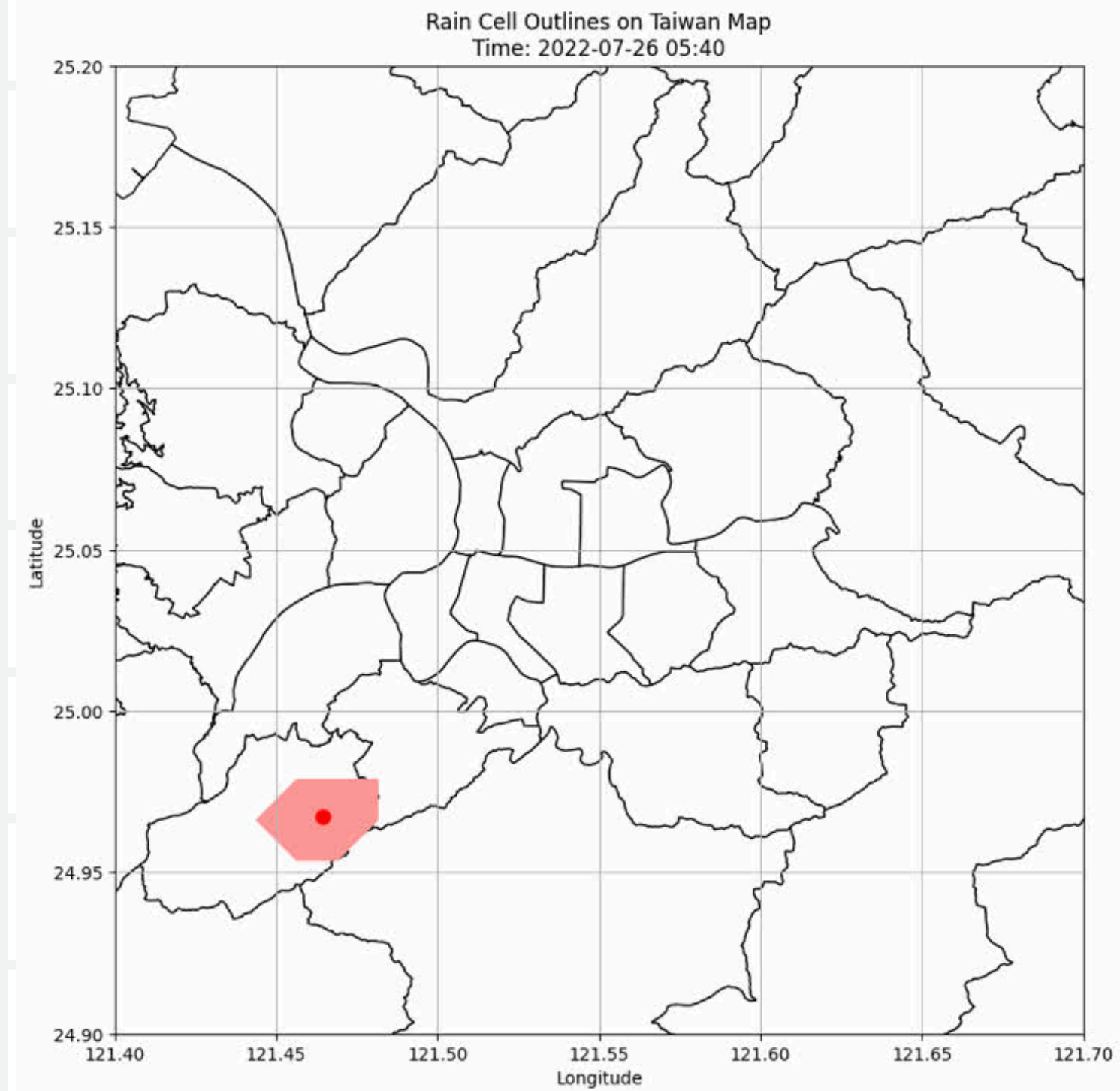
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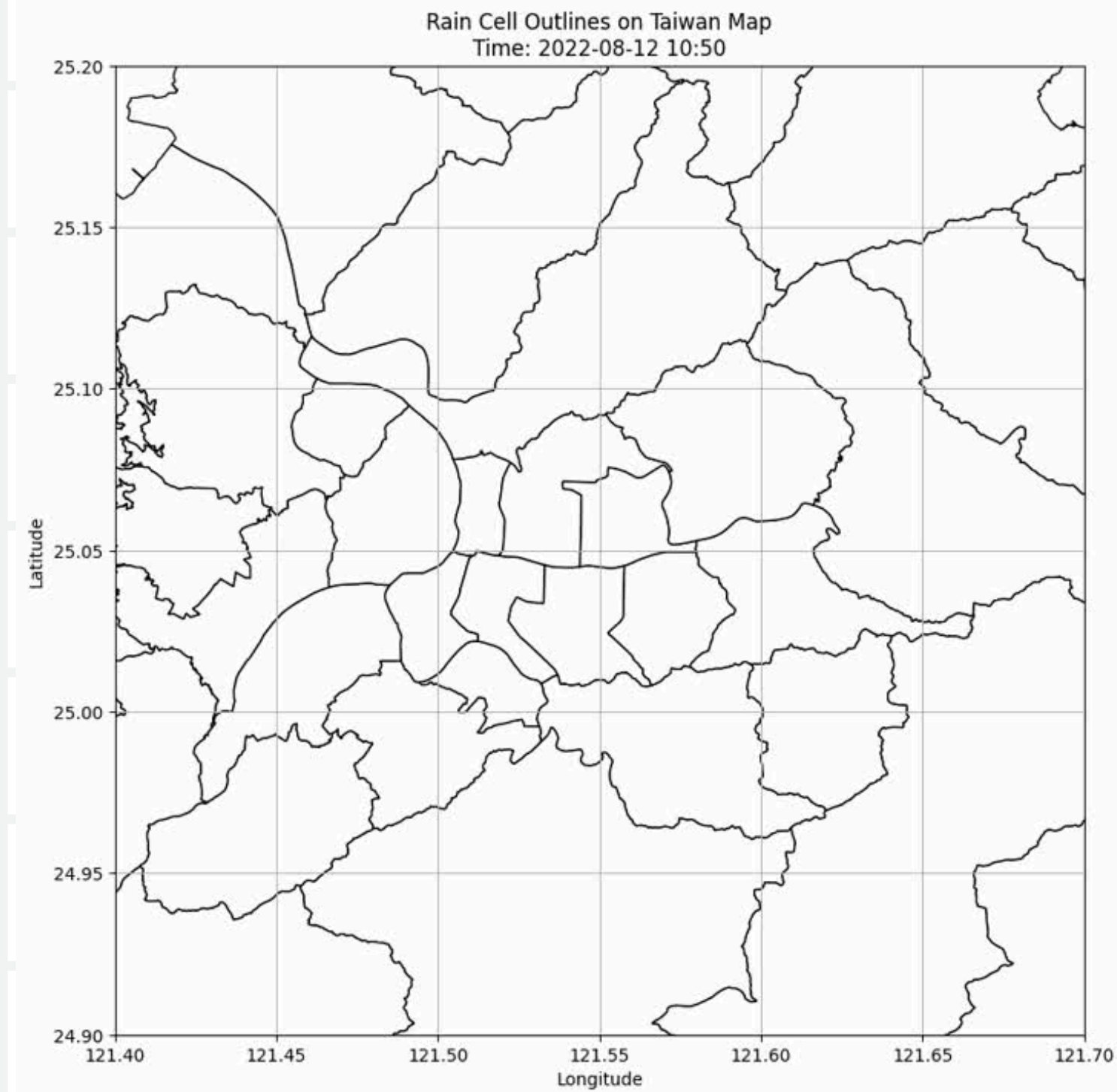
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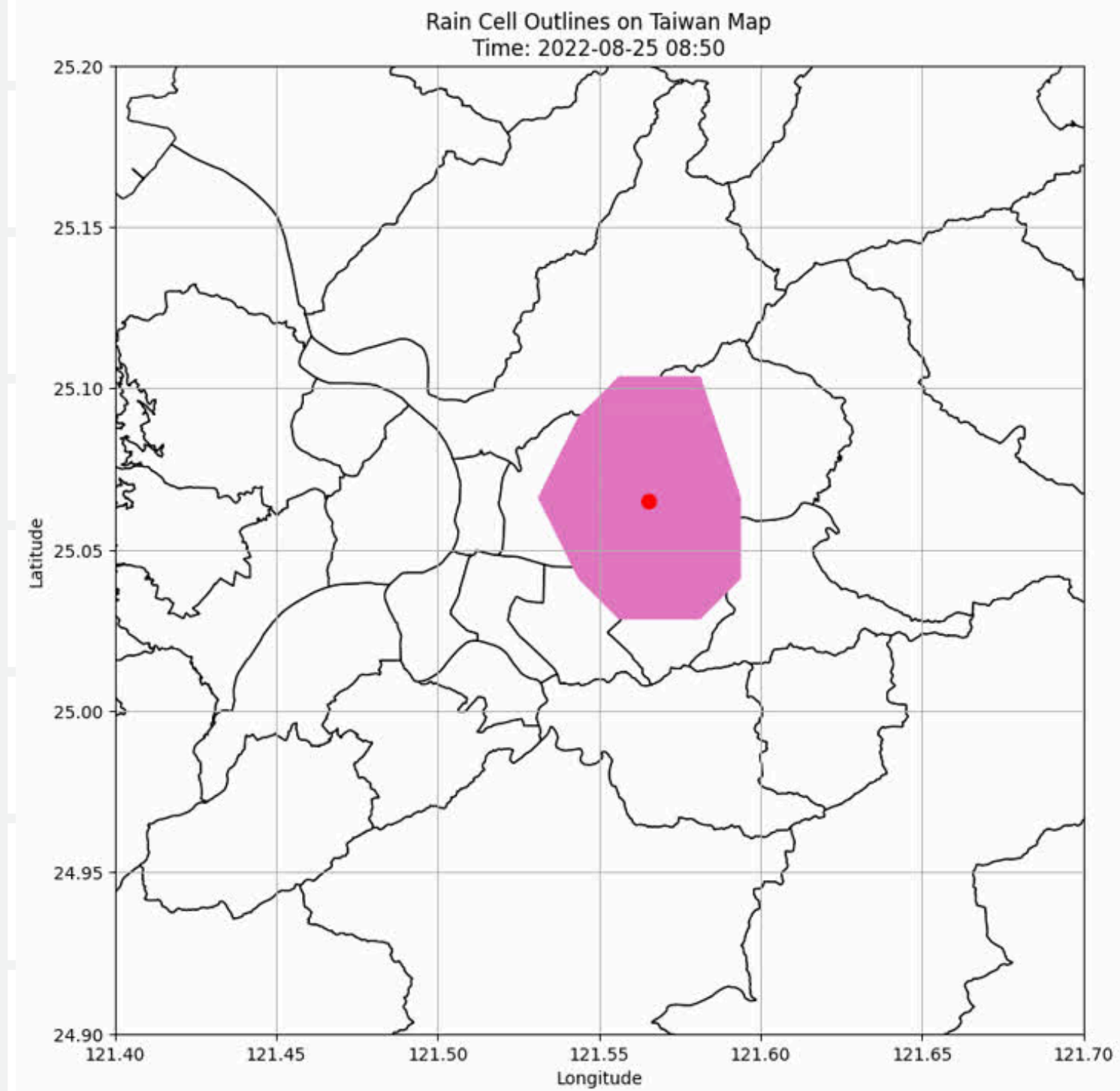
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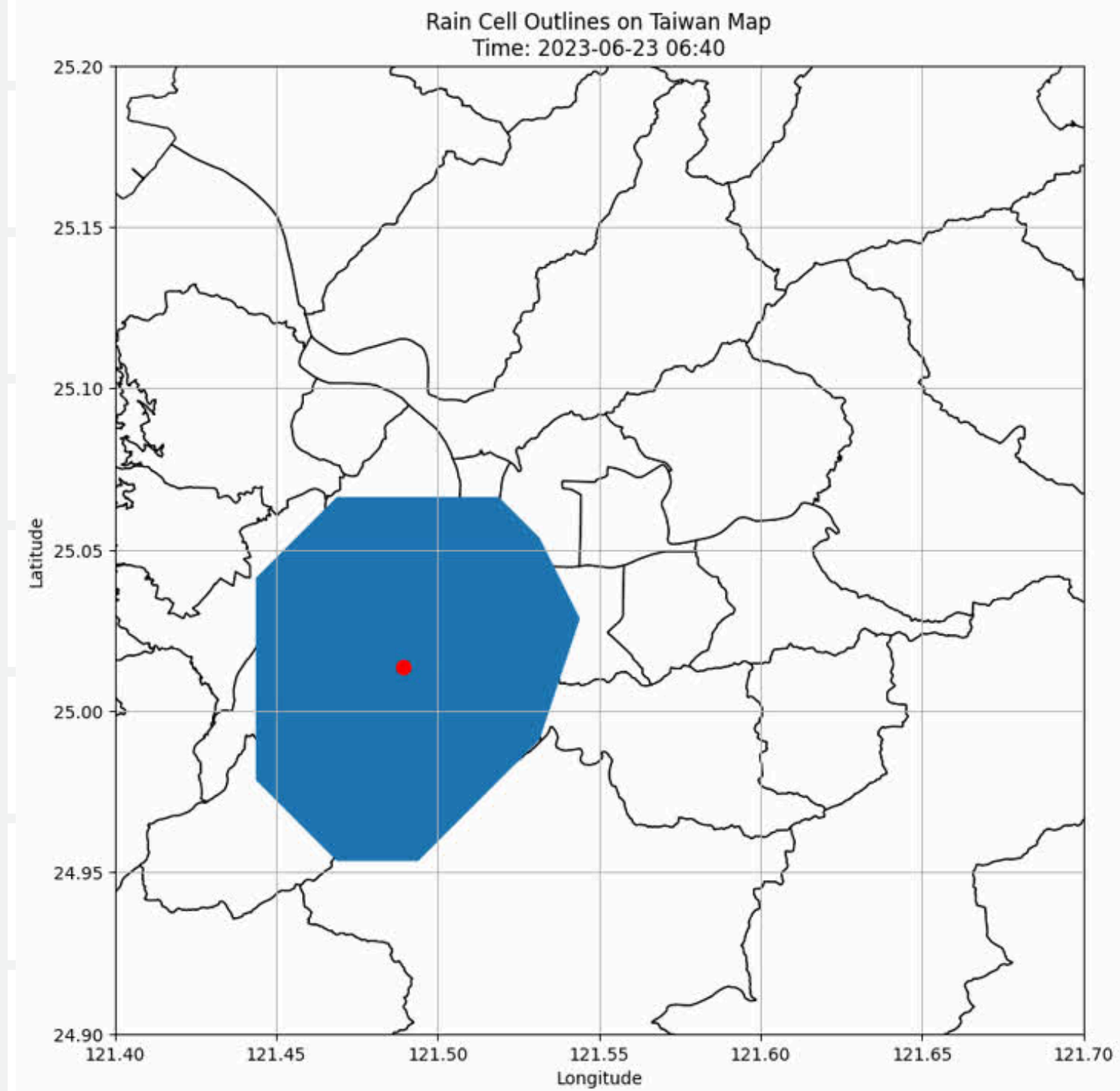
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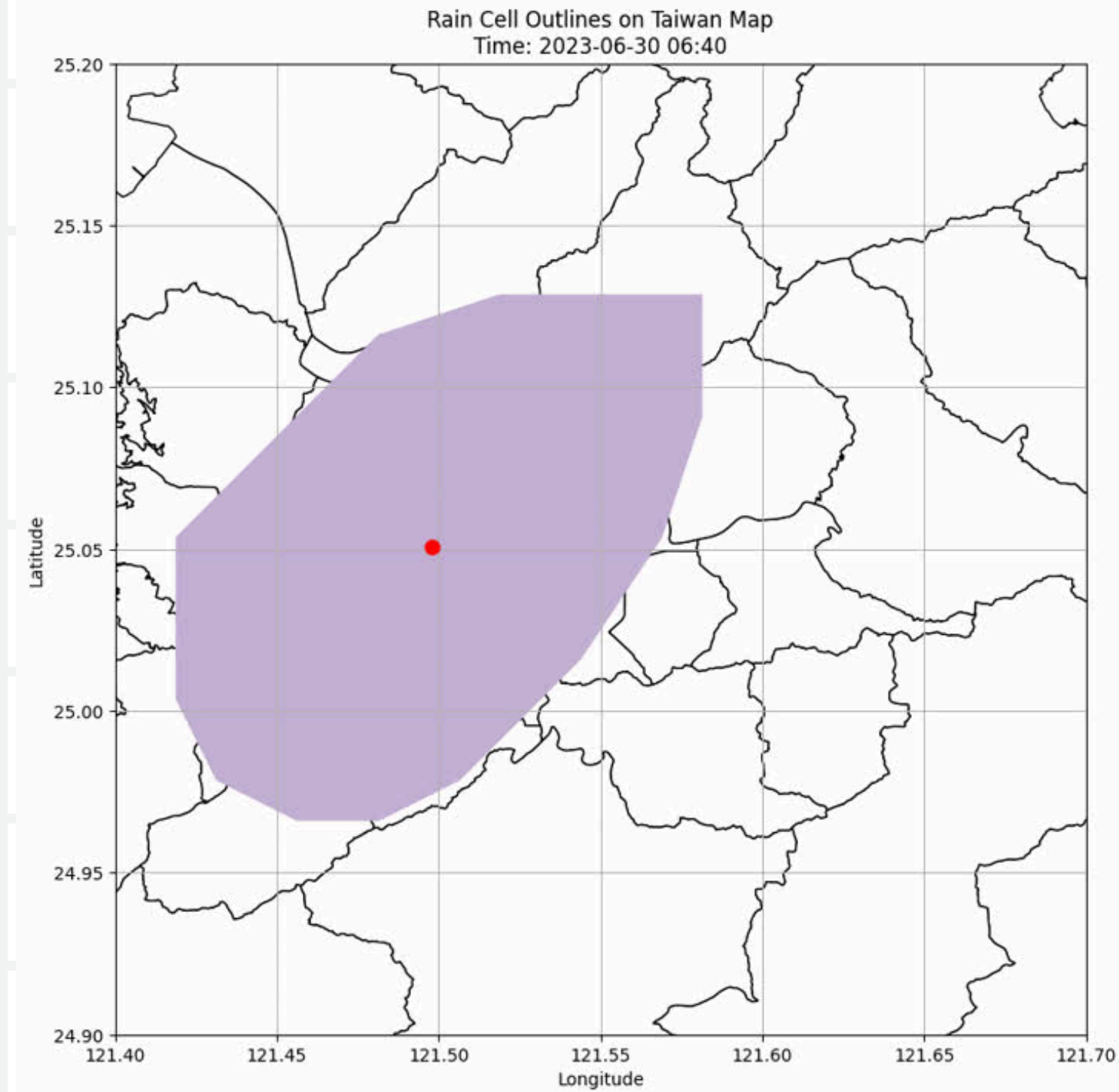
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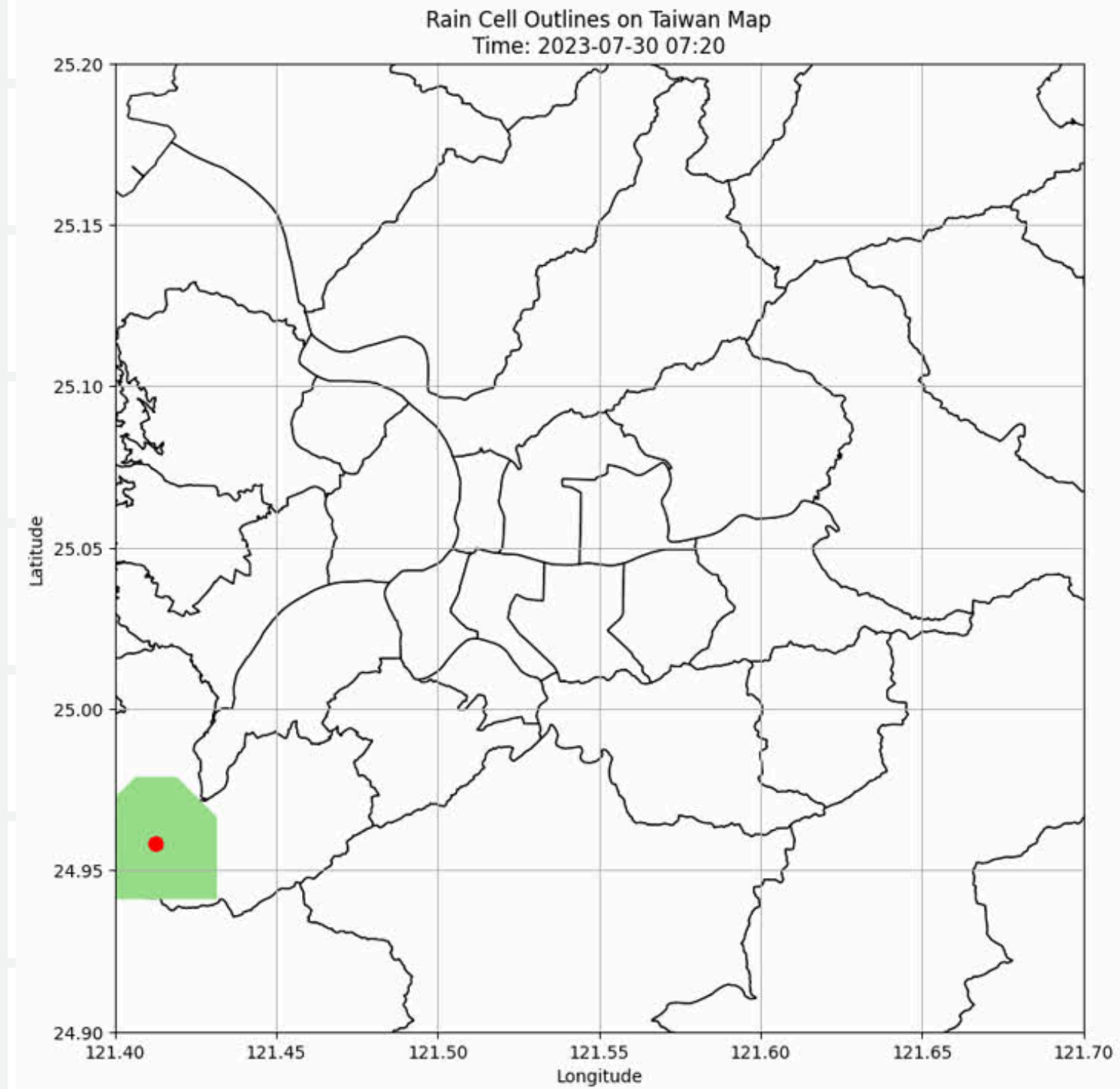
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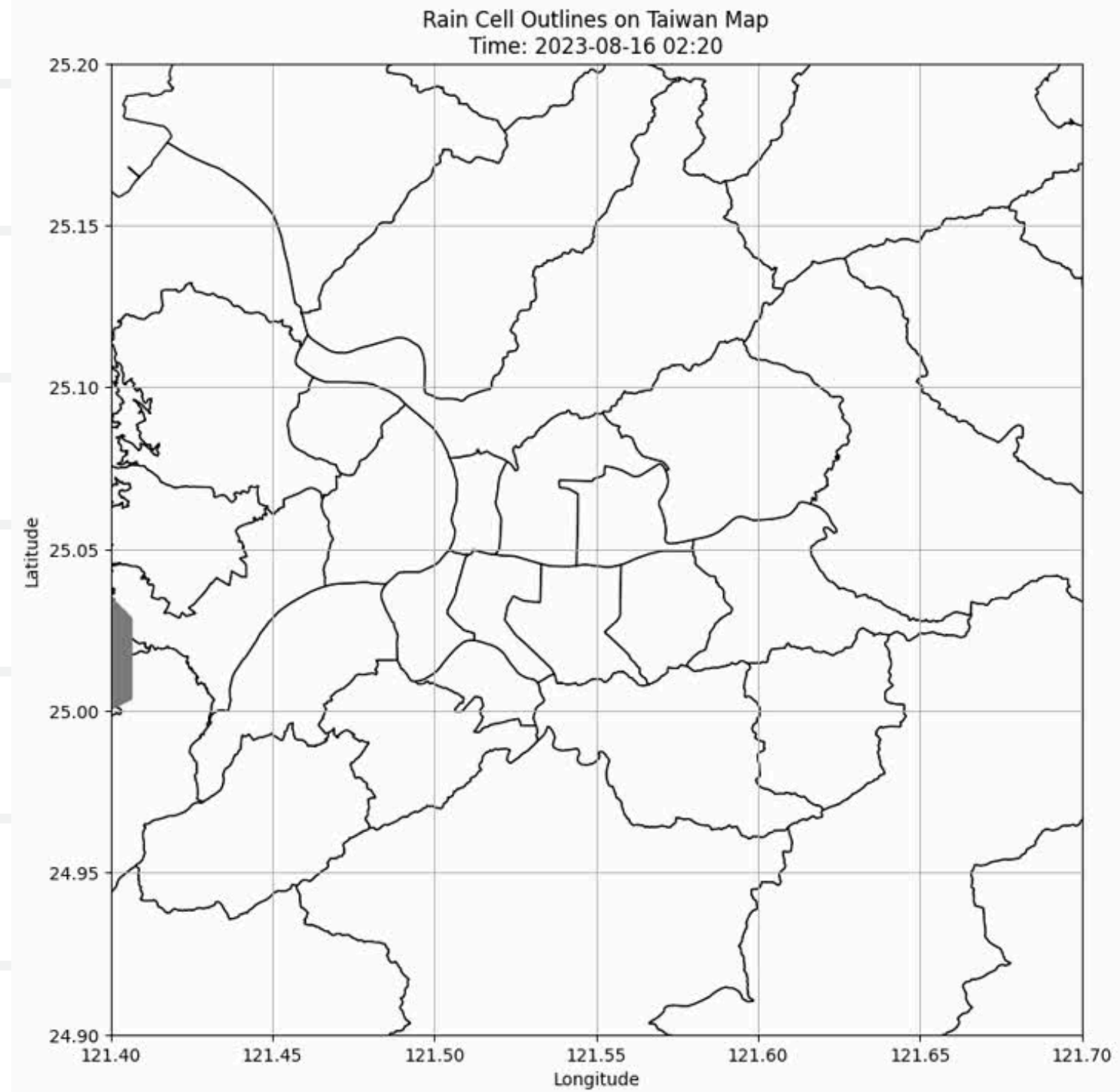
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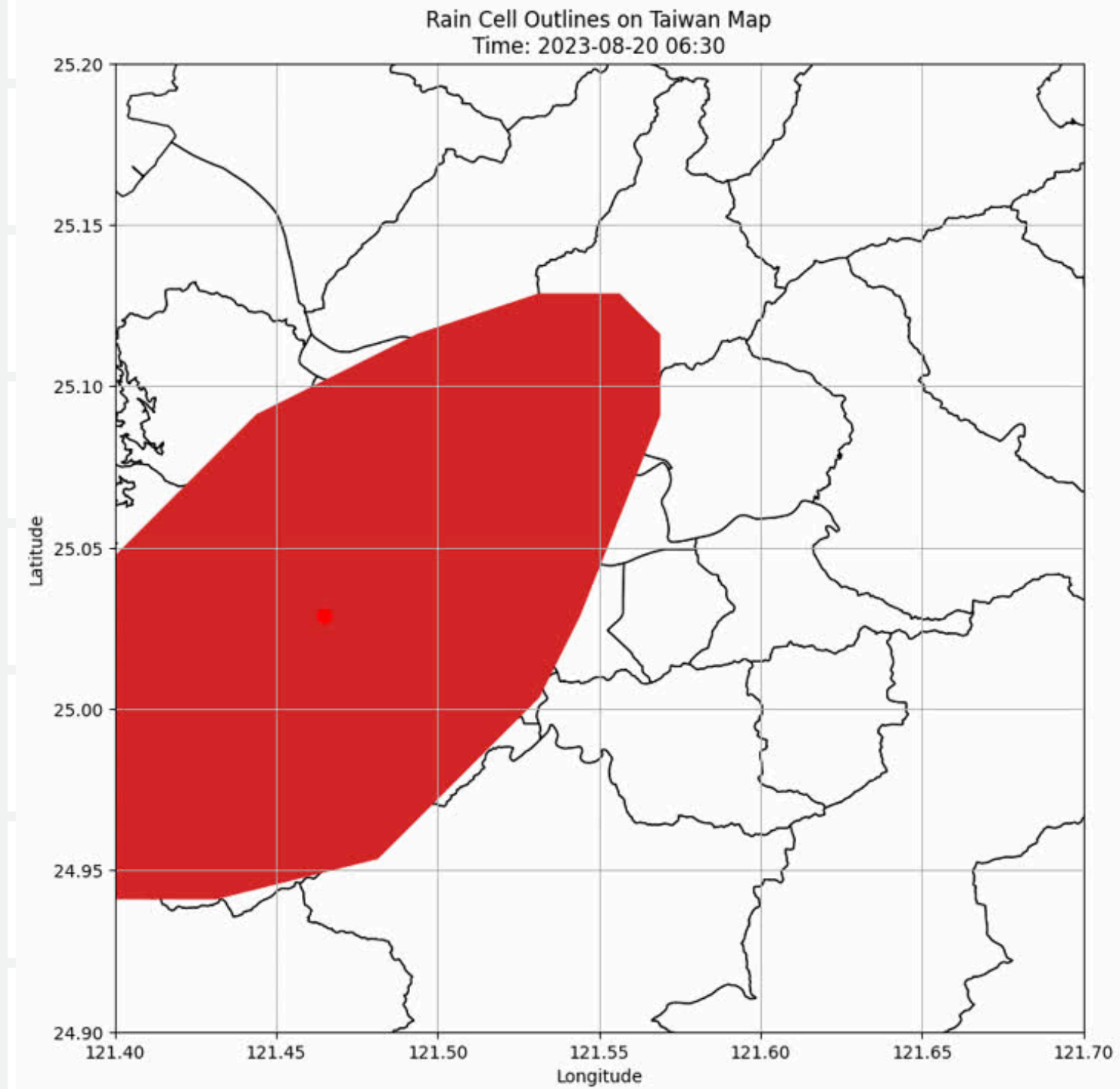
RAIN CELL GIF



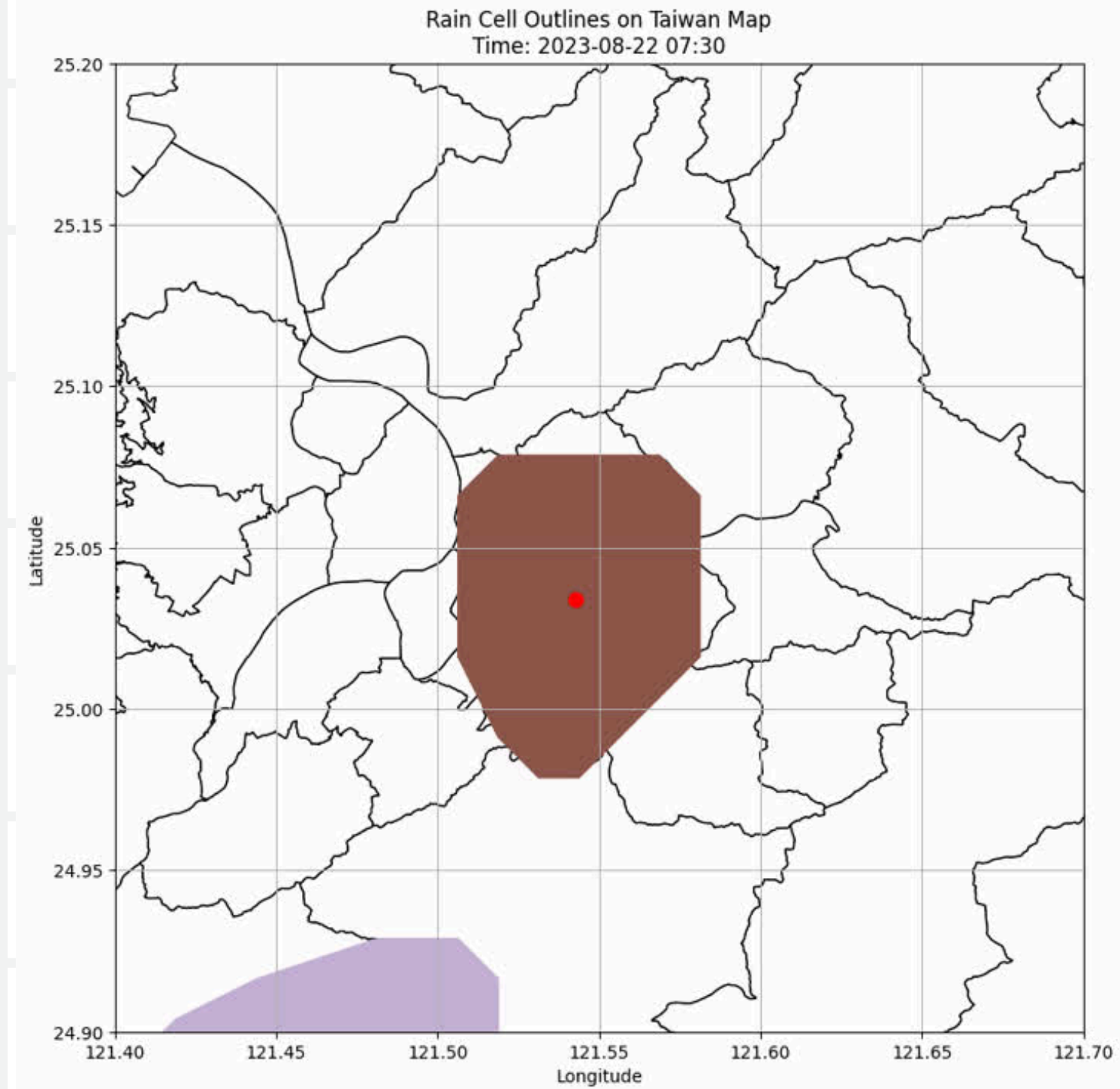
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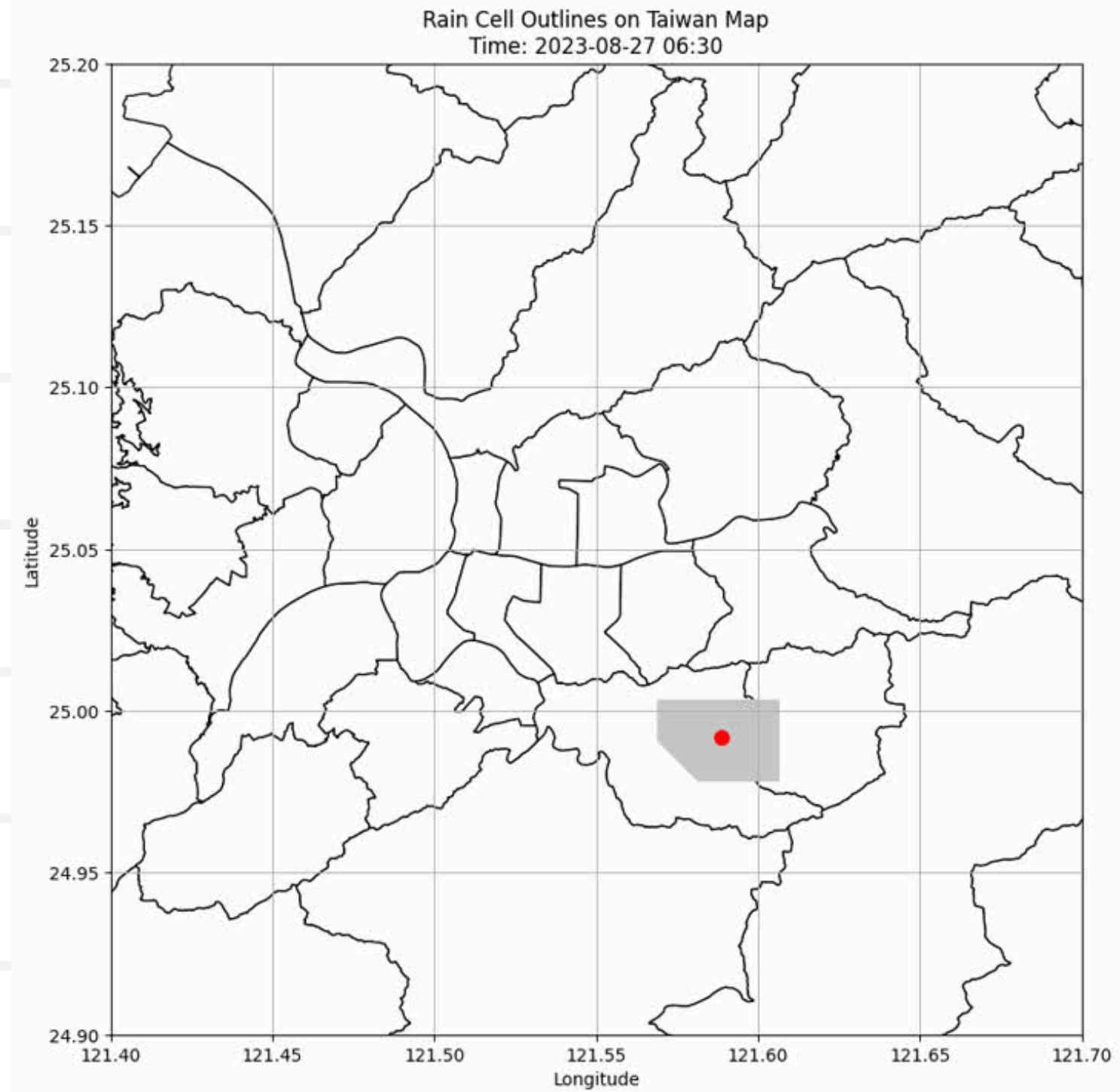
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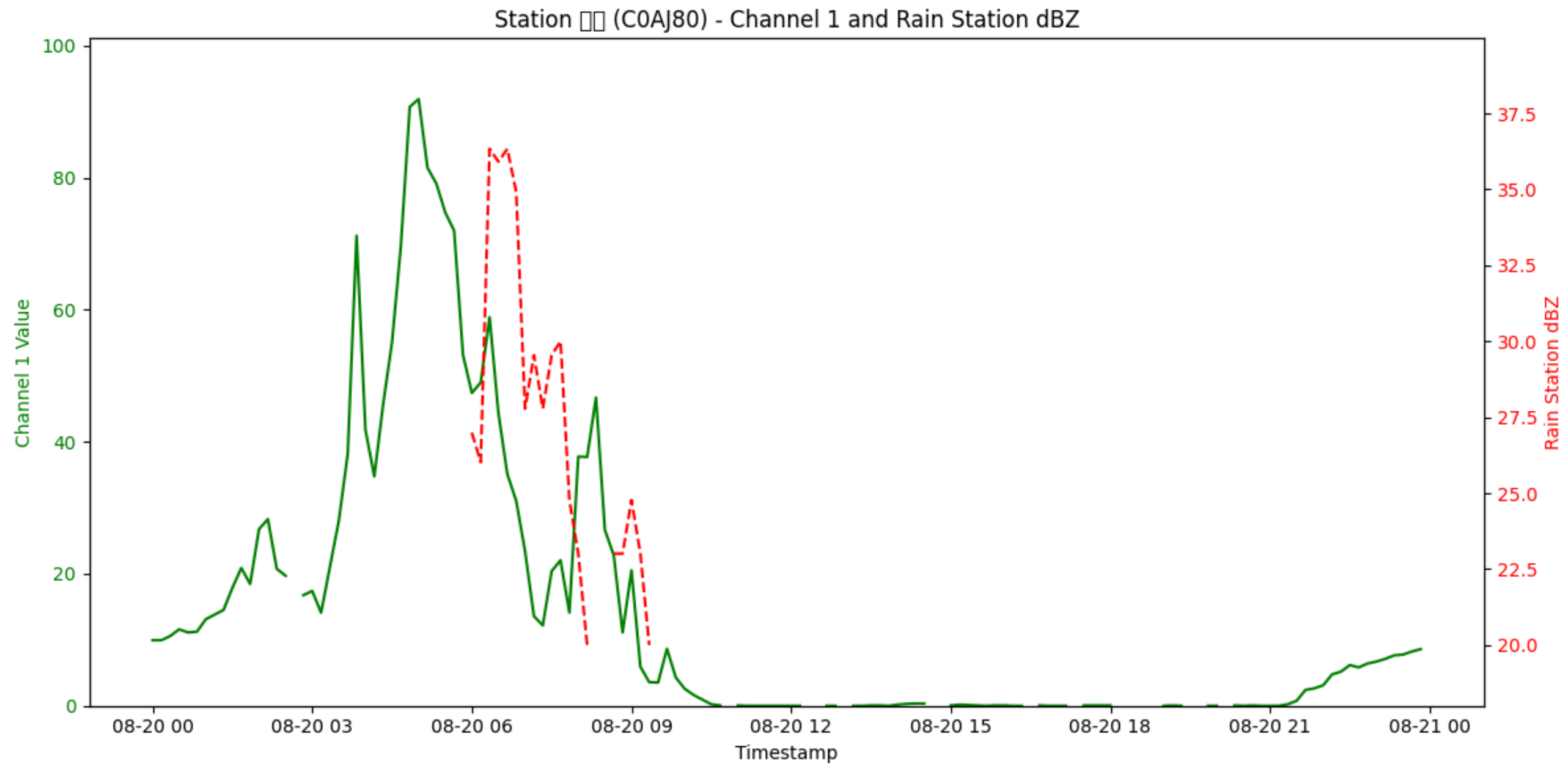
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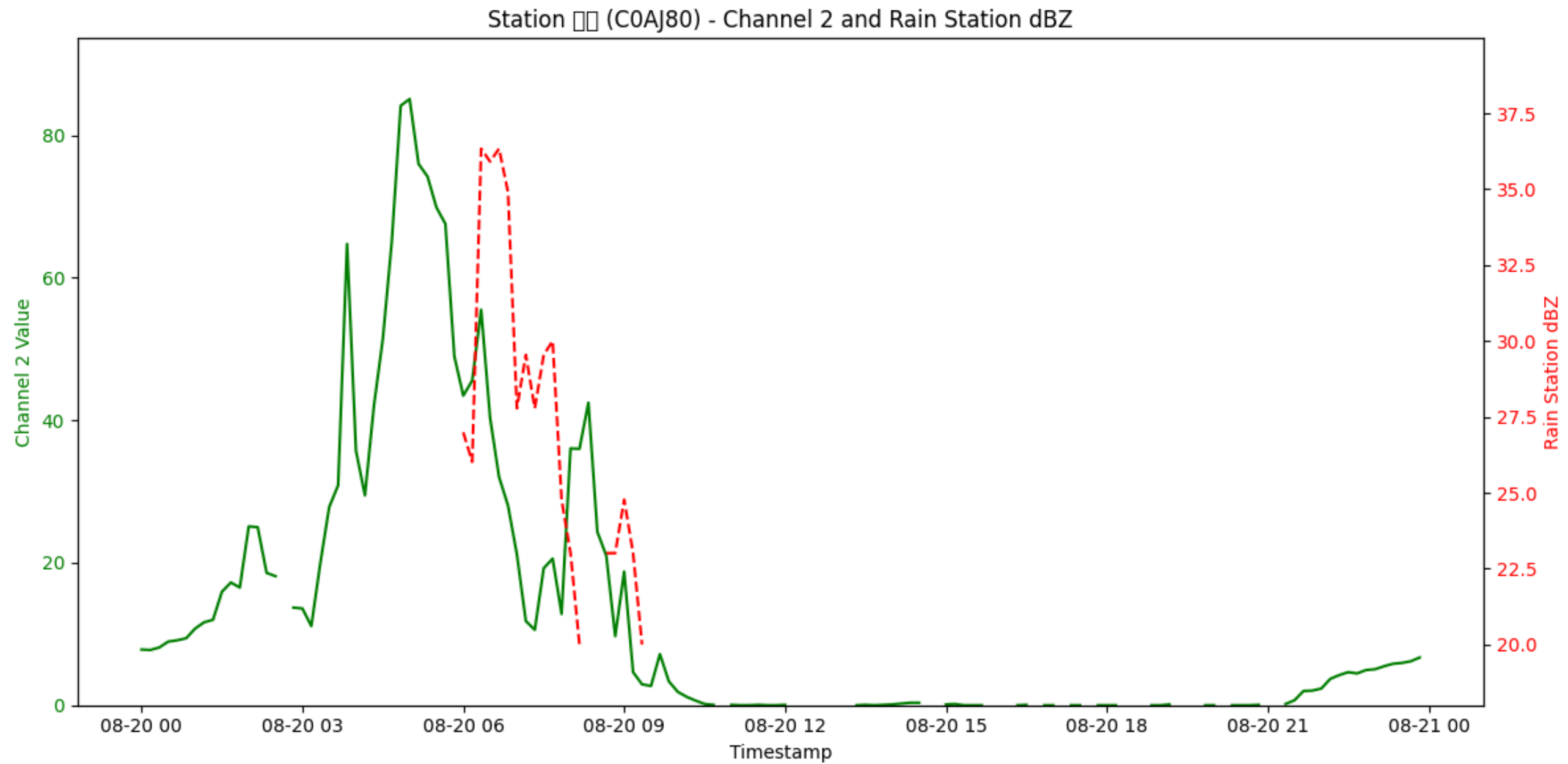
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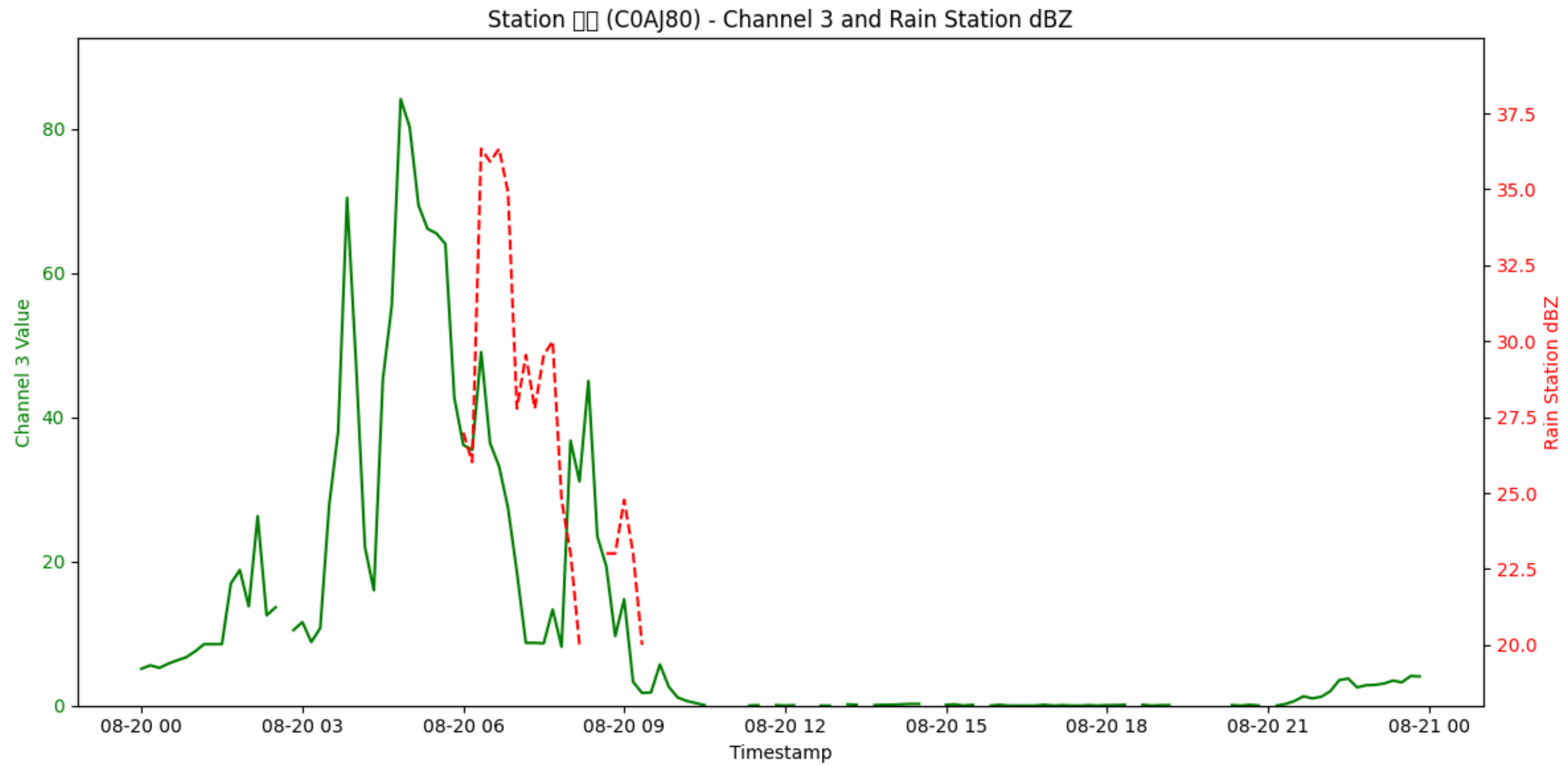
RAINFALL VS CHANNELS



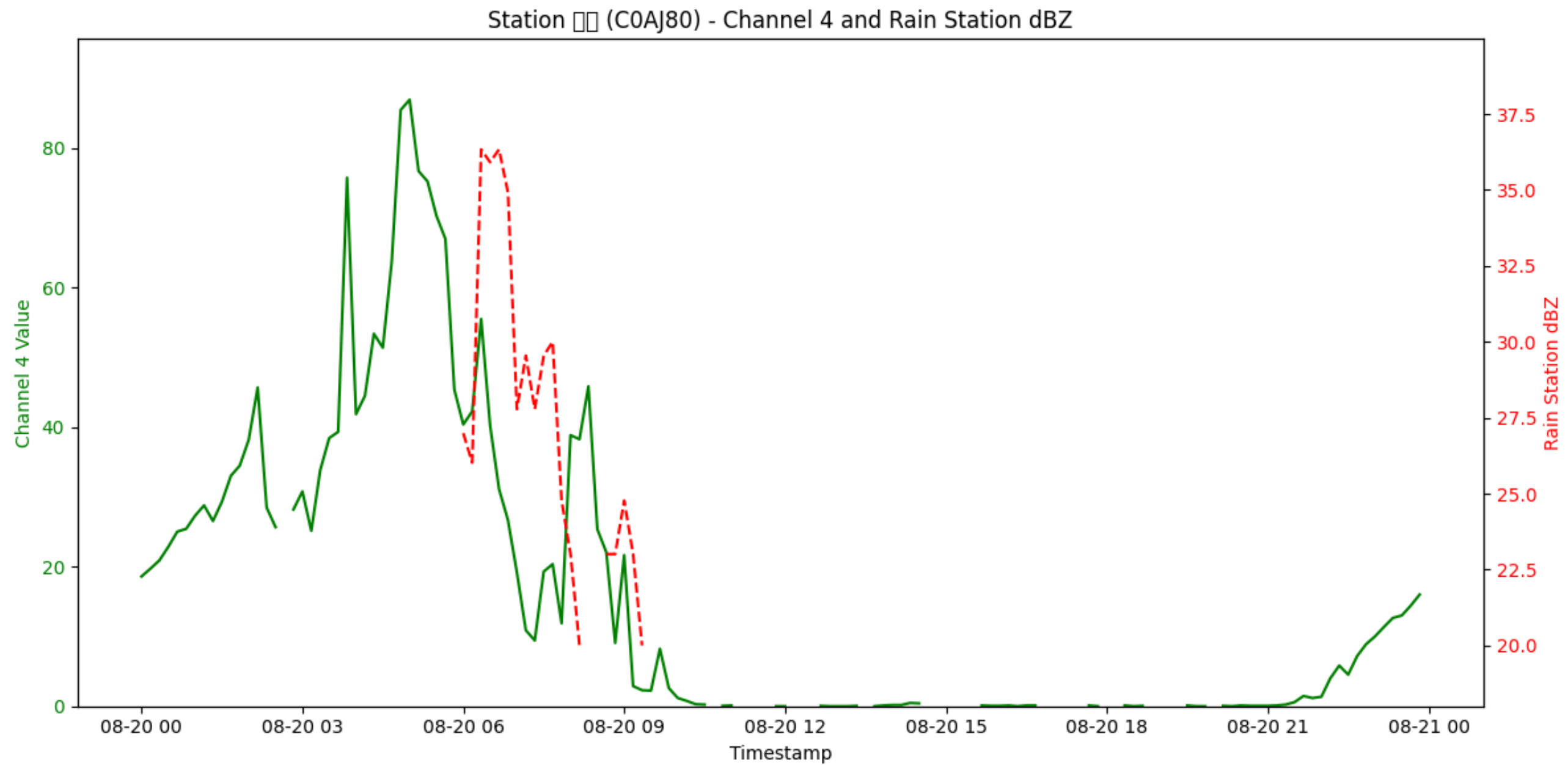
RAINFALL VS CHANNELS



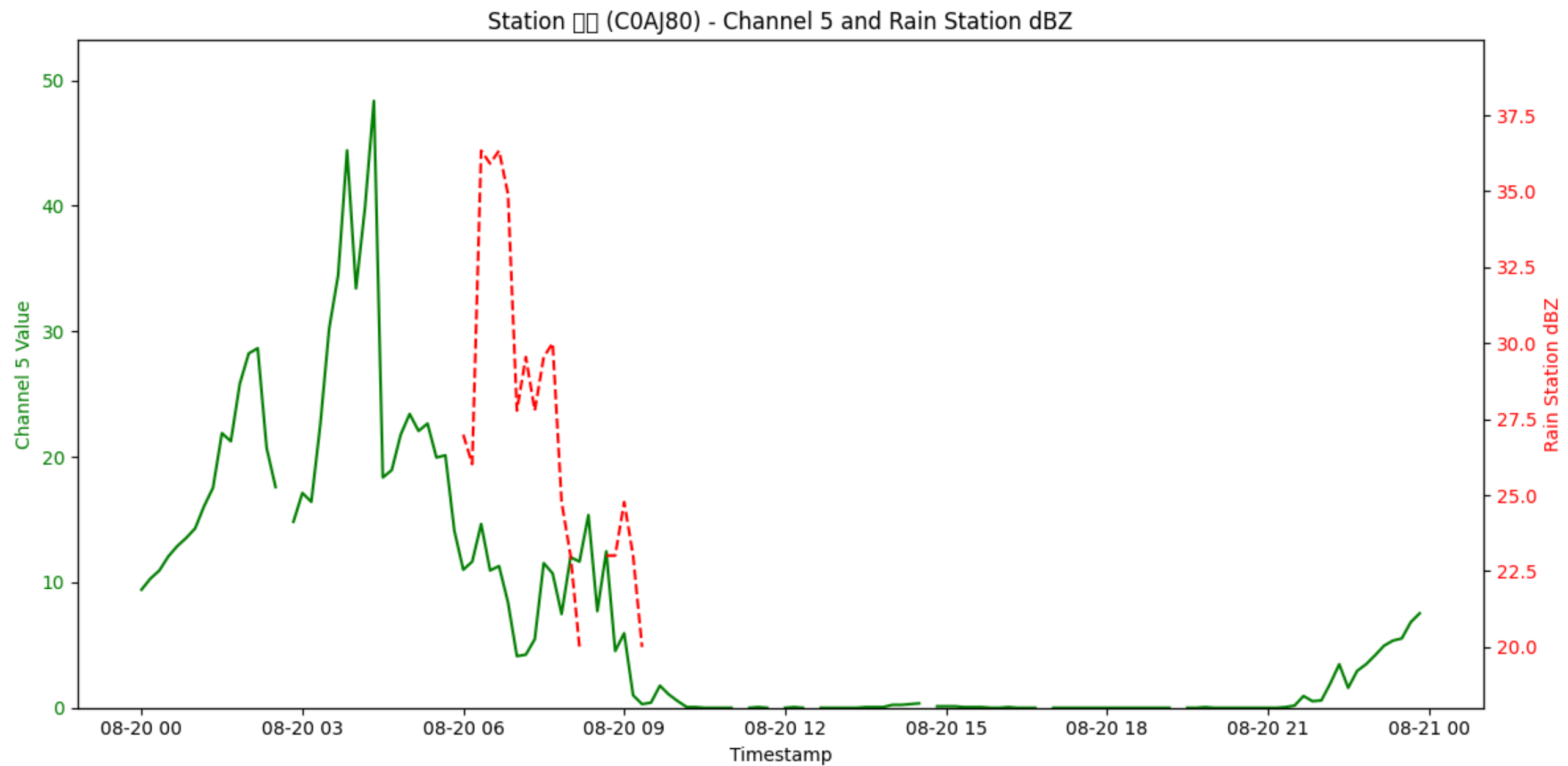
RAINFALL VS CHANNELS



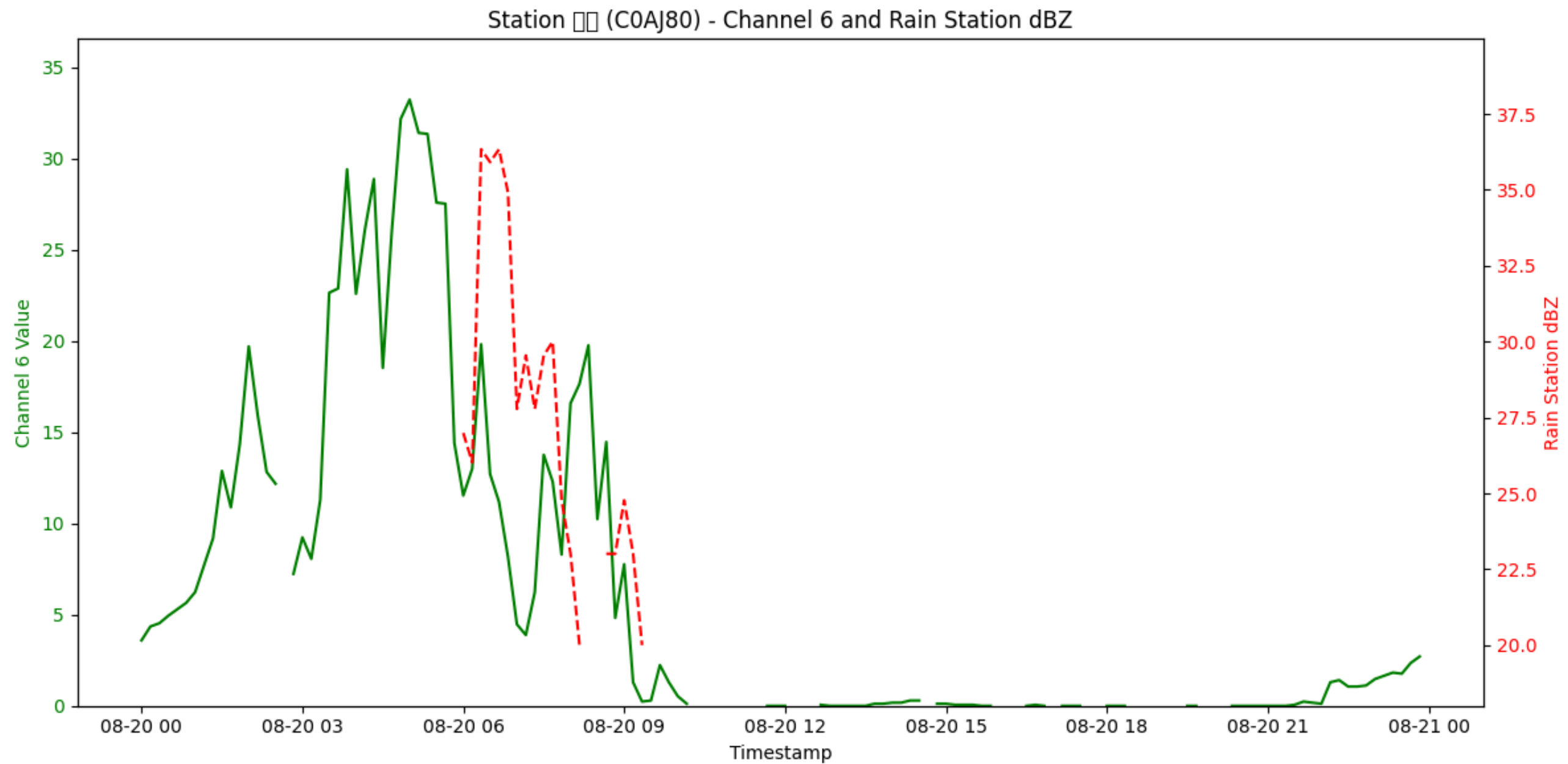
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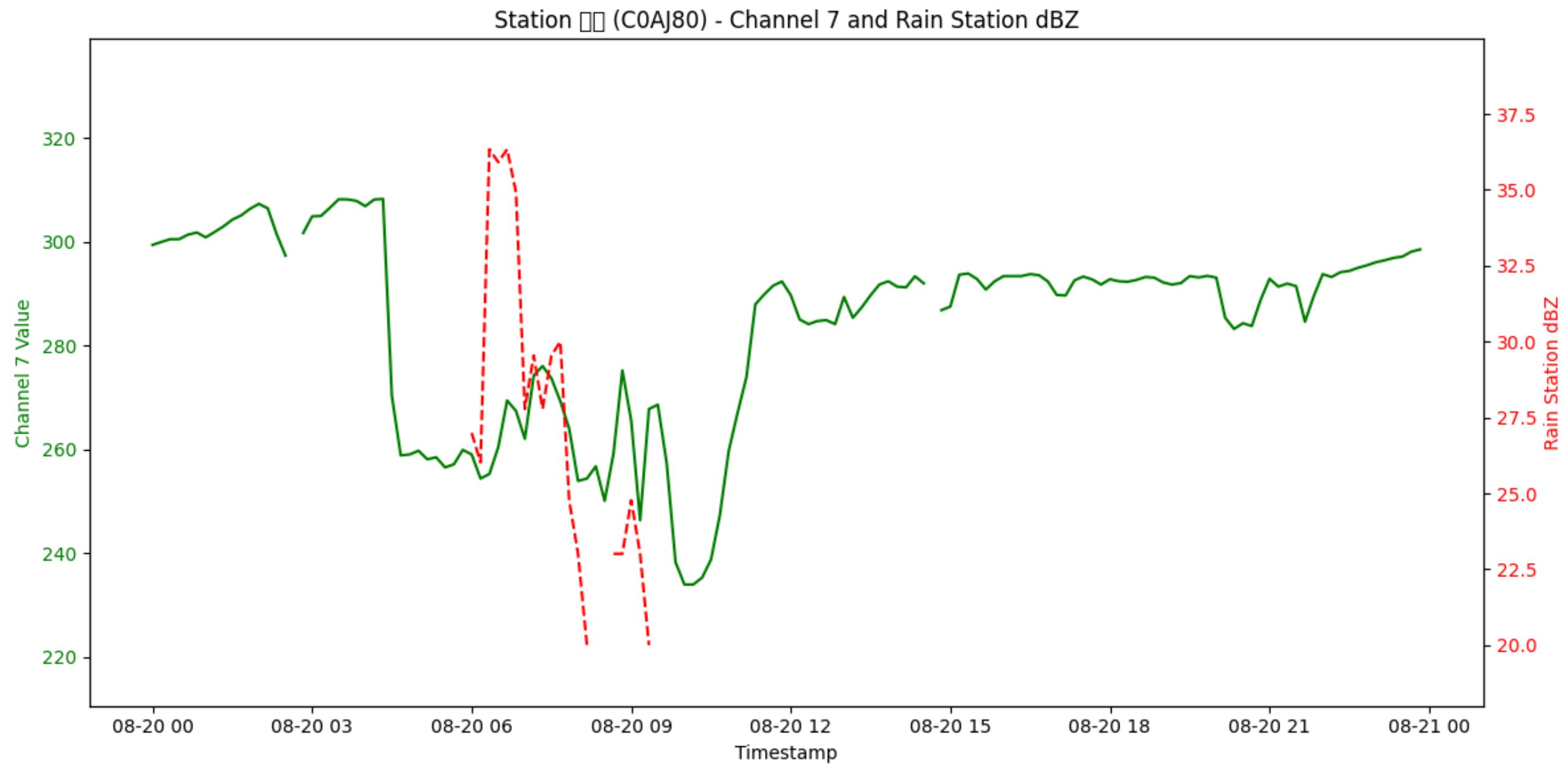
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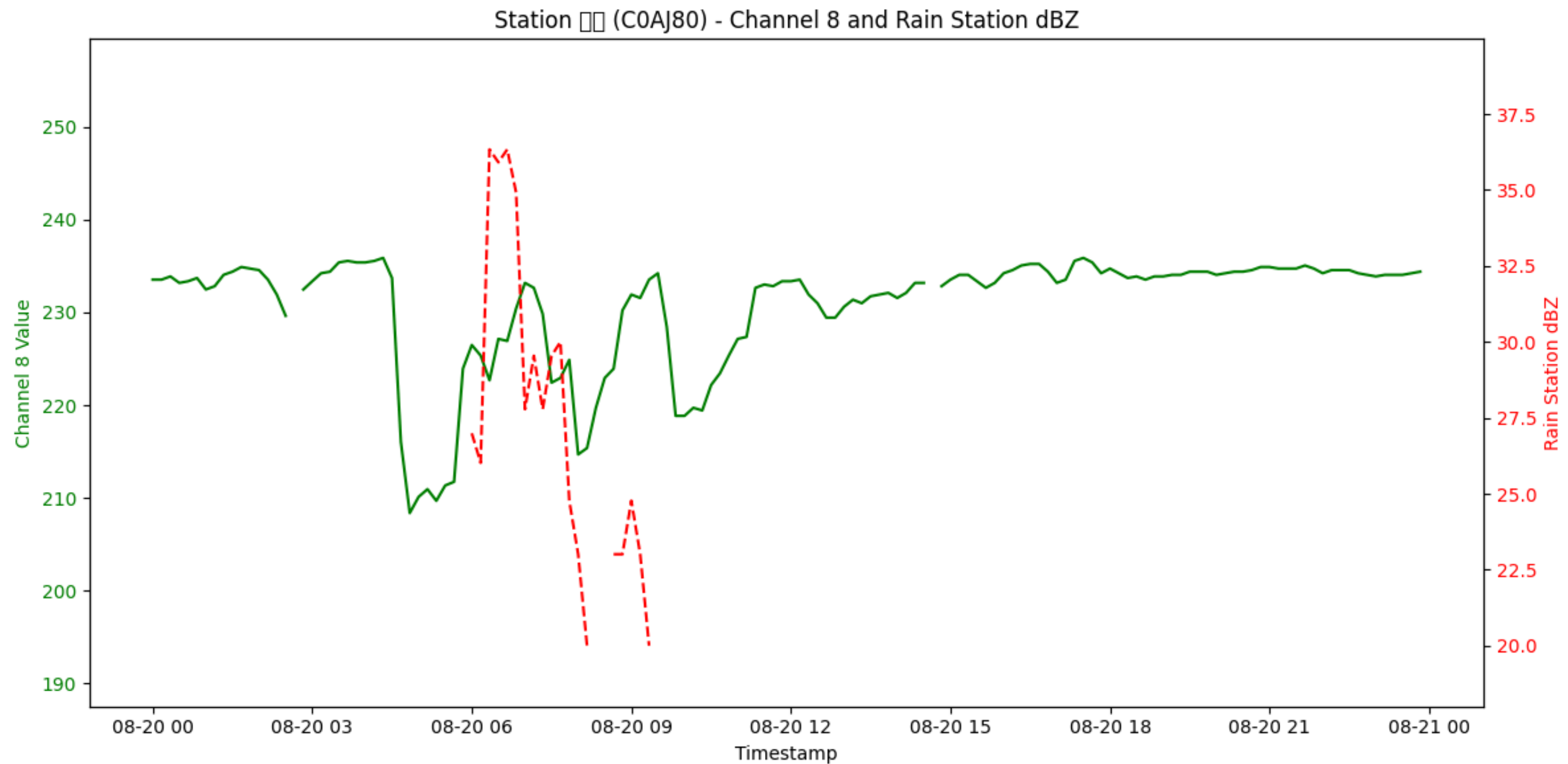
RAINFALL VS CHANNELS



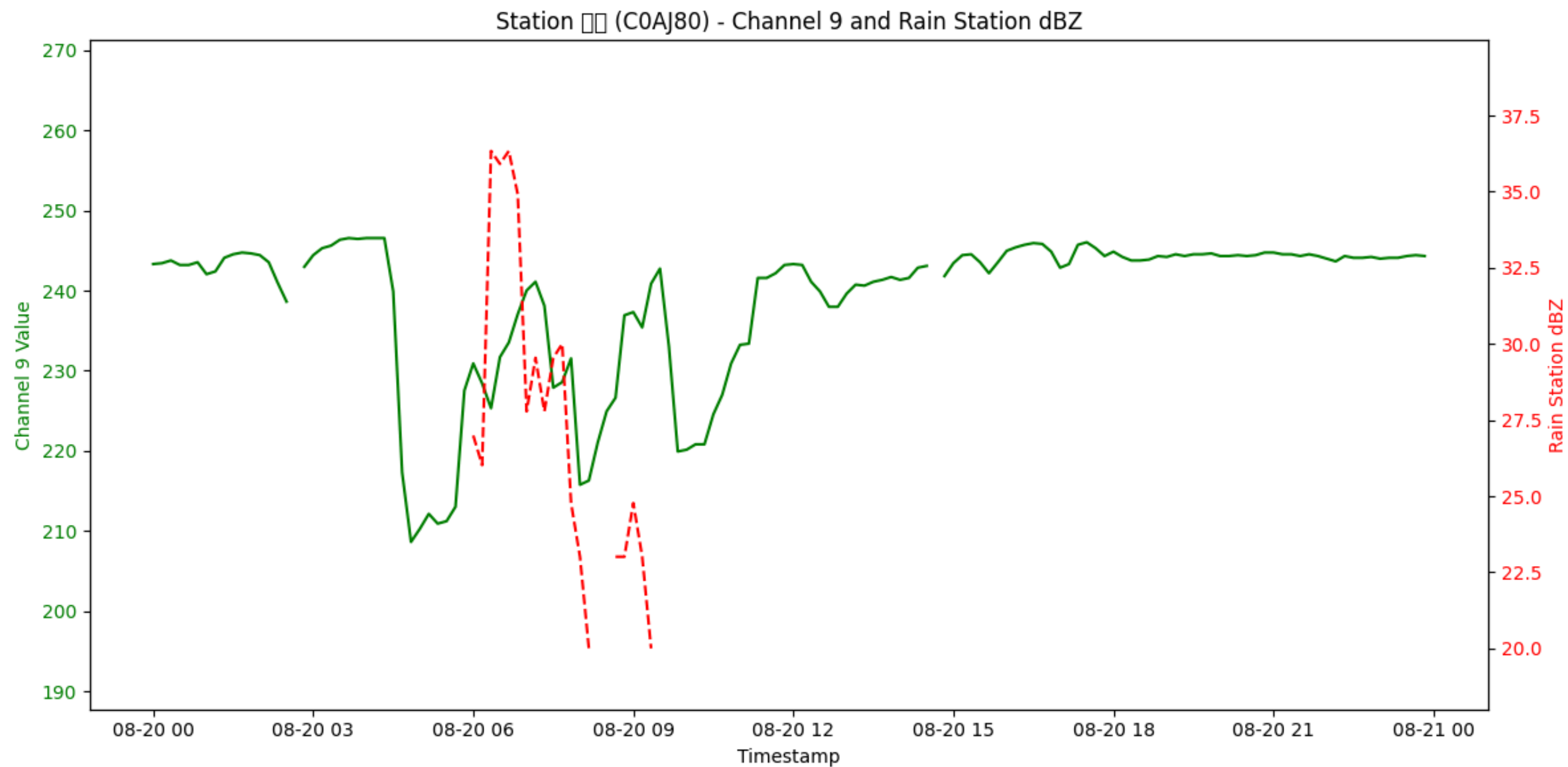
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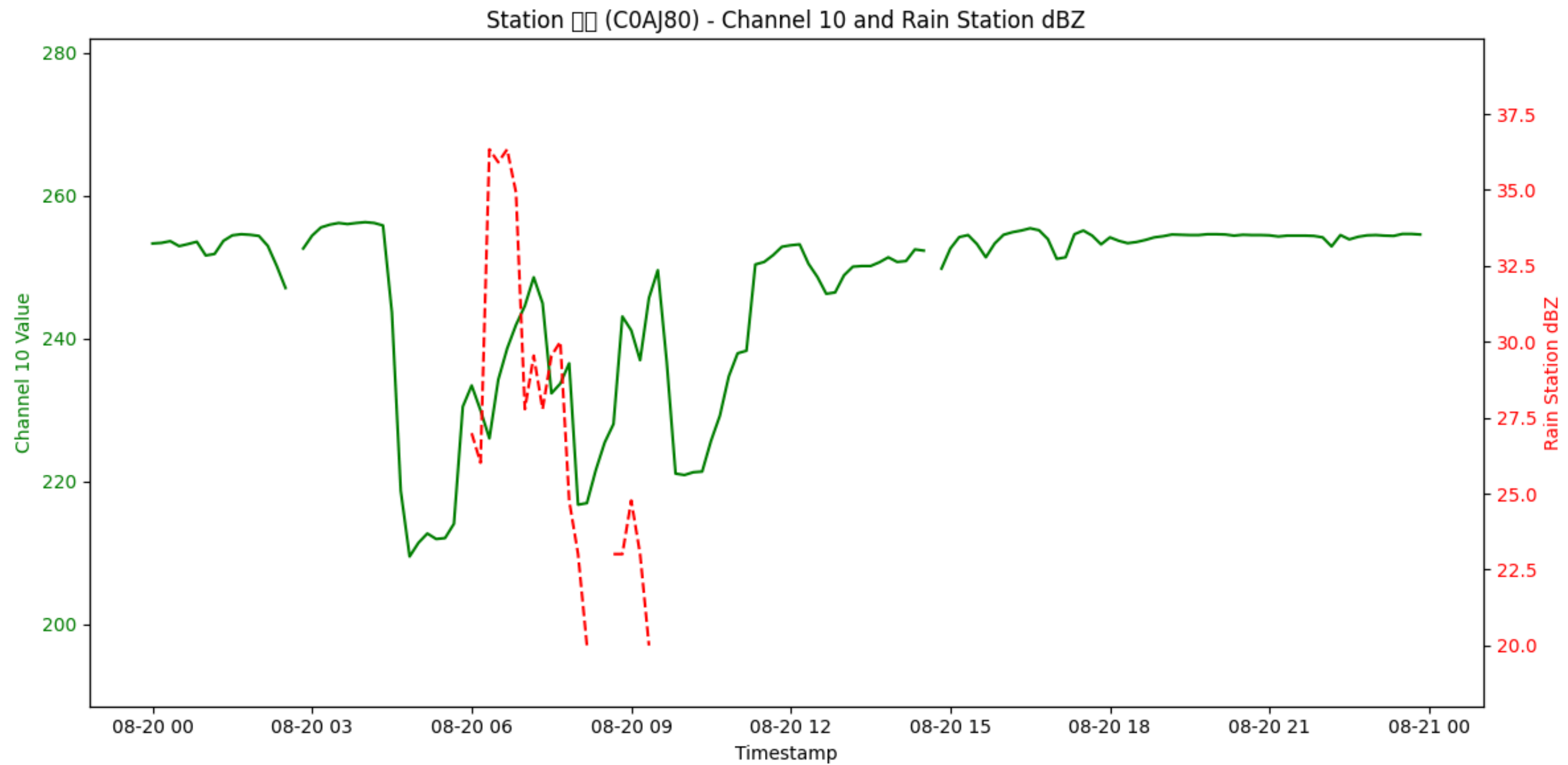
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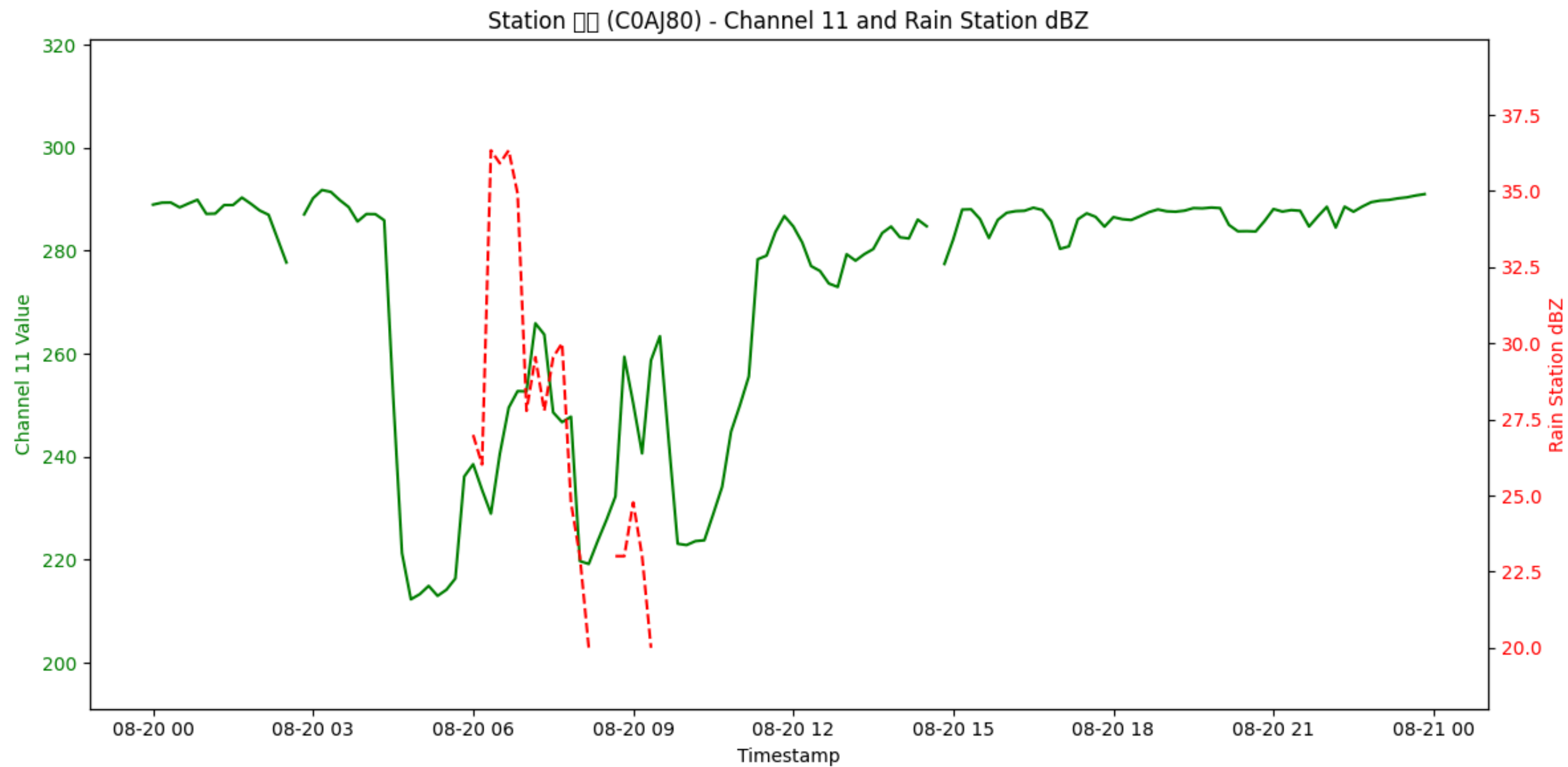
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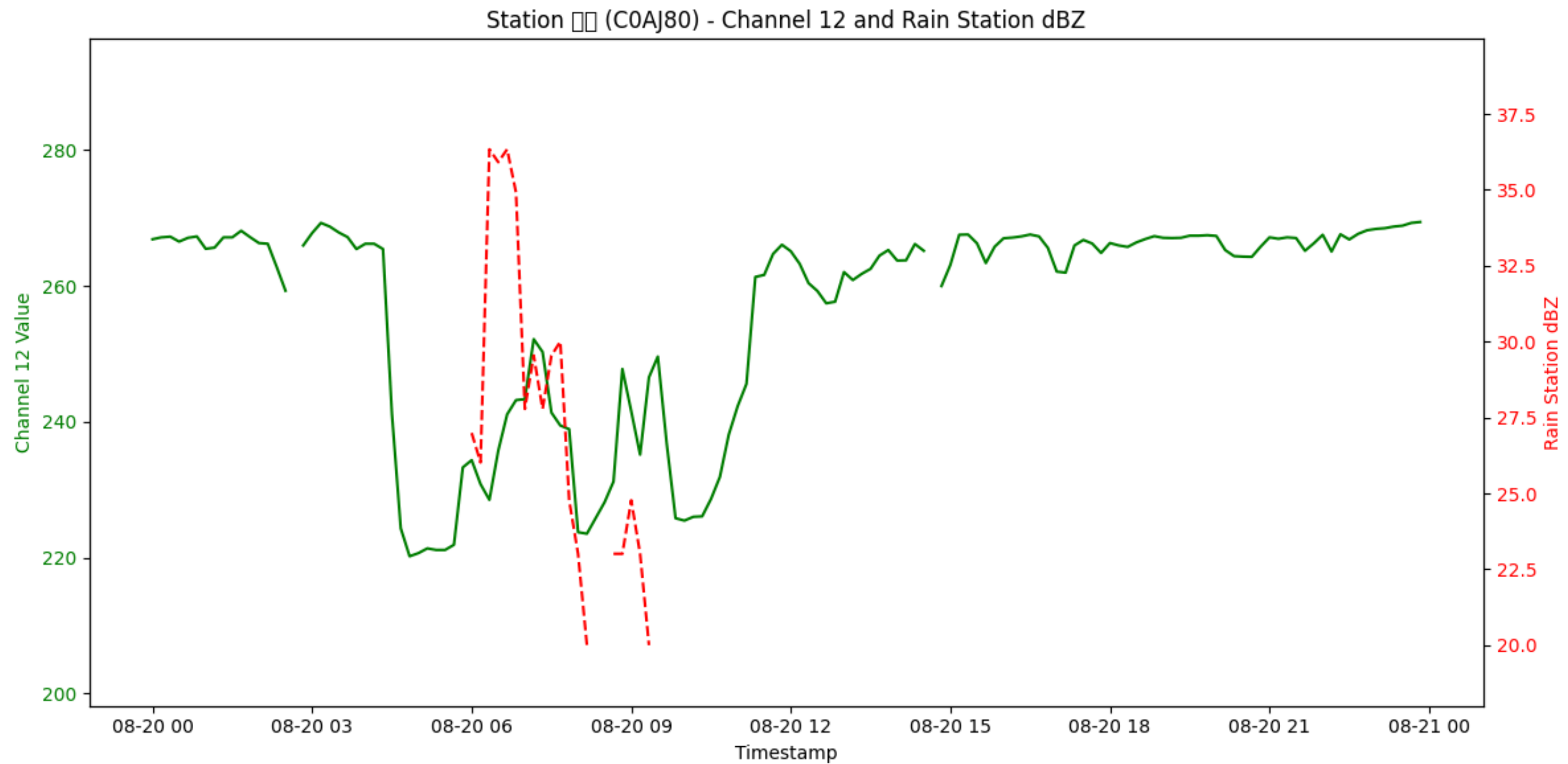
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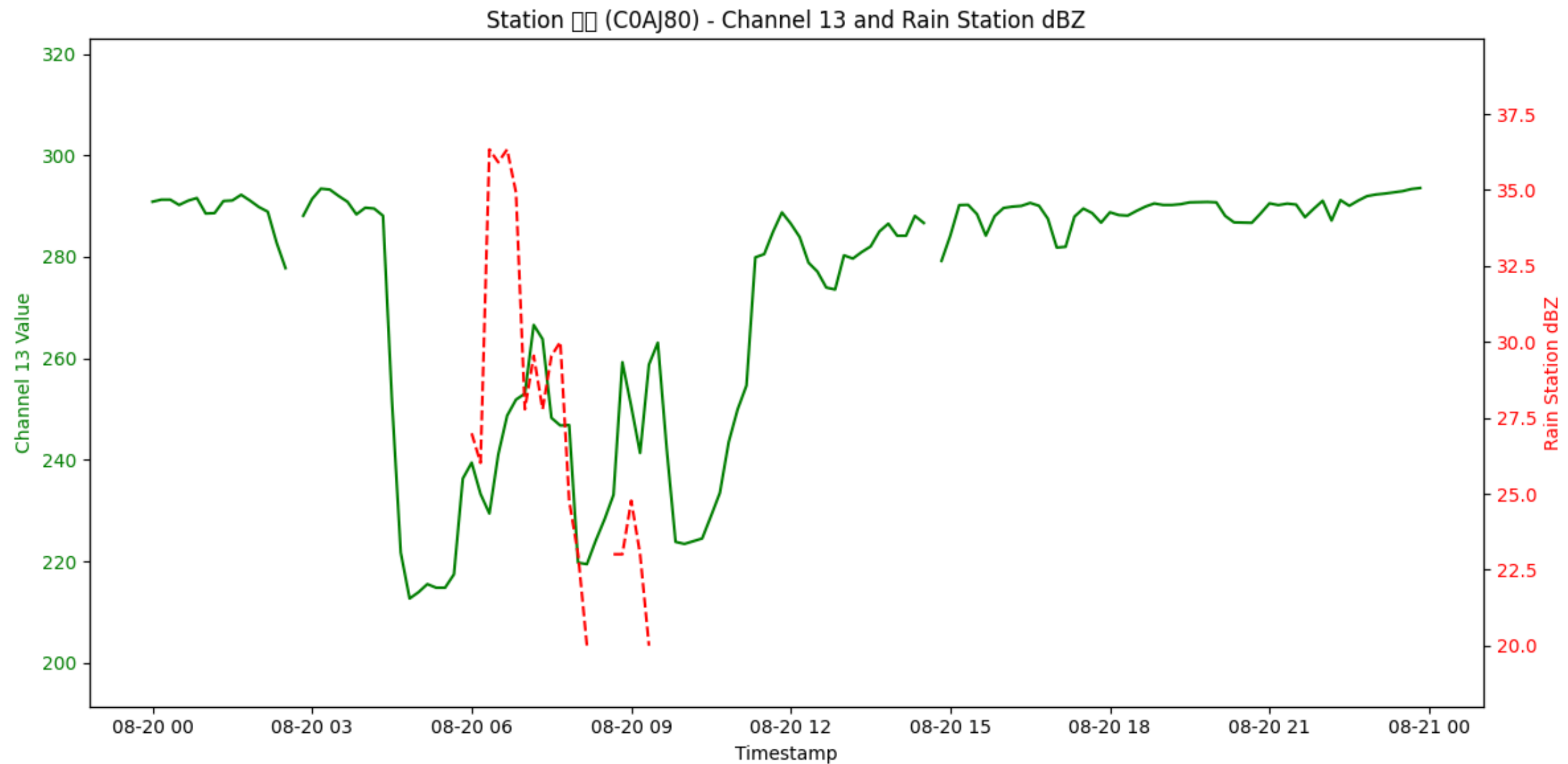
RAINFALL VS CHANNELS



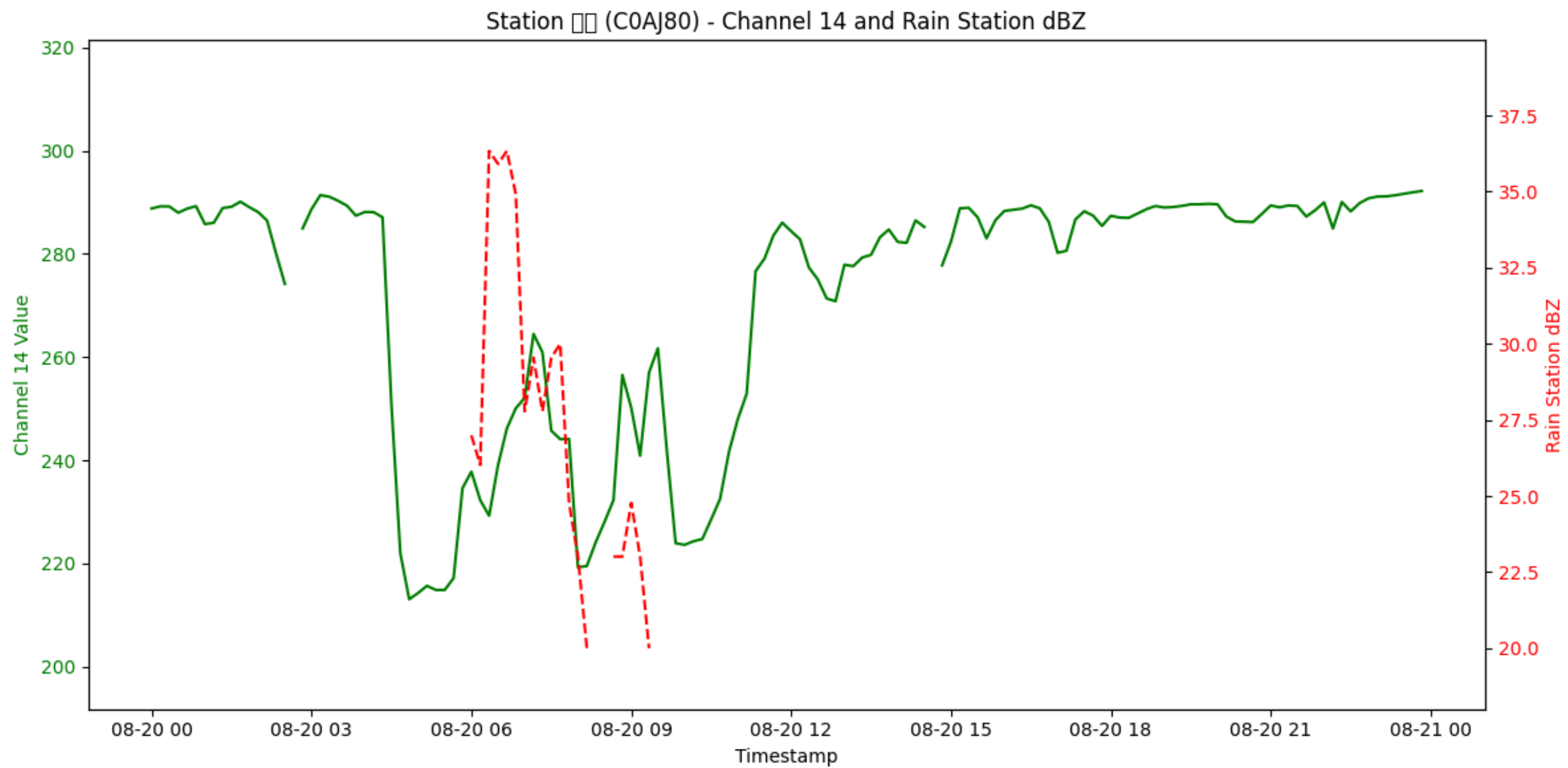
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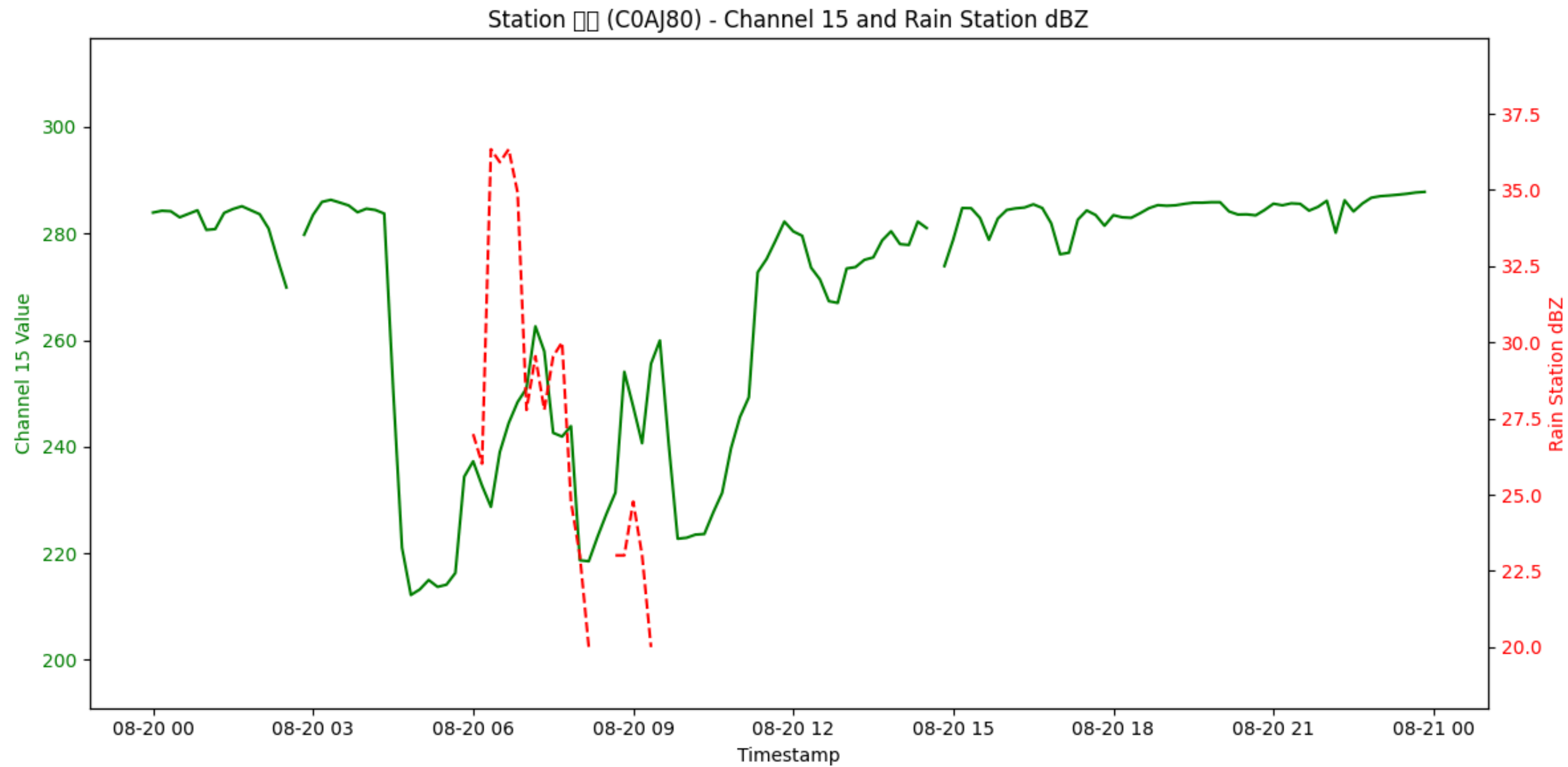
RAINFALL VS CHANNELS



RAINFALL VS CHANNELS



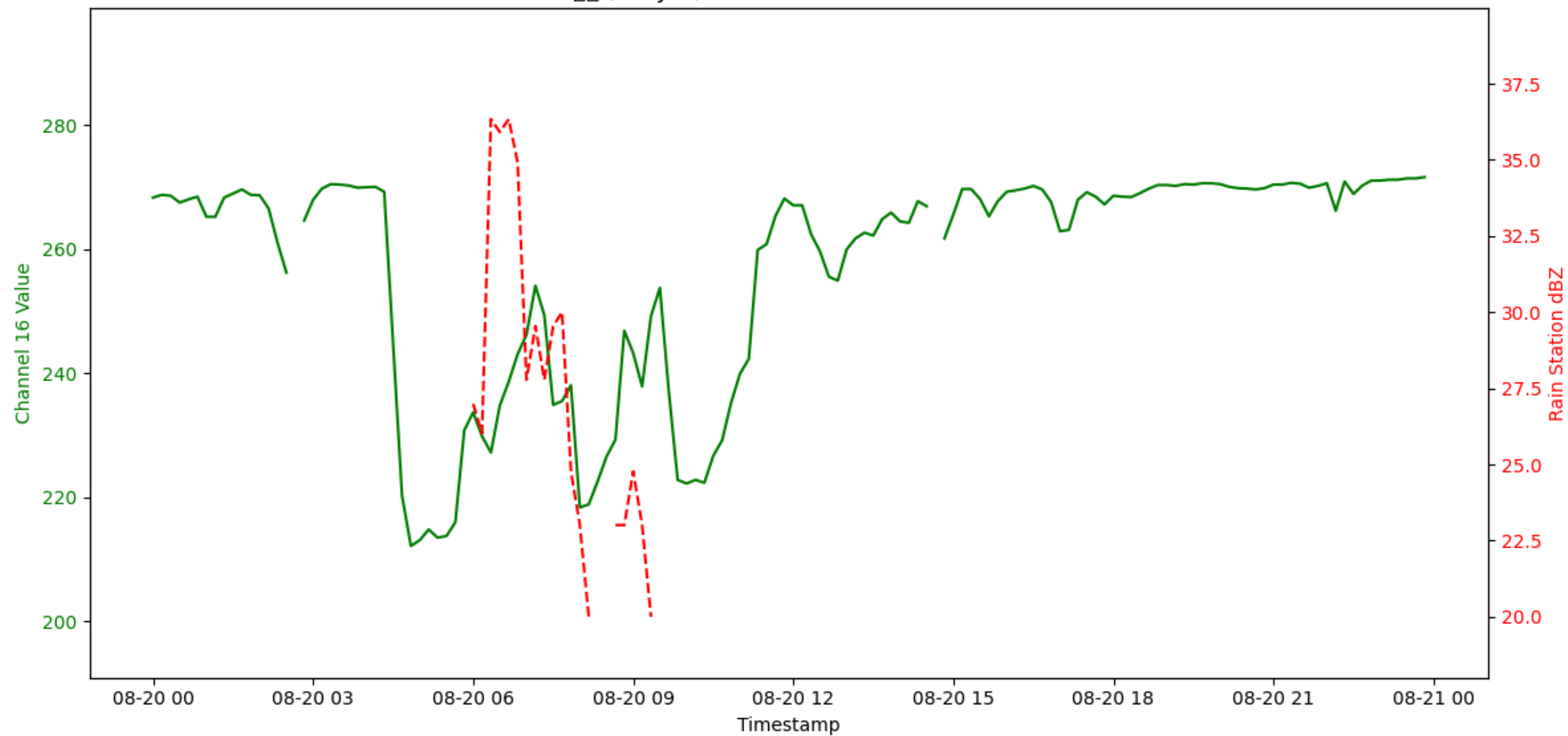
RAINFALL VS CHANNELS



RAINFALL VS CHANNELS



Station □□ (C0AJ80) - Channel 16 and Rain Station dBZ



RAINFALL VS CHANNELS



Channel	波段範圍	名稱
1	0.47	藍光通道 (Visible Blue)
2	0.64	紅光通道 (Visible Red)
3	0.86	近紅外 (NIR Vegetation)
4	1.37	近紅外 (NIR Cirrus)
5	1.6	雪冰通道 (Snow/Ice)

RAINFALL VS CHANNELS

Channel	波段範圍	名稱
6	2.2	雲粒徑 (Cloud Particle)
7	3.9	中紅外 (Shortwave IR)
8	6.2	水汽上層 (Upper-level Water Vapor)
9	6.9	水汽中層 (Mid-level Water Vapor)
10	7.3	水汽低層 (Lower-level Water Vapor)

RAINFALL VS CHANNELS

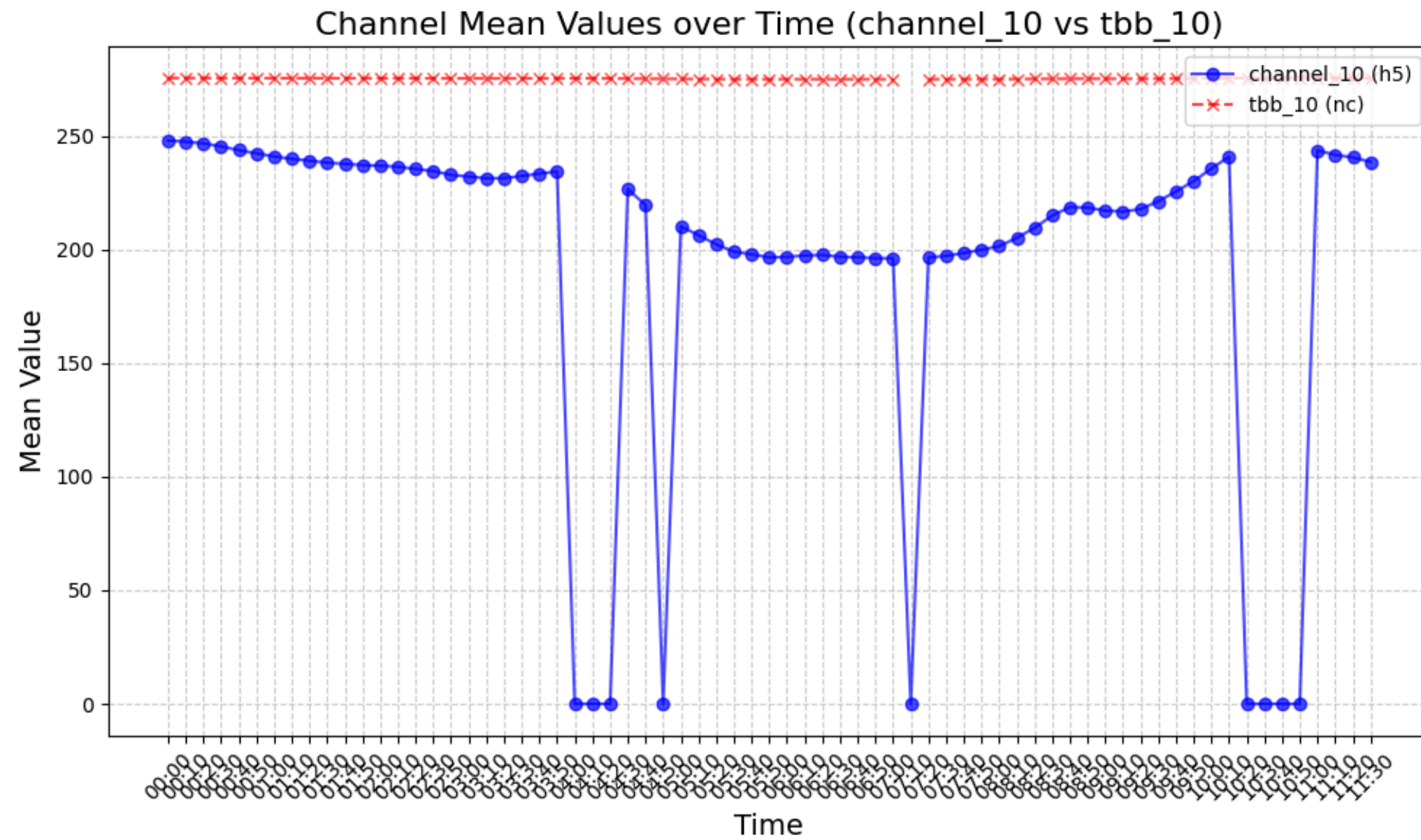
Channel	波段範圍	名稱
11	8.4	硅酸鹽吸收 (Cloud-Top Phase)
12	9.6	臭氧 (Ozone Band)
13	10.3	窗區紅外 (Clean IR Window)
14	11.2	長波紅外 (IR Window)
15	12.3	微量氣體 (Dirty IR Window)

RAINFALL VS CHANNELS



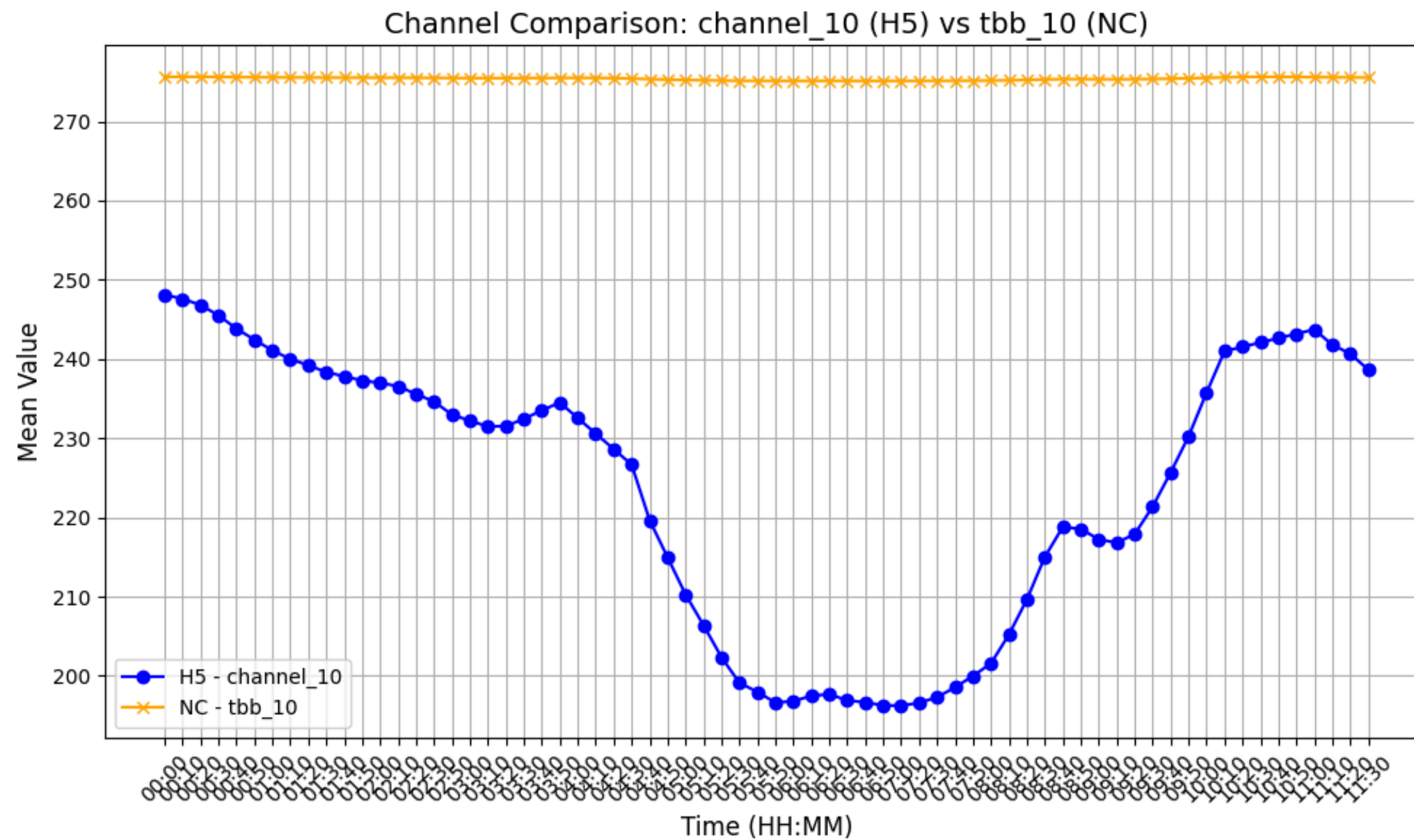
Channel	波段範圍	名稱
16	13.3	二氧化碳（CO ₂ Absorption）

NCDR VS 氣象廳

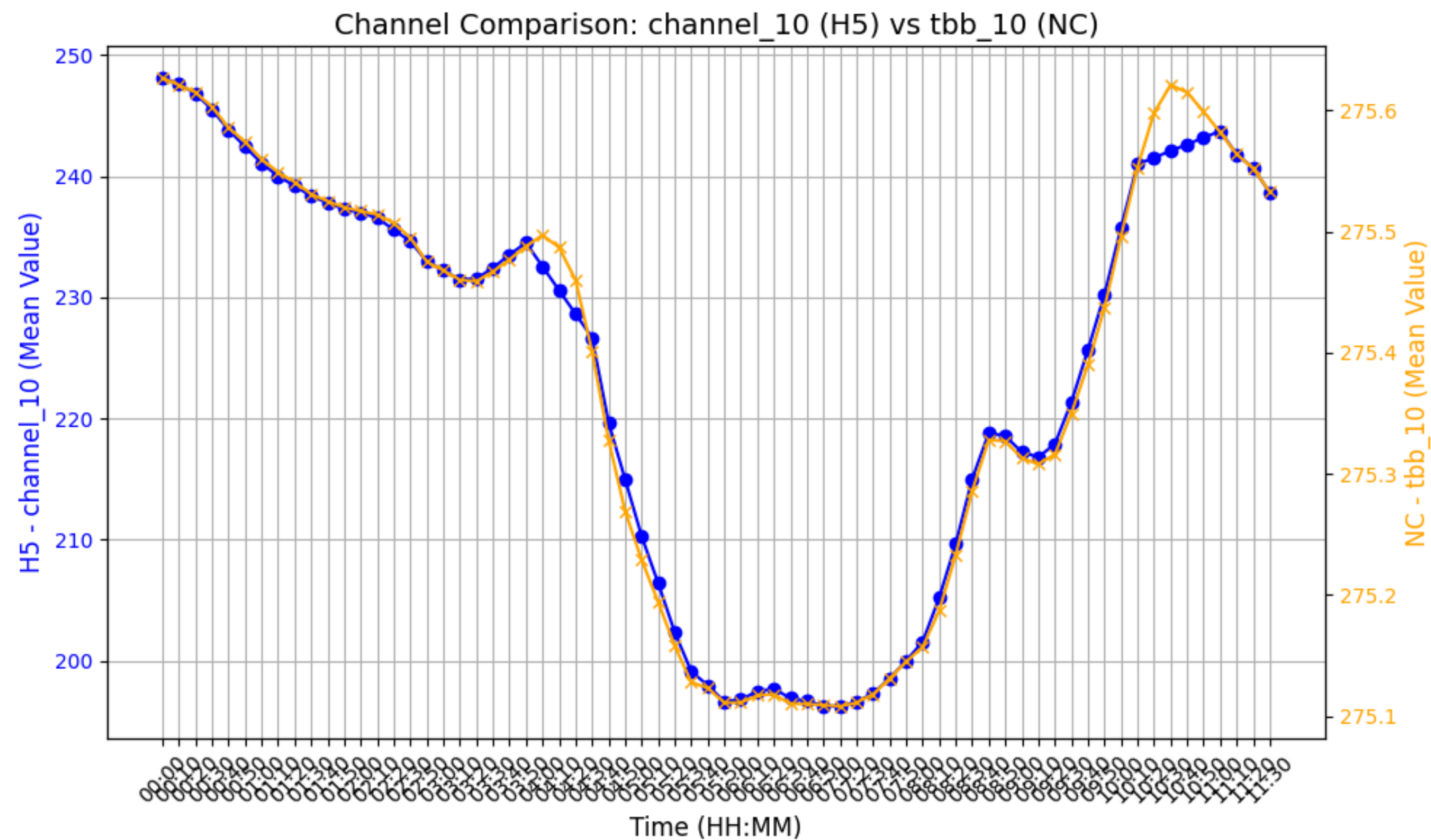


藍色為**NCDR**資料(尚未補缺值)
紅色為氣象廳資料

NCDR VS 氣象廳



NCDR VS 氣象廳



藍色為**NCDR**資料(內插補缺值)
黃色為氣象廳資料
用不同座標軸來看變化趨勢

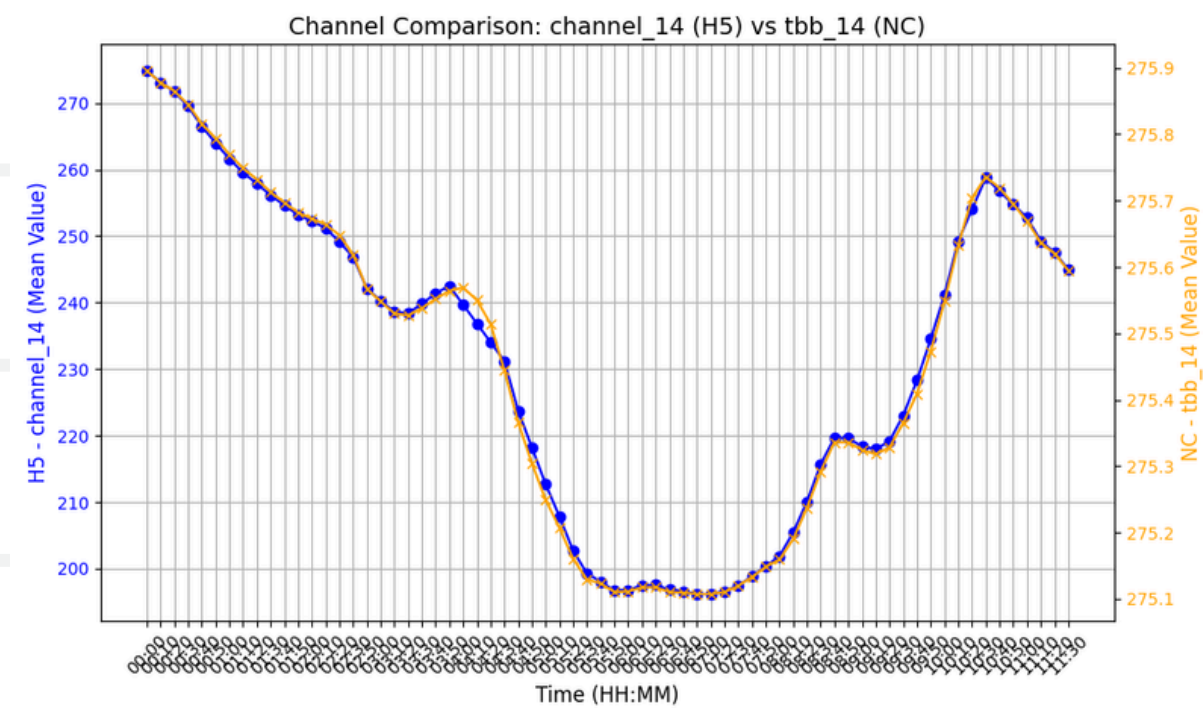
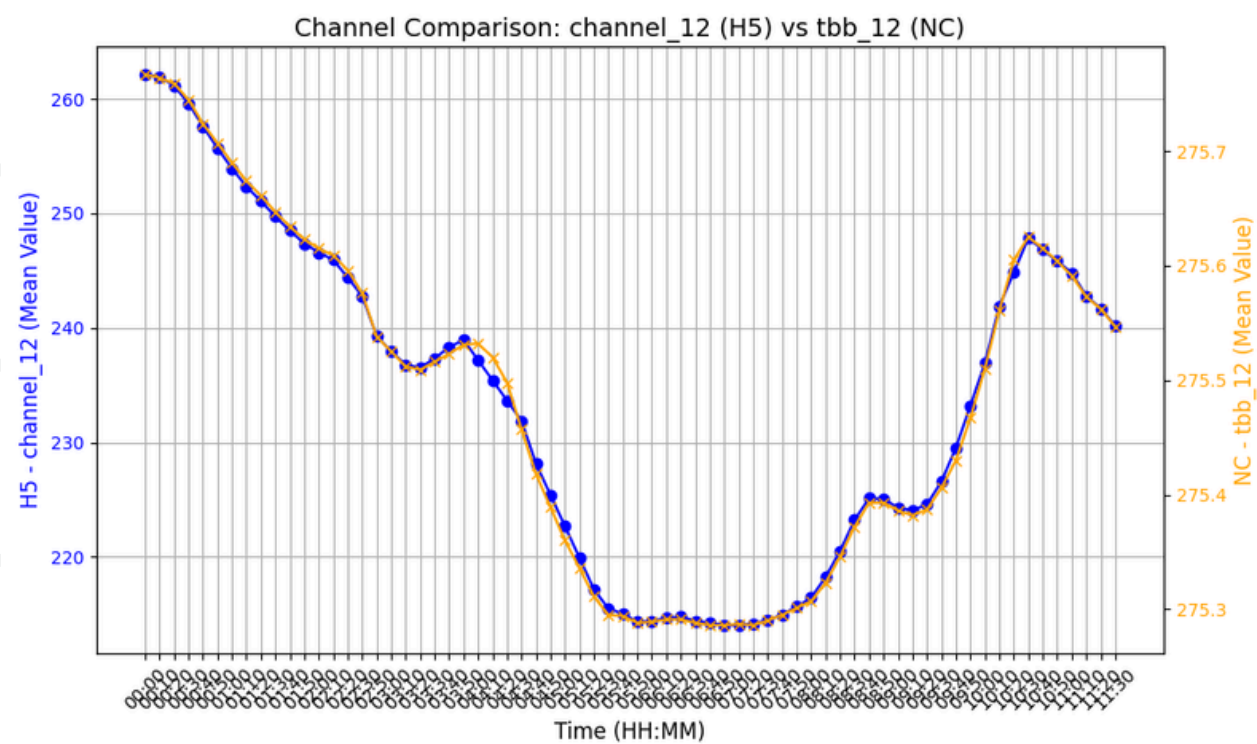
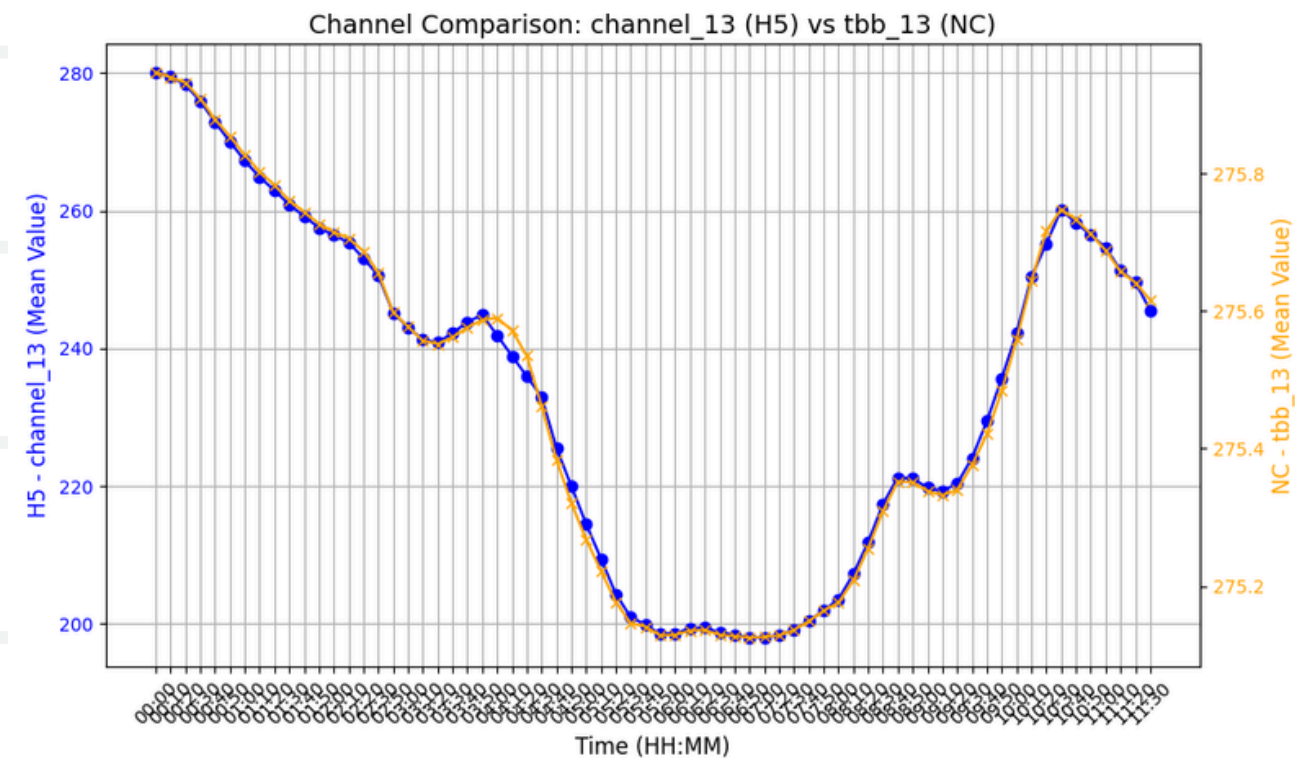
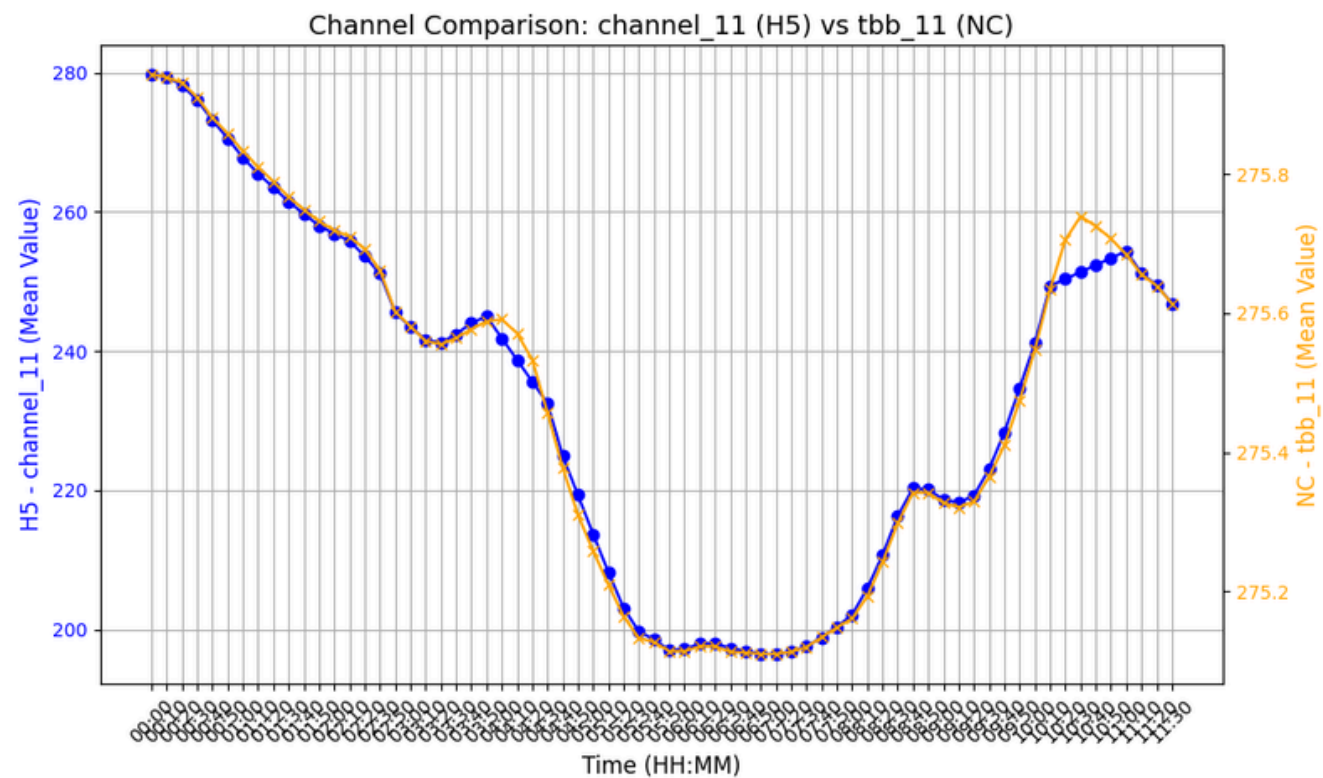
NCDR VS 氣象廳



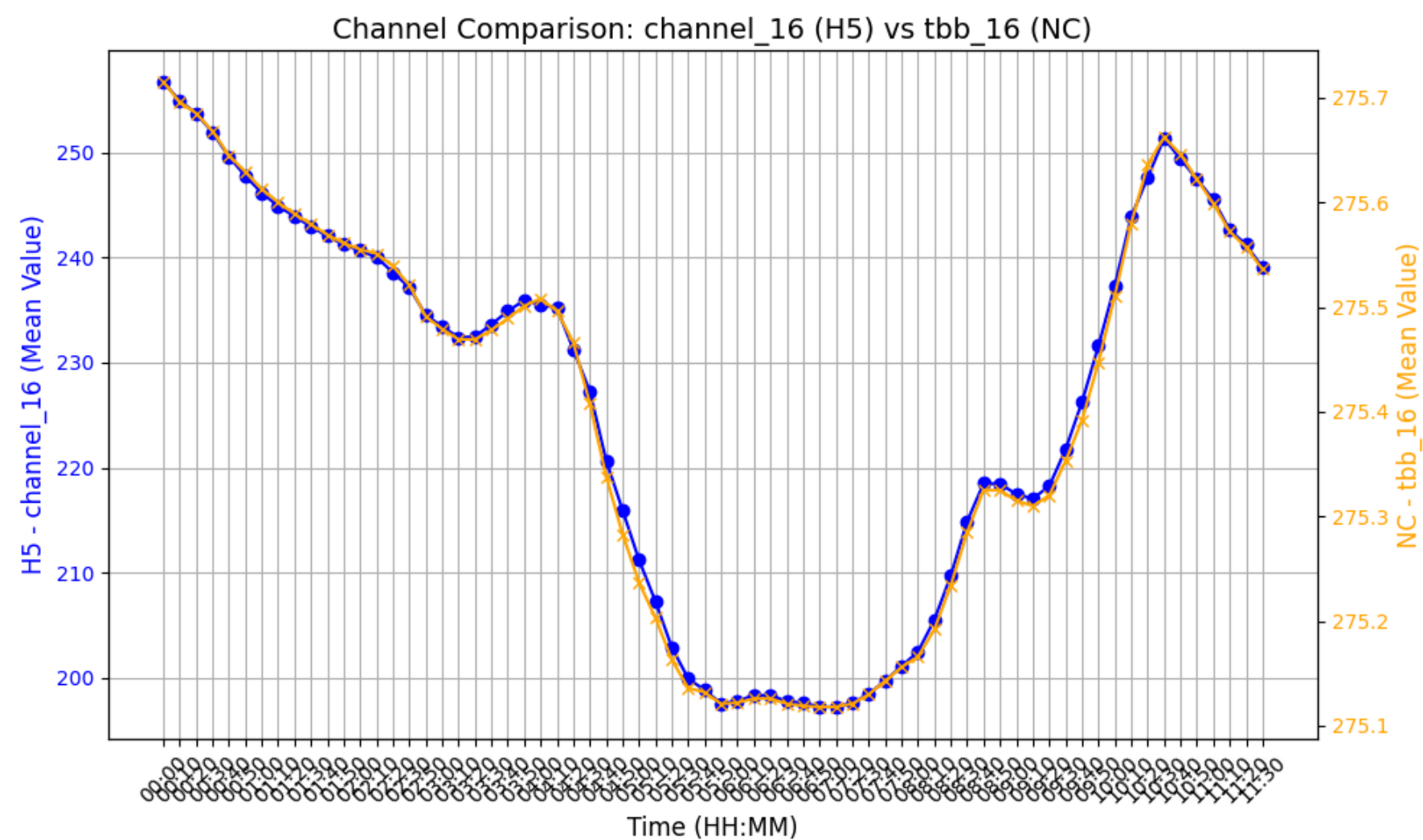
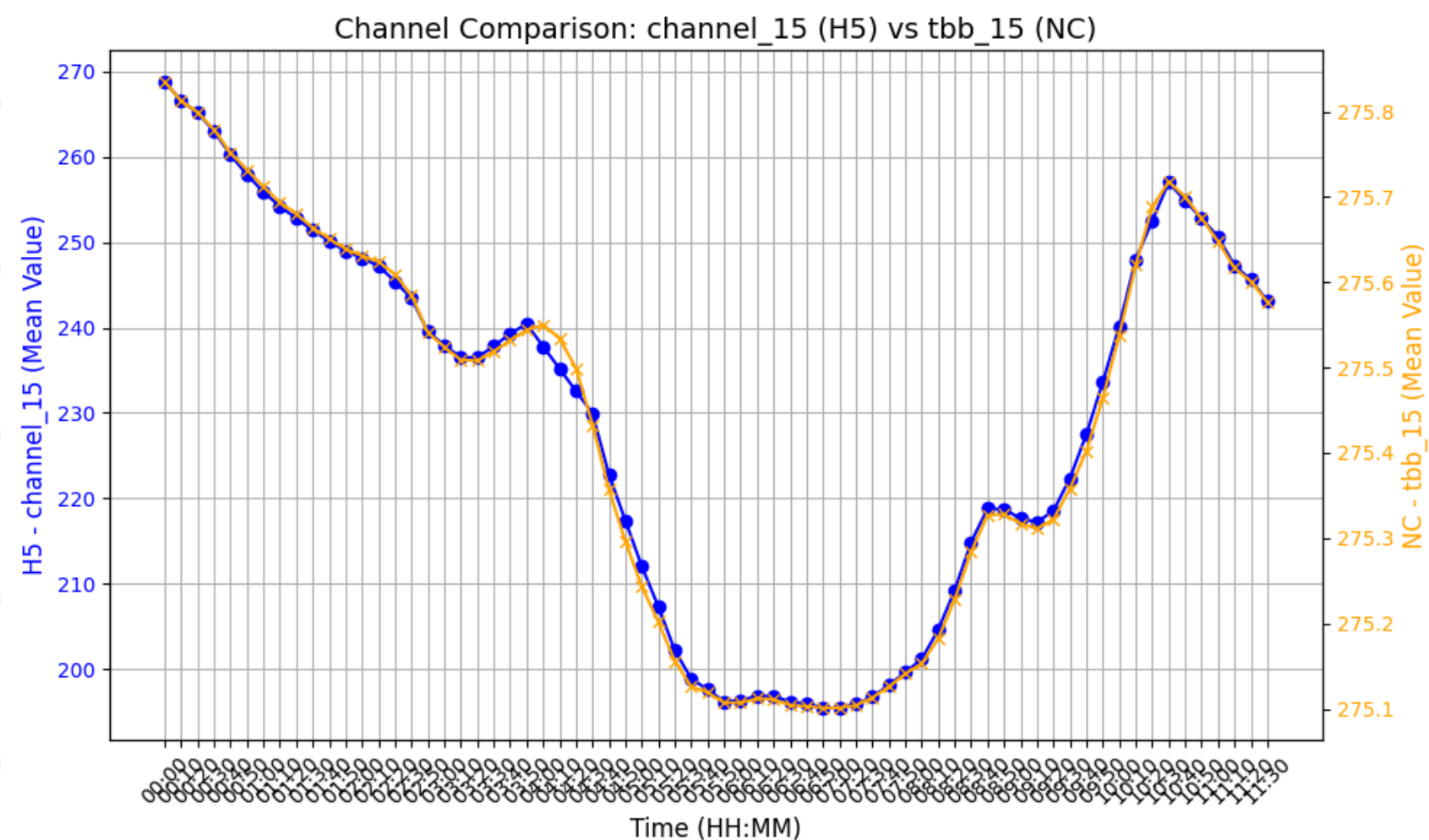
目前遇到問題:

- 氣象廳與NCDR的資料雖然變化趨勢相同，但測量值的區間不同(同一頻道)
- 氣象廳資料裡可見光波段測量值皆為0
- NCDR缺值較多
- 紅外線波段(TBB)不同頻道的範圍不同，但變化趨勢幾乎一樣

NCDR VS 氣象廳



NCDR VS 氣象廳





2025/06/10~



目前遇到問題:

- **Warning: Zero weight for featId:**
- 在 `__recurseTracksStartingFromFeature()` 裡 追蹤某個細胞的「前向關聯 (forward association)」時出現的錯誤警示

原因:

- 影像解析度太低，分裂後的細胞被誤判為新物件
- 資料某一幀 **JSON** 不存在，或 **FeatureCollection** 為空。
- 特徵面積異常小或空集合
- 子細胞移動距離太遠，兩幀之間距離超過 `axmax (fcmater.py)`

ZERO WEIGHT FOR FEATID



解決辦法1: **fallback** 對每個權重不夠的子雨胞採用平均權重分配

優點:

- 補救效果好、可控制、明確標記 **fallback_used**

缺點

- 精度不穩、可能誤連軌跡

ZERO WEIGHT FOR FEATID



解決辦法2: 放寬 **axMax** 限制

優點:

- 不需要改主流程、可以搭配 **cost** 決策過濾

缺點

- 容易配錯，系統整體追蹤品質變差、不易發現失誤



FALLBACK



選擇使用方法1(fallback)

原因:

- 保有原模型的嚴謹性
- 只讓極少數無法追蹤的細胞使用
- 後續品質分析（因為 **fallback** 都會標記）
- 不希望放寬判定（避免影響體精度）

AXMAX



何時會使用方法2(axMax)?

- 大多數細胞都移動很快
- cell 面積都小，但移動量大 → axMax 很小 → 很容易 fail
- 想提高回傳結果的量，不怕誤配、追完整比較重要
- 放寬 axMax 可能成功率提高，但精度下降

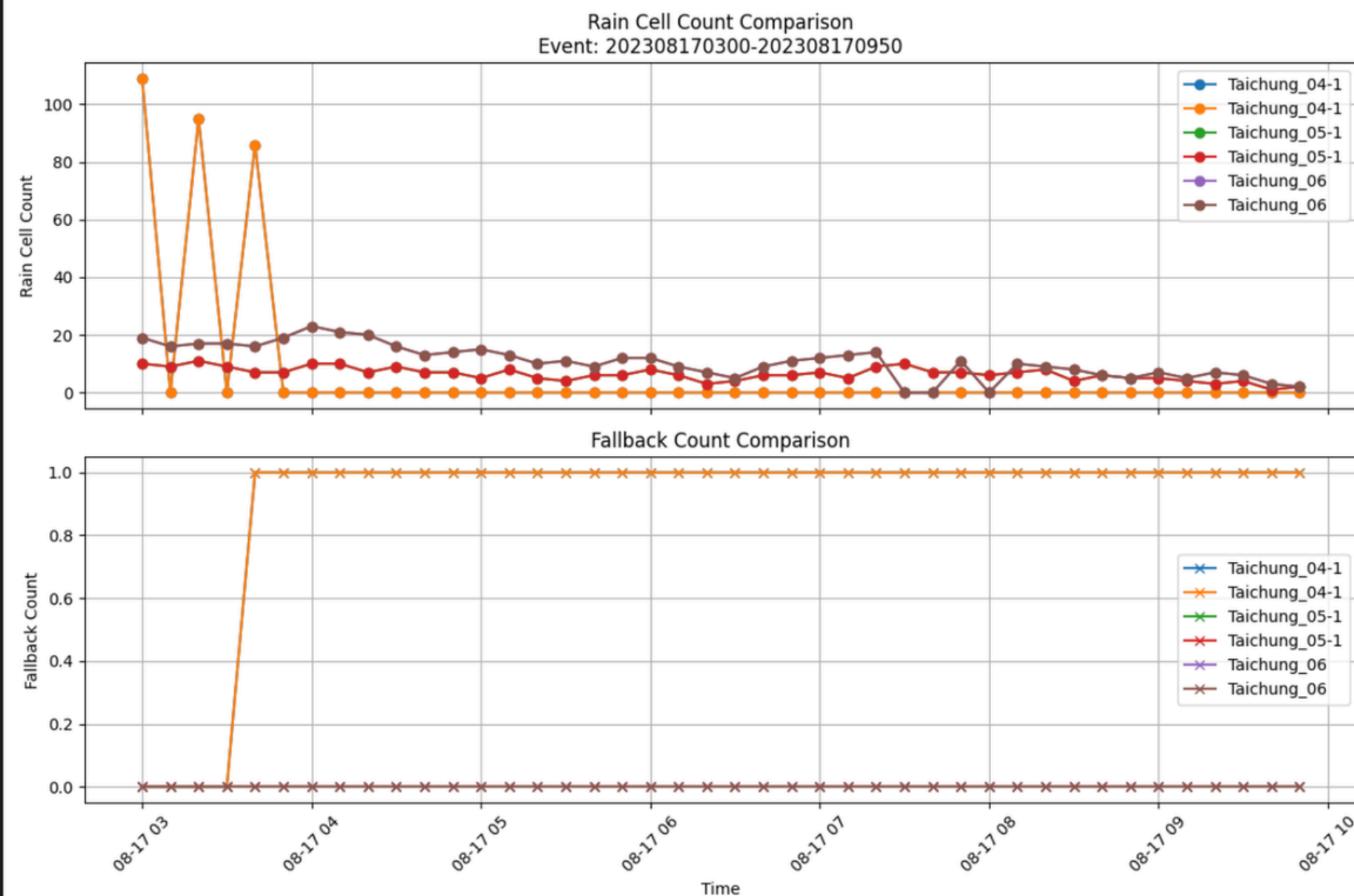
FALLBACK流程



1. 當前時間的雨胞做路徑推測，看看他在上一個時間點有每有可能有母雨胞
如果沒有可能歸類到新雨胞
2. 等全部分類好後，在去上一個時間點對每個雨胞做路徑預測，預測他在下一個時間的有沒有可能有子雨胞，並跟上一步的結果match
3. 有一個雨胞被歸類在子雨胞(回推時被分類)，但在前推時(上個時間點做預測)沒有任何雨胞是他的母雨胞，觸發fallback

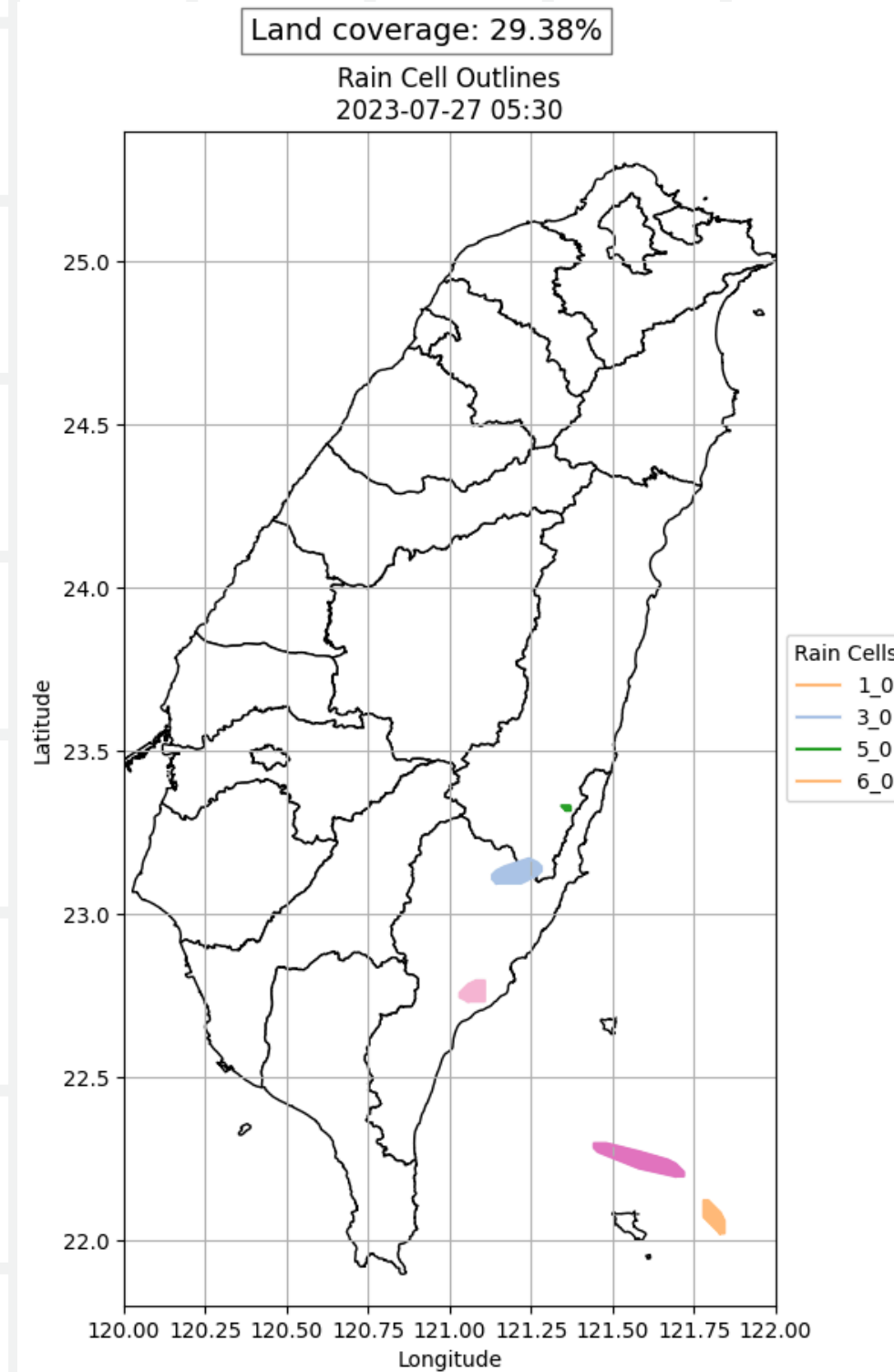
比較結果

- 對不同參數做雨胞數量
(回報率)、**fallback**次
數(路徑品質)的比較圖
找到最優參數



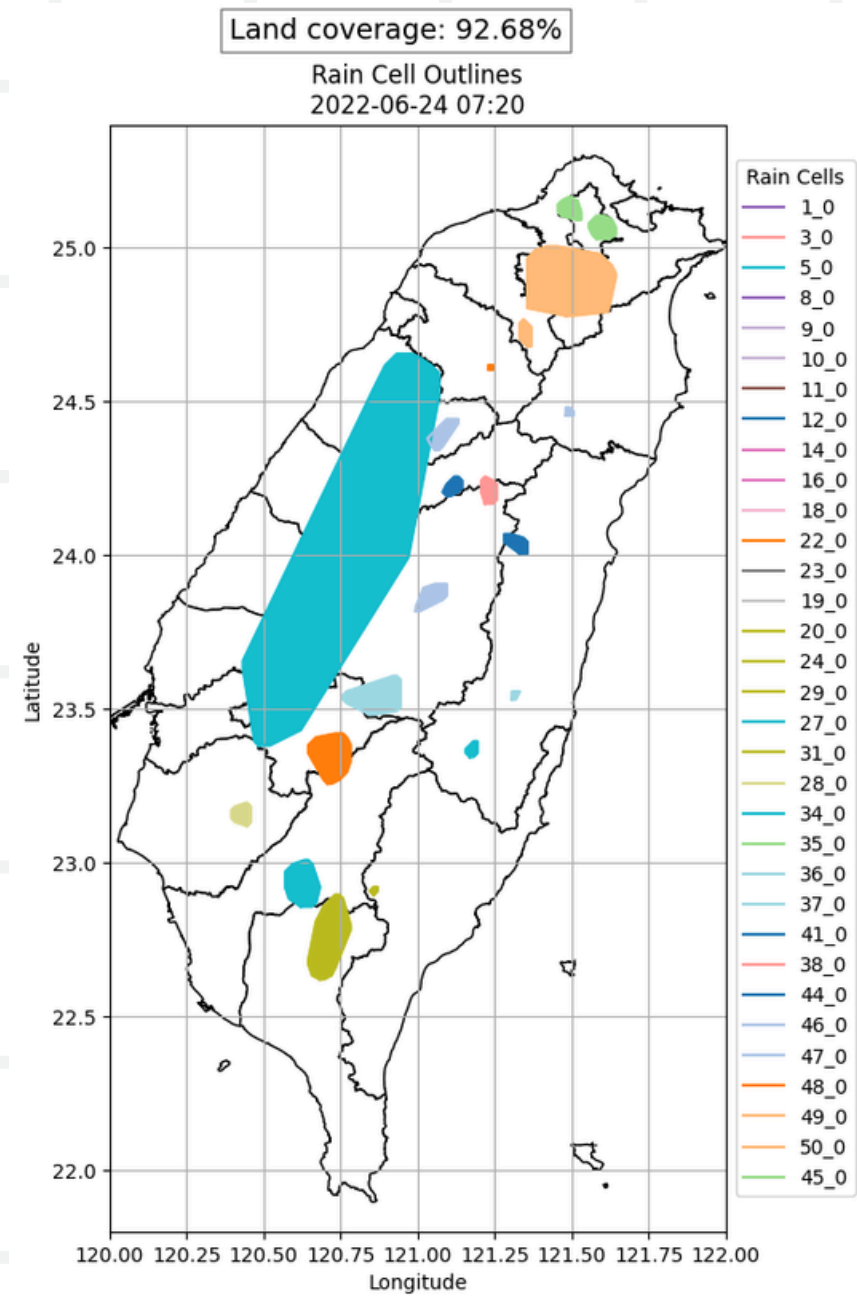
比較結果

- 另外在投影雨胞到地圖上時，計算雨胞占地面積，篩選出由陸地產生的雨胞事件

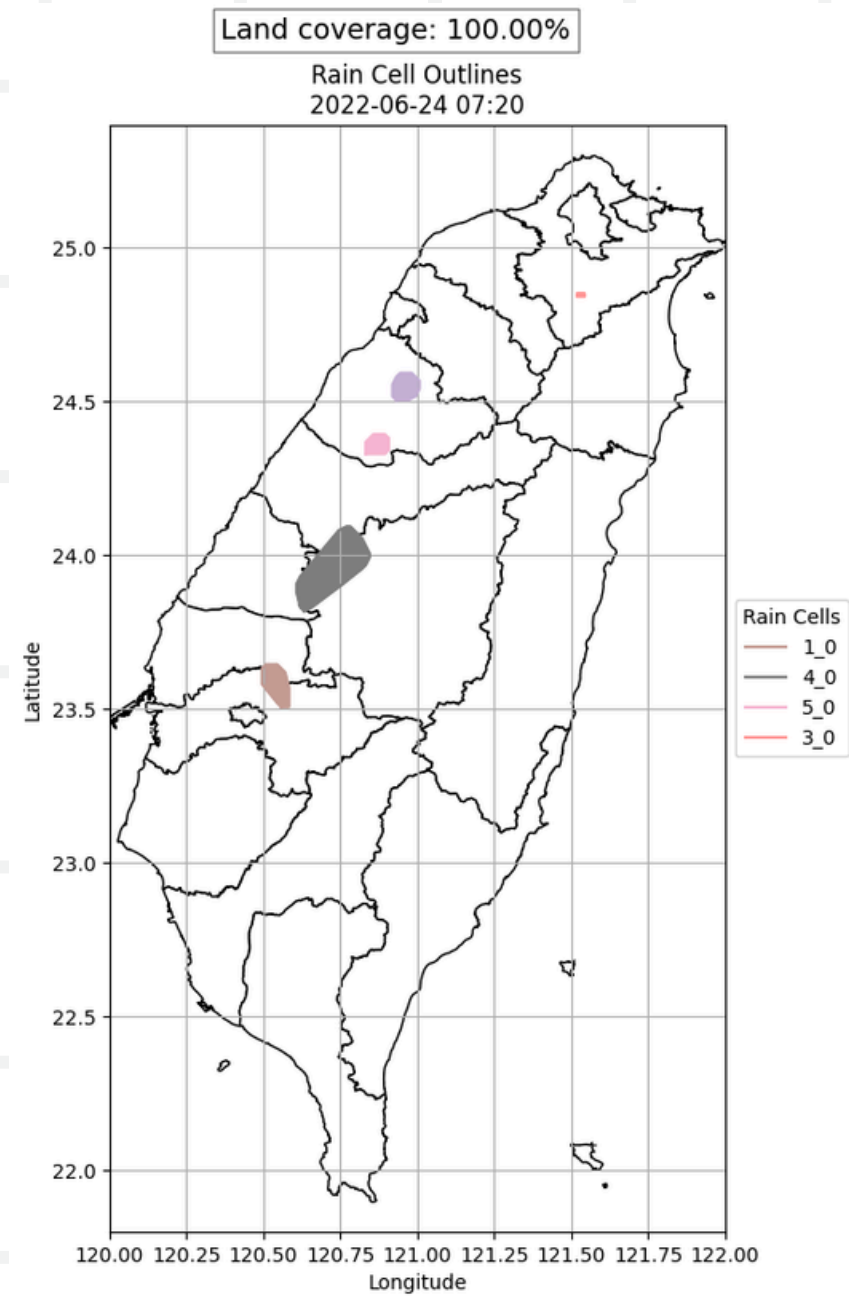


比較結果

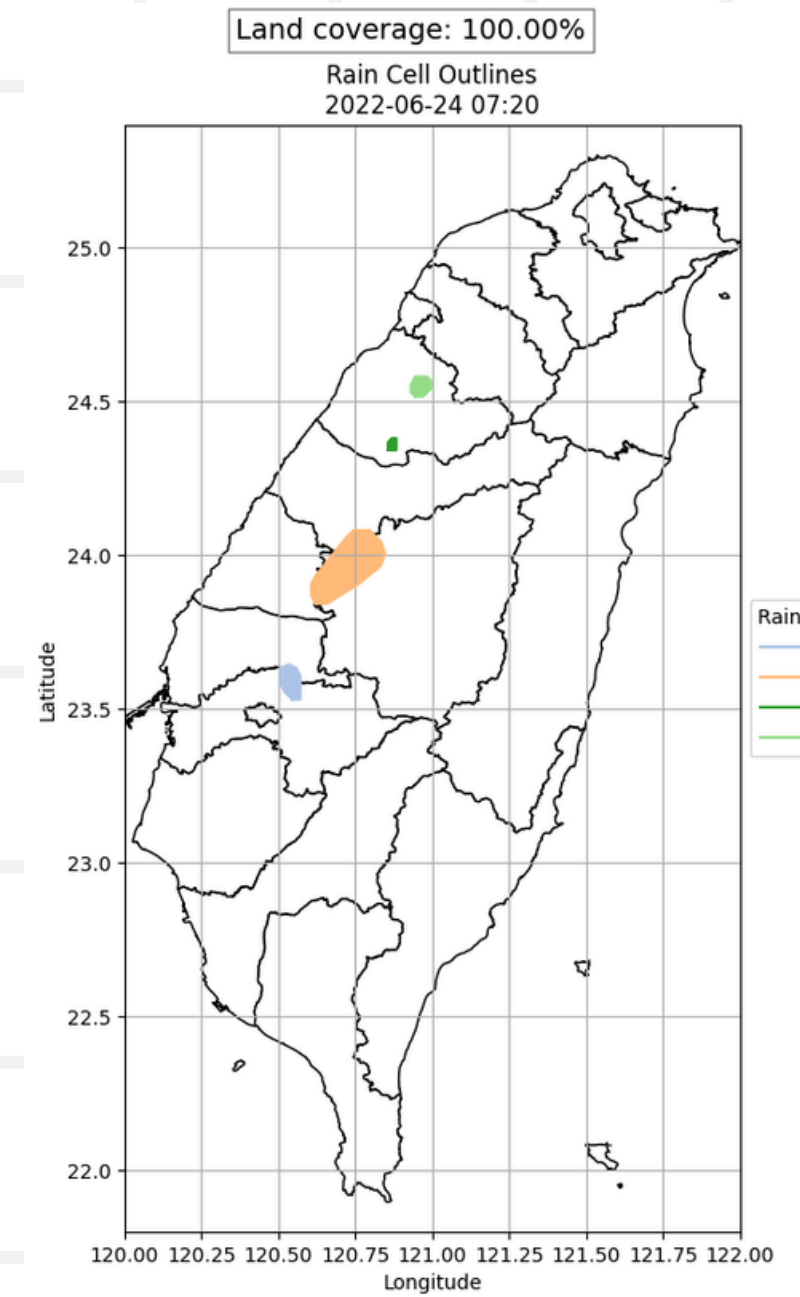
Setting 7



Setting 6



Setting 5



比較結果



Setting 7

```
Param VIL  
Min area: 4  
Smooth width: 1.0  
Min thresh: 1  
Max thresh: 80.0  
Bin width: 3.0  
fall back yes
```

Setting 6

```
Param VIL  
Min area: 4  
Smooth width: 2.0  
Min thresh: 5  
Max thresh: 80.0  
Bin width: 2.0  
fall back yes
```

Setting 5

```
Param VIL  
Min area: 6  
Smooth width: 3.0  
Min thresh: 5  
Max thresh: 80.0  
Bin width: 5.0  
fall back yes
```



遇到問題



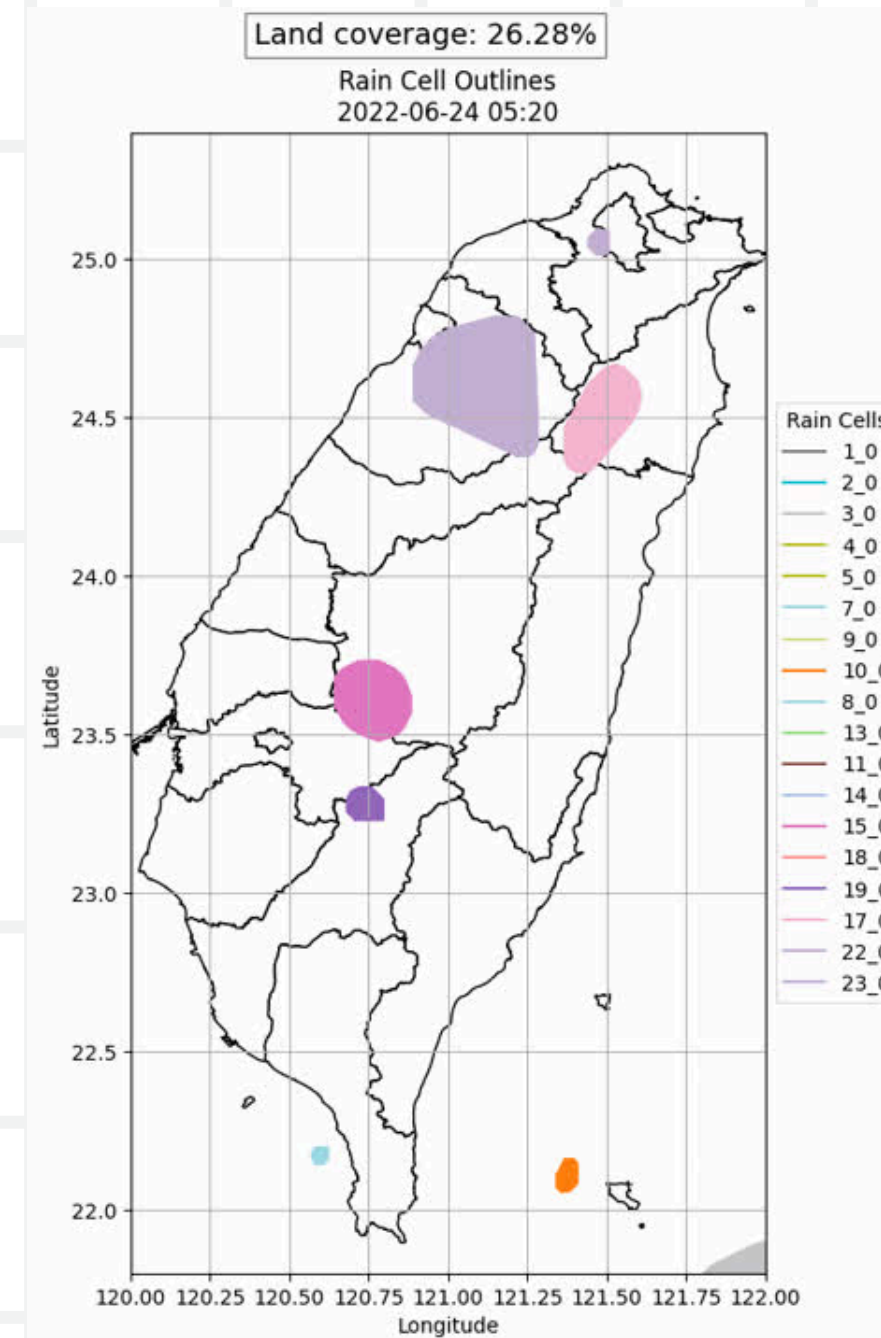
- 雨胞占地面積大致偏小
- 難找到同時符合在陸地生成、在陸地消散的雨胞
- 時間解析度太低，難找到雨胞精確的消散地點

比較結果

Setting

Param VIL
Min area: 2.5
Smooth width: 3
Min thresh: 0.25
Max thresh: 80.0
Bin width: 3.0
fall back yes

VIL 0.25



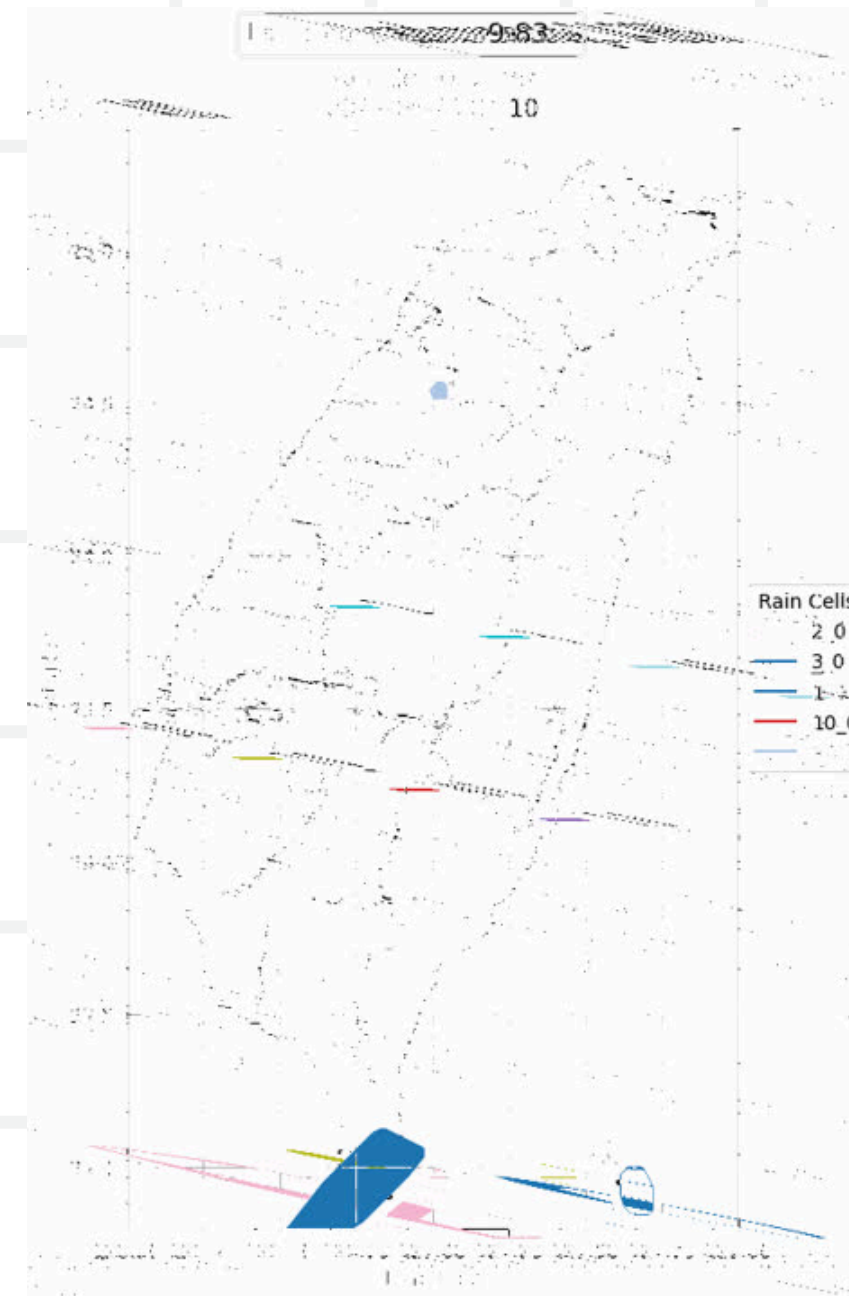
比較結果



Setting

```
Param VIL  
Min area: 2.5  
Smooth width: 3  
Min thresh: 0.5  
Max thresh: 80.0  
Bin width: 3.0  
fall back yes
```

VIL 0.5

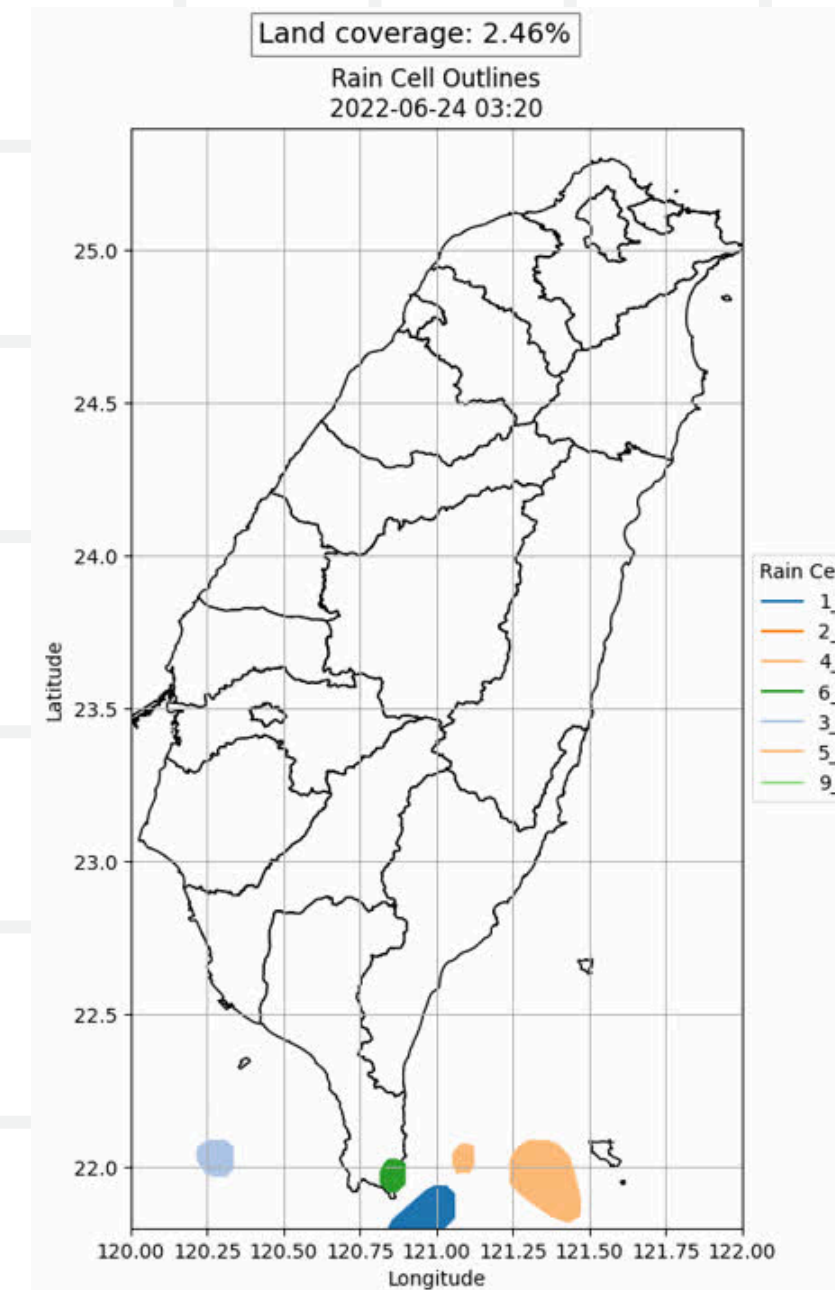


比較結果

Setting

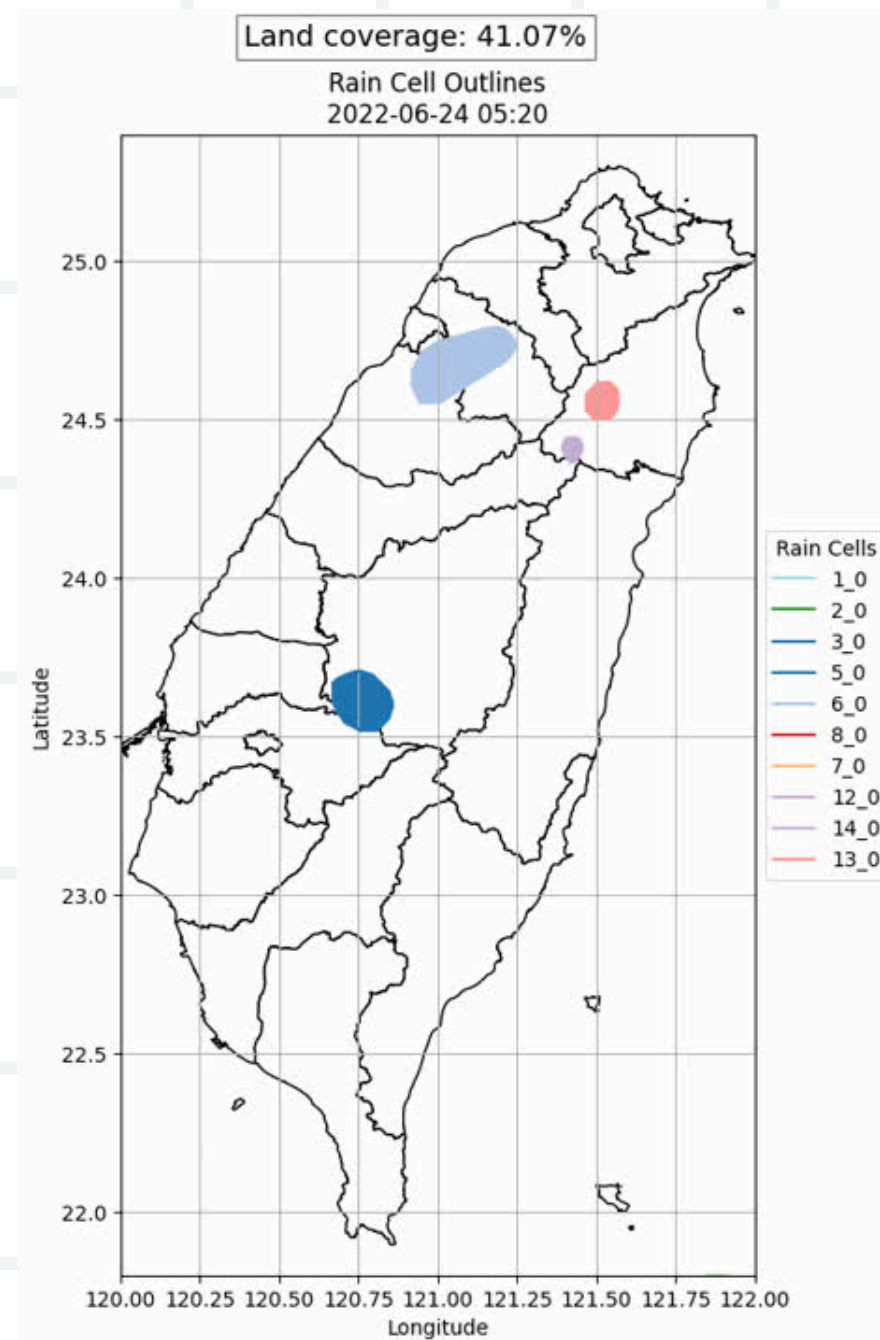
```
Param VIL  
Min area: 2.5  
Smooth width: 3  
Min thresh: 0.75  
Max thresh: 80.0  
Bin width: 3.0  
fall back yes
```

VIL 0.75

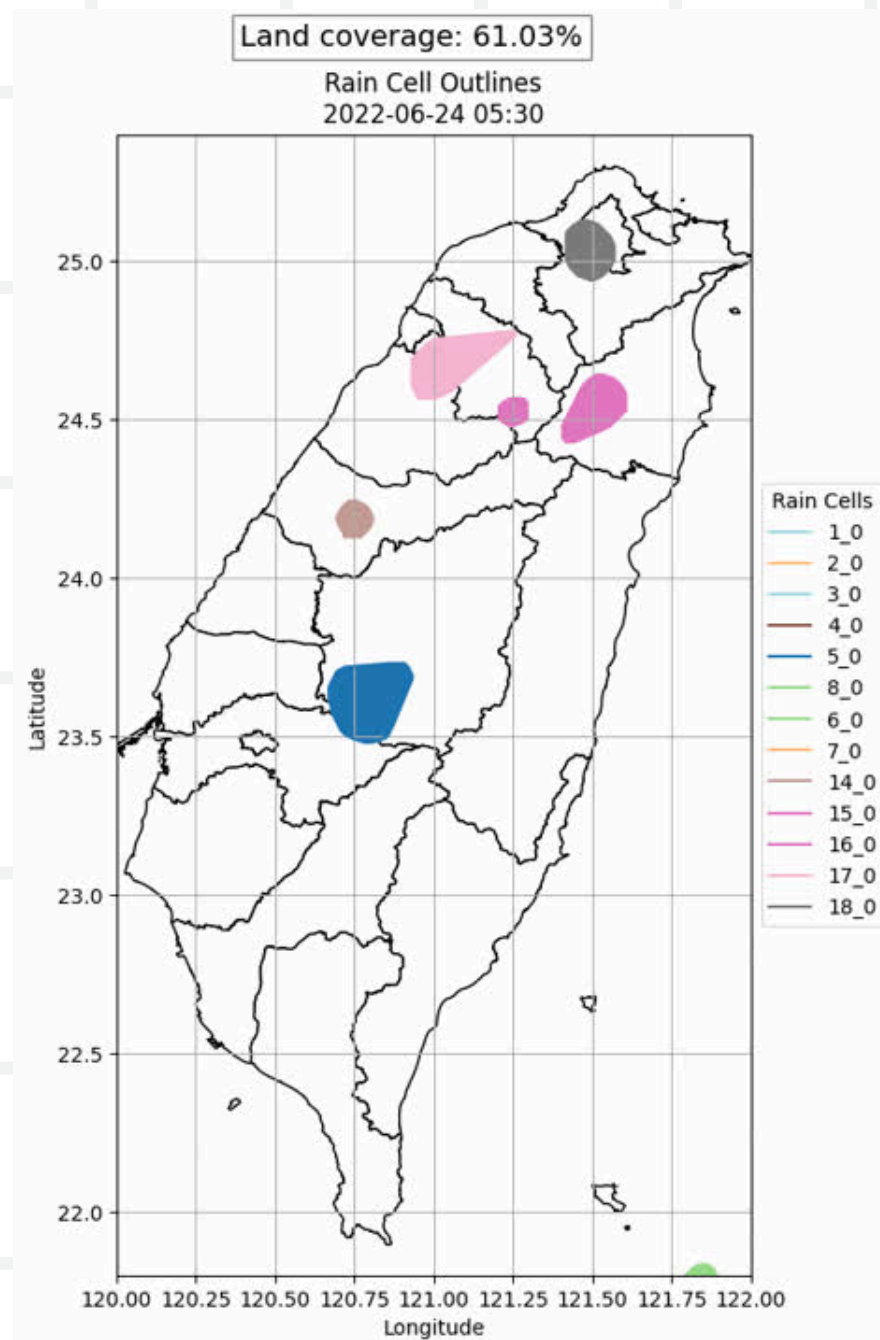


比較結果 VIL 0.75

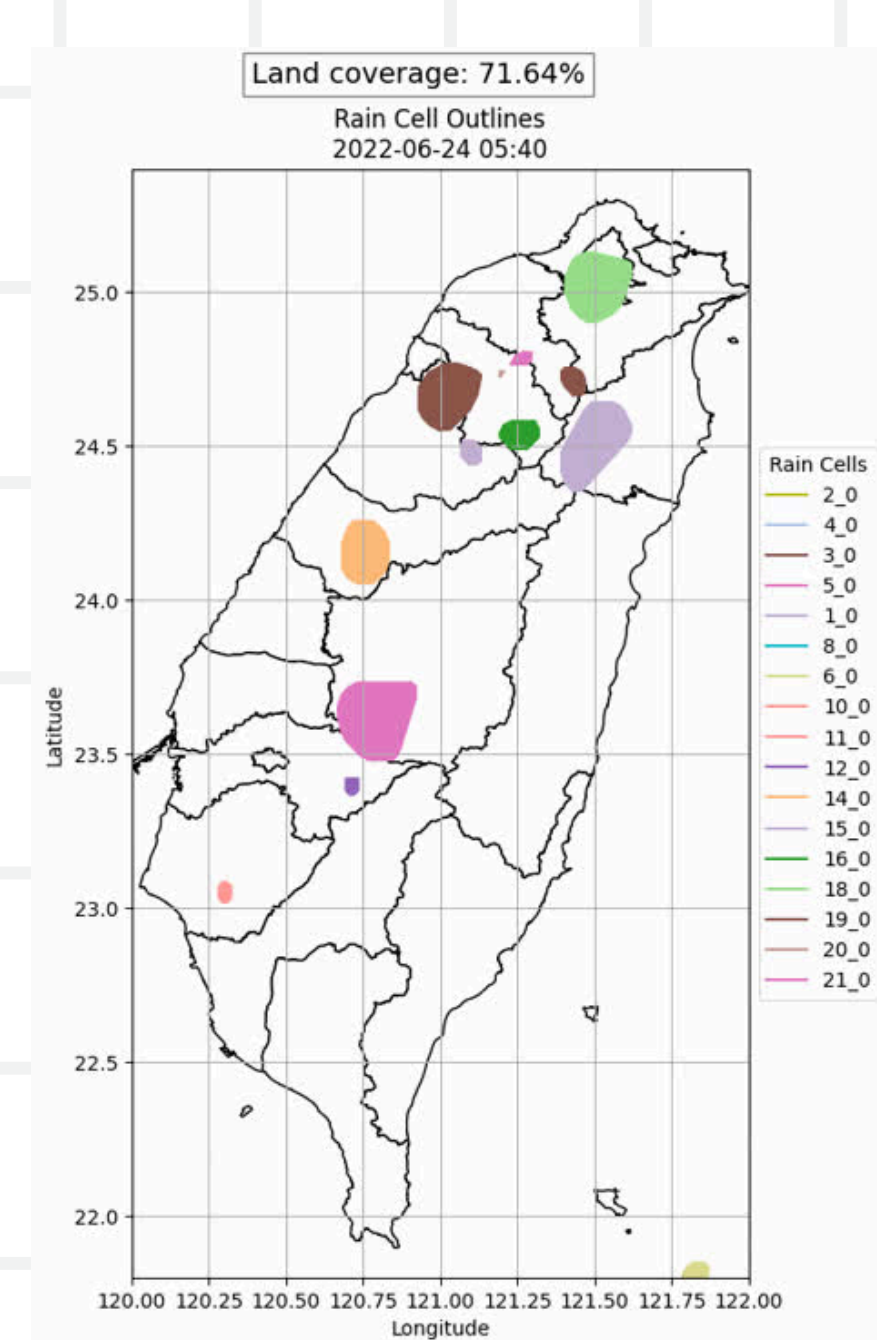
ax * 1.2



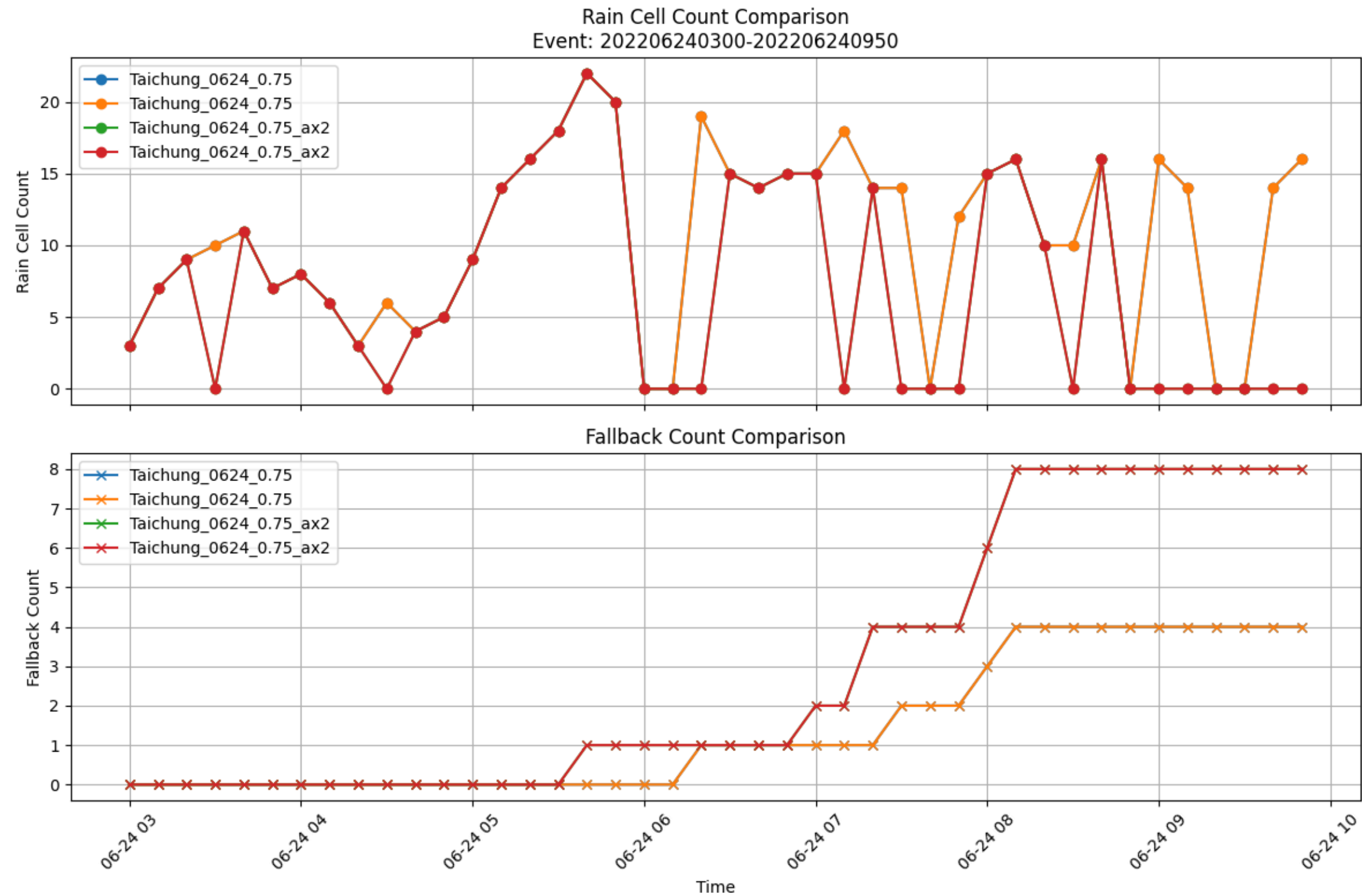
ax *1.5



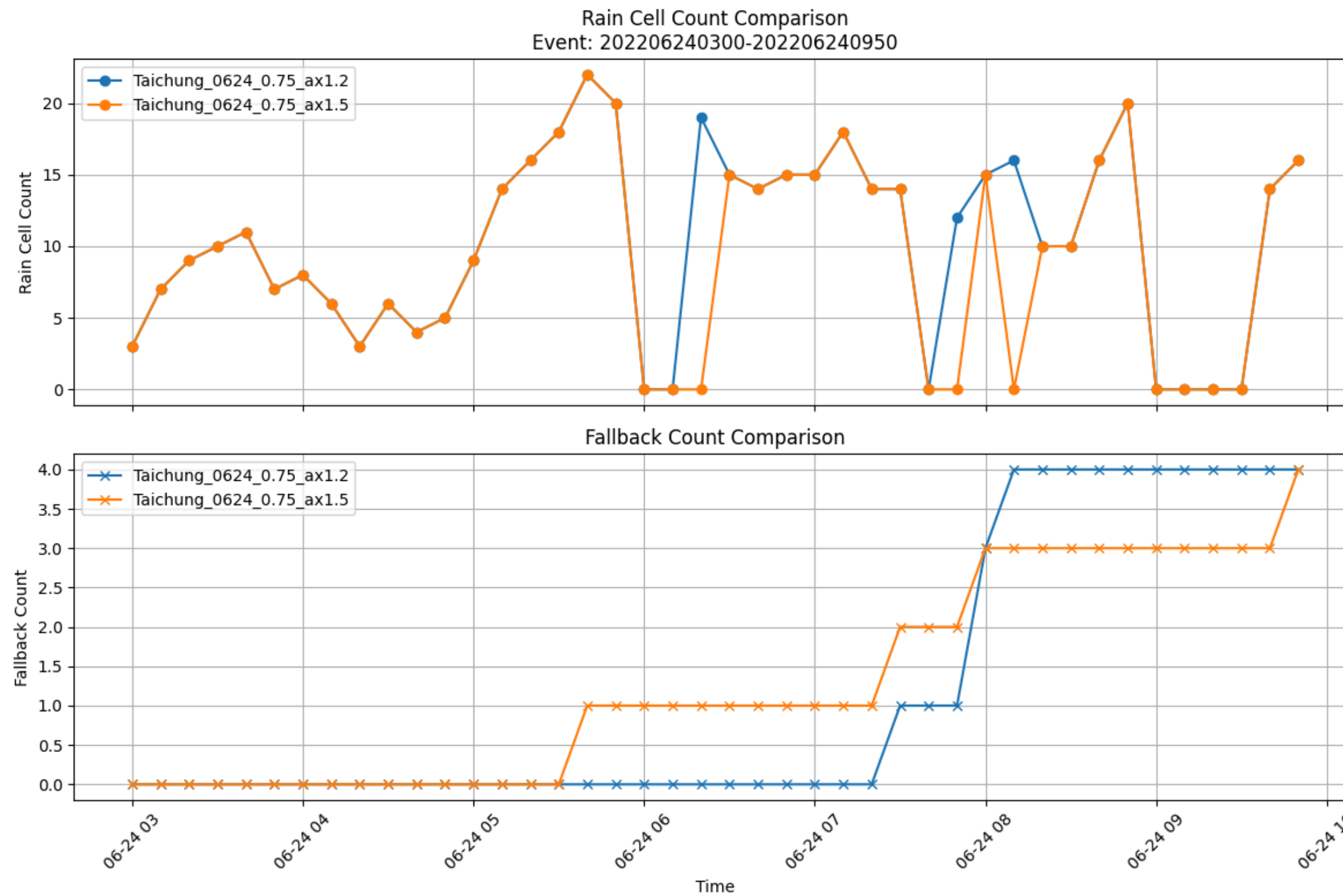
ax *1.7



ax1 vs ax2



ax1.2 vs ax1.5



ax1 vs ax1.2

